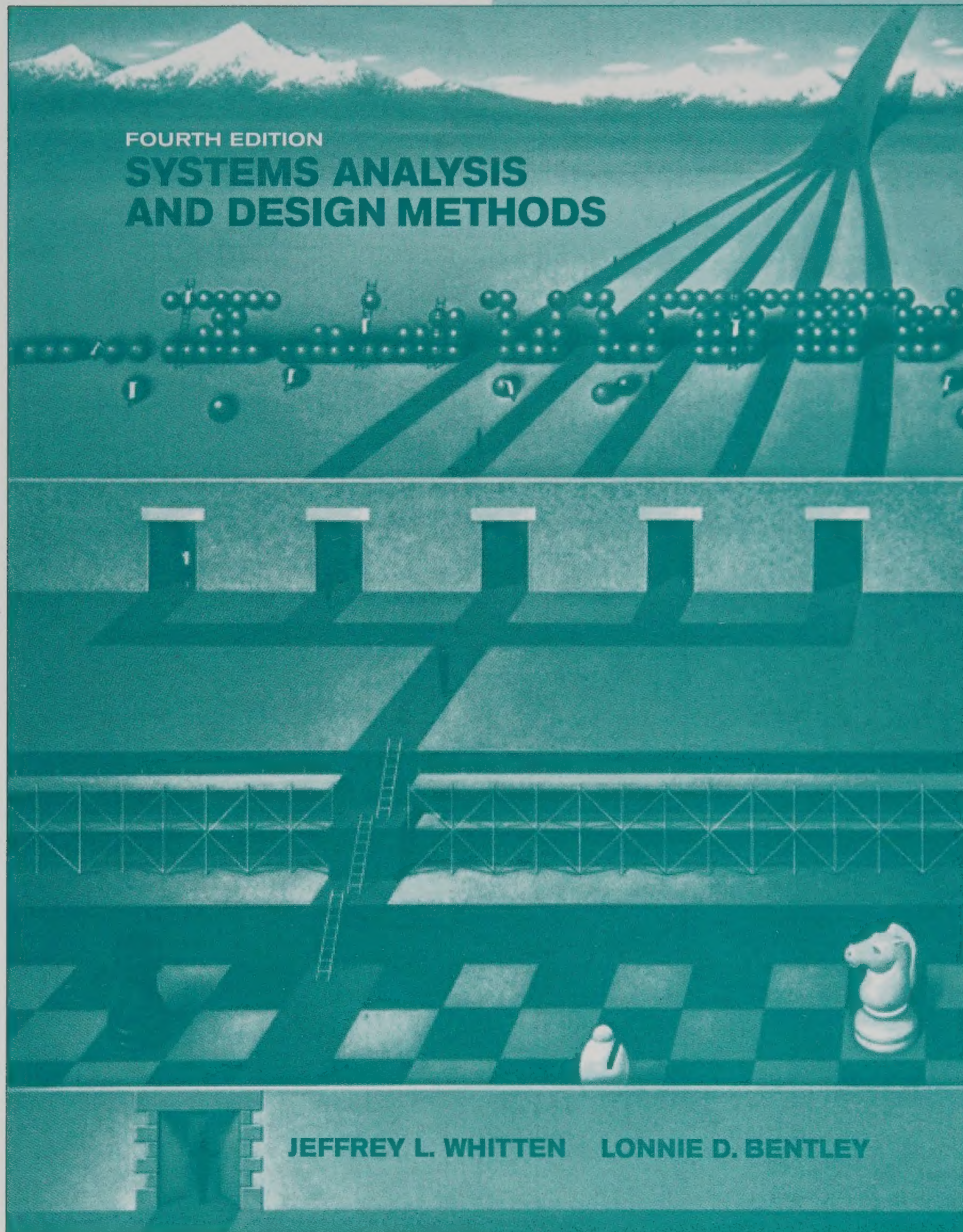
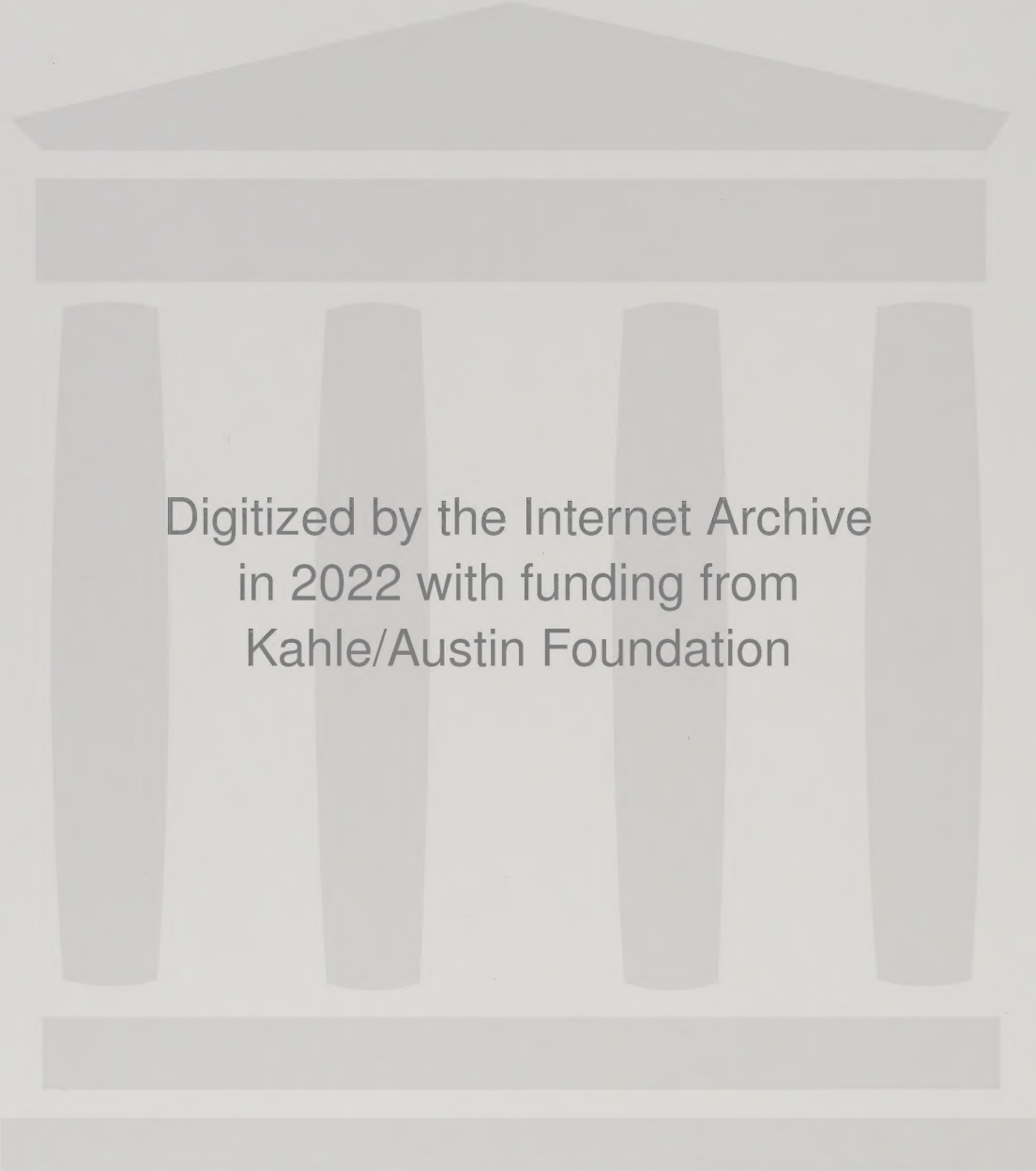


Application Cases in Systems Analysis and Design

for use with



Prepared by
Kevin Dittman



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Application Cases in
Systems Analysis
and Design
for use with
Systems Analysis
and Design Methods

Fourth Edition

Jeffrey L. Whitten
Purdue University

Lonnie D. Bentley
Purdue University

Prepared by
Kevin Dittman
Purdue University



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Application Cases in Systems Analysis and Design
SYSTEMS ANALYSIS AND DESIGN METHODS

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SYSTEMS ANALYSIS AND DESIGN METHODS

Jeffrey L. Whitten and Lonnie D. Bentley

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From the Author

The purpose of Projects and Cases is to provide students with a practical, first experience in systems analysis and design within the academic environment.

One of the significant difficulties encountered by instructors in a first systems analysis and design course is the development of practical assignments that will train students in the tools and techniques of the profession. Systems analysis and design courses are not like programming courses. Exercises for a programming course are very similar in nature. While the objectives change for each programming project, students perform similar activities for every program - planning, modular decomposition, logic design, coding, debugging, testing, etc.

On the other hand, because the skills taught in a typical systems analysis and design course are much more diverse, the activities required to complete each exercise are dissimilar. For example, the data flow diagram assignment is very different from the output design assignment. Also, there is a need for a greater number of projects or exercises in a systems course than in a programming course. Finally, to fully appreciate the nature of systems work, most instructors have learned that they need to interrelate the exercises in the form of a single project to help the student appreciate the integration of the tools and techniques and their applications.

This Casebook offers both the case approach and another alternative that has been class tested and refined. It offers the following options:

- ◆ **Running Case.** This option is a typical case study. Most of the necessary information has been provided. Your instructor may modify these cases or request that you do so. I have also controlled the size and scope of the project. Two different cases have been provided for your use.
- ◆ **Build Your Own Case.** This option is a controlled approach that allows students to construct their own systems analysis and design project from their own work experience. This work experience does not have to be in data processing. The students' familiarity with the business environment alone increases their interest and motivation. Perhaps more important, the instructor doesn't have to develop the case and can spend more time helping the students with problems. Mechanisms are provided to control the size and scope of projects to ensure fairness to all students. Additionally, as an instructor, I have found the diversity of cases make teaching the course more interesting and invigorating. Enthusiasm from the instructor is contagious to the student - that is true of any course!

For both of these options, I have divided the students' projects into milestones. Milestones are cross-referenced to the Whitten/Bentley text.

Each milestone is described in terms of synopsis, objectives, prerequisites, project, and deliverables. Most milestones also include suggested references for additional information in the text listed above.

For those courses that require or offer a joint technical or professional report writing activity with their systems analysis and design course, I offer guidelines and requirements for organizing systems specifications into a formal report. These are often offered as the advanced option of a particular milestone.

The projects can be done individually or by teams. The growth in requirements is not linear since systems integration imposes additional requirements on team members to coordinate their

efforts. It is highly recommended that examinations also be used for courses that use the team approach.

Finally, for the instructor and students' convenience, I also provide the following tools:

- ◆ Blank forms and templates are not only provided in the casebook, but also can be downloaded from the textbook's web site [<http://www.tech.purdue.edu/textbooks/sadm/>]. You have my permission to duplicate them (contingent upon student purchase of this casebook). I feel this approach is preferable to forcing the student to pay for multiple copies of the forms and charts, since many students either have access to similar documents or instructors have personal preferences concerning forms and charts.

I ask your cooperation in not duplicating these materials for courses which have not adopted this casebook. Thank you.

Time Constraints

Different schools have different time constraints for projects. This is especially true for those schools on the quarter system as opposed to the semester system. At Purdue, we successfully cover the Whitten/Bentley text and casebook in a single semester.

One additional time issue is in order. Each of the case options is based on a true business problem that the author has experienced and results in a realistic case. However, the requirements to complete the case will not result in a complete, systems solution. Avoid the temptation to expand the requirements. Great care has been exercised to ensure that project size is reasonable for teams of different sizes (as well as individual projects), especially considering the impact of a milestone on future milestones.

I welcome suggestions for improving this casebook. They should be communicated, in writing, to the publisher.

Acknowledgements

I would like to thank Jeffrey Whitten and Lonnie Bentley for giving me the opportunity to share my experiences with other faculty and students. Without their constant motivation, guidance, and support, I sometimes wondered how I would ever survive. To Rick Williamson and the Irwin/McGraw-Hill staff, thanks for your patience and understanding. I would also like to thank all my Purdue Computer Information Technology Students, especially Erica Web and Lori Morris, for all their help in producing the case studies. And to my dear wife Diana and sons Brandon, Bryan, and Andrew, thanks for putting up with my endless hours of working and pounding on the computer until way past the midnight hour. I'm done for awhile. It's time for dad and the family to go to Disney World.

HOW TO USE THIS CASEBOOK: PROJECT DESCRIPTION AND SCHEDULE

Introduction

Systems analysis and design are applied disciplines. What is the best way to learn an applied discipline? Consider another applied discipline with which most of you are already familiar - computer programming. You learned programming concepts, techniques, and syntax from your textbook and lectures. But did you really learn everything you needed to know from the textbook and lectures? Not really! You really learned computer programming by writing programs and making mistakes - you "learned by doing." This is the approach I will use in this casebook.

Unfortunately, systems analysis and design projects present some difficulties not encountered in programming course projects. A meaningful systems project has a greater scope than a typical programming project. Part of learning systems analysis and design is learning to deal with the size of systems themselves. On the other hand, a typical system can easily become so large as to overwhelm you and discourage the learning experience. Therefore, I must impose constraints to limit the size, but not so much as to eliminate the "systems" flavor of the work.

Systems analysis and design can be factored into at least three skill subsets: systems analysis and design concepts, systems analysis and design tools and techniques, and the process of doing systems analysis and design. The first two skill sets can be taught and demonstrated in a textbook and lectures. The last can only be appreciated in an applied project environment. That is what this casebook is all about. There are two popular approaches to a first systems project:

- ◆ A "live" project with real clients. This approach usually leads to projects that are too small to reinforce the "system" flavor needed in the course. Why? The client needs to get something out of the project - usually, working programs. By adding programming to the course requirements, the systems analysis and design aspects are slighted and the project's systems scope is reduced in order to complete the project within one term. Additionally, you are forced to deal with political and behavioral aspects at the same time as you are trying to learn the concepts, tools, and techniques. That really isn't fair to you. Therefore, I cannot suggest this approach for a first course project in systems analysis and design.
- ◆ A "simulated" project. This approach is usually based on a case study that is either provided to you or developed by you. The instructor can control the size of the project and restrict it to exercises that reinforce systems analysis and design material. This is the approach that I will use.

Project Overview

In your programming course(s), specifications for programming projects were probably supplied by your instructors (or by authors of programming textbooks). Where do these programming specifications come from in the "real world"? They are usually prepared by a systems analyst or

programmer/analyst. In this project, you will perform a series of activities that result in a superset of such programming specifications, this time for a typical information system.

I should probably prepare you for a surprising revelation. You probably have never received programming (or system) specifications that are as detailed and precise as those that you will prepare in this project. Before you unfairly criticize your programming instructors for not providing such specifications, consider the amount of time that you will have put into this project. Few instructors, with their many teaching, research, and administrative responsibilities, have time to complete such specifications. And few publishers are willing to dedicate the number of pages required for such specifications in the typical programming textbook. Nonetheless, this project represents the "ideal" specification for systems and the programs included in those systems.

There are two alternative case approaches in this casebook. You must select a project from one of these two alternative approaches. However, your instructor may restrict you to one of the alternatives, depending on his/her preference.

The Build Your Own Case Project Alternative

In this alternative, you select a business and application from your personal background. While it might be nice to select this business and application from a real data processing experience, it is not a requirement. Most of you have work experience outside of data processing and, in some sense, that work experience may provide a better foundation on which to base a business and application.

After your business and application have been approved, you will build a case around them. In the early stages of this project, you will probably tend to make your scope too large, not realizing the implications until later in the project, usually too late. To prevent you from becoming overwhelmed, we will control the size of your case study so that it both demonstrates all of the characteristics of a modern information system and remains small enough that you get an opportunity to practice all of the necessary systems analysis and design skills.

The advantage of this case study approach is that the business and application are at least somewhat familiar to you. You don't waste time trying to understand an instructor's or author's business and application. At the same time, you are not burdened with the politics and behavioral components of a "live" project.

Some students pick projects that they wish to implement at a later date. If you choose to do this, I must at least insist that you add any components to the case study that I require to demonstrate features of modern information systems, whether those features are needed or not. How far beyond those minimum requirements you wish to go is between you and your instructor.

The RUNNING Cases Project Alternative

In this alternative, two case projects have been provided for you in this workbook. You must choose one of the cases, unless your instructor has a preference.

I have controlled the size and scope of these projects so that they will not become too large and overwhelm you. Yet, both cases include characteristics common to all good information systems.

You will have the opportunity to practice all of the necessary systems analysis and design skills covered in the textbook.

The advantage of this approach is that since several people are doing the same project, you can learn from other students. Additionally, the instructor only has to become familiar with one project and can provide help to all students in the class. After selecting from one of the two project options, refer to the guidelines provided throughout this casebook's project milestones for that project option.

One characteristic common to all cases in this book is that the resultant project is realistic but controlled. The requirements are abbreviated so that the project can be completed within a reasonable amount of time. Avoid the temptation to do more than we require. Take our word for it - the project, in order to exhibit reasonable systems characteristics, is substantial. Increasing the requirements in one milestone will amplify the requirements in subsequent milestones and eventually overwhelm you.

Project Objectives

It is practically impossible for full-time students (or part-time students with jobs) to perform both systems analysis and systems design within the scope of a single-term project - not unless you drastically reduce the scope of the project and, subsequently, the "systems" nature of the project. This is, first and foremost, a learning experience. If you want to accomplish anything, you must not try to accomplish everything.

The specific objectives of the project are as follows:

1. After completing the project you should have a better understanding of the information system: its users, its business purpose, capabilities, and components. The systems analyst analyzes, designs, implements, and supports such systems.
2. After completing the project you should be able to apply the tools and techniques of systems analysis and design, with little supervision, to a "live" project.
3. After completing the project you should have better business and technical communications skills. The analyst is essentially a mediator between the business user and computer programmer. Systems analysis and design is as much a communications framework as it is a process of building systems.
4. After completing the project, you should be a better programmer. What? That's right. You will be a better programmer if you understand how to read and modify systems specifications prepared by an analyst. Programmers frequently find themselves re-doing incomplete or inadequate design specifications.

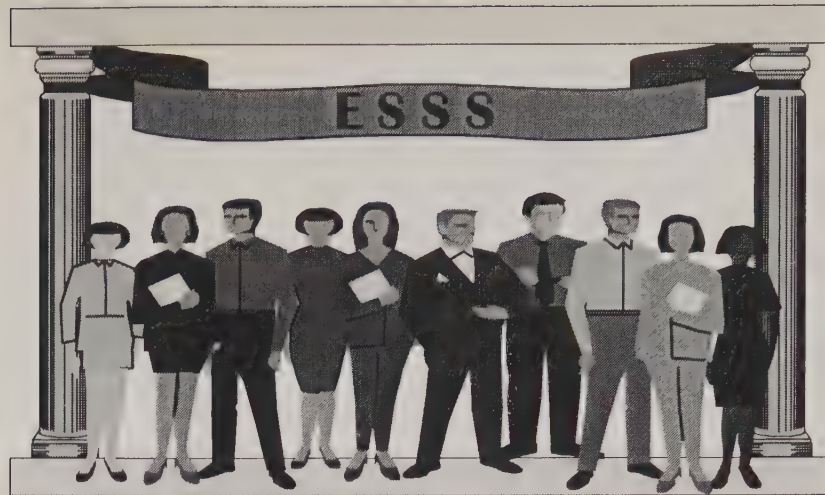
Assignment and Schedule

Your project is divided into milestones. Some milestones may be completed as an in-class exercise. Other milestones may be completed by your instructor (to save time) and reviewed in class. Most milestones will be completed by you as project assignments.

The Value of This Project to You

Before we begin the project, I want to encourage you to do your very best work. Put together a truly professional, "spit and polished" document - you'll realize an unexpected benefit. In a job search, you'll discover that the document serves as a "portfolio" of your skills and potential as a systems analyst. It can communicate professionalism and the pride that you take in your work. Don't be surprised if you get some systems documentation responsibilities as a part of your first job or a summer job - we have repeatedly discovered this to be true!

Good luck and enjoy the project!



EMPLOYEE SELF SERVICE SYSTEM

INTRODUCTION

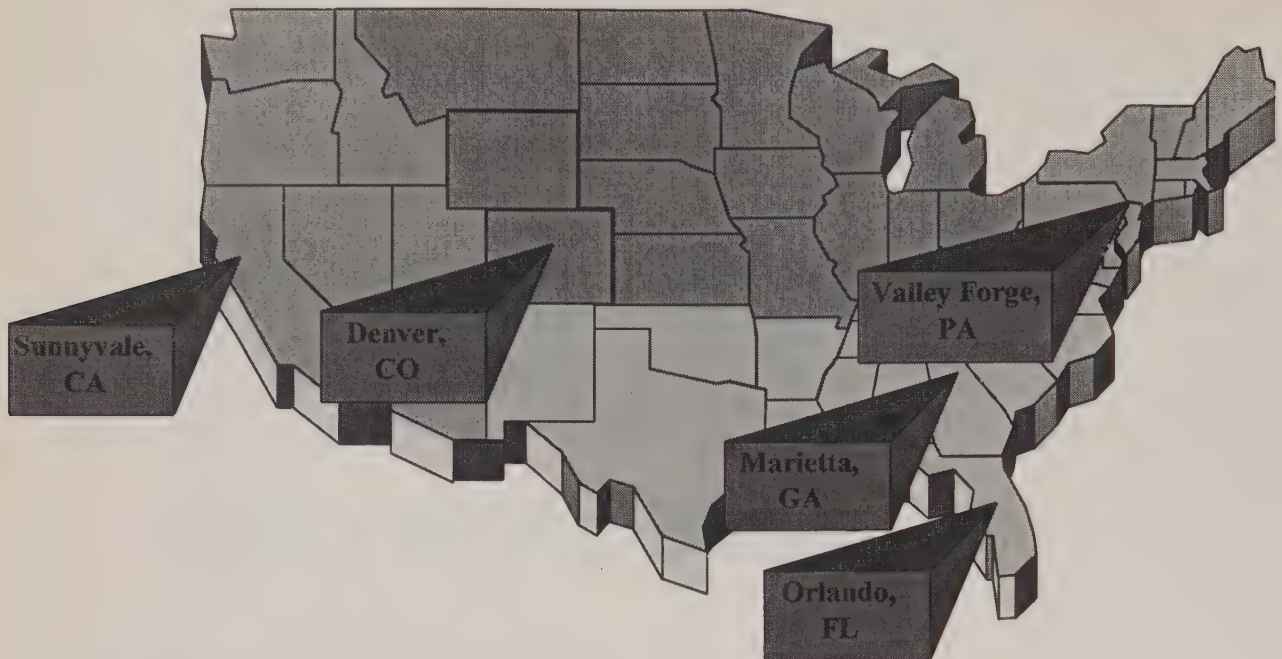
In this section you will learn the necessary background information to be able to complete the twelve milestones of this case study. This information includes a history of the business, a description of the business's current facilities, and the descriptions of the problems that triggered the project.

Case Background

A-1 Information Systems headquartered in Orlando, Florida employs more than 4,000 employees throughout the United States. A-1 IS provides leading edge technologies, distributed computing, mainframe, micro, communication, and consulting services to its parent company A-1 Corporation, headquartered in Bethesda, Maryland as well as to external customers including the U.S. Government. In addition A-1 IS is responsible for the development and support of all their internal systems that support their day-to-day business processes and operations.

CASE STUDY 1 - EMPLOYEE SELF SERVICE SYSTEM

A-1 IS currently operates in five sites across the nation and they are as follows:



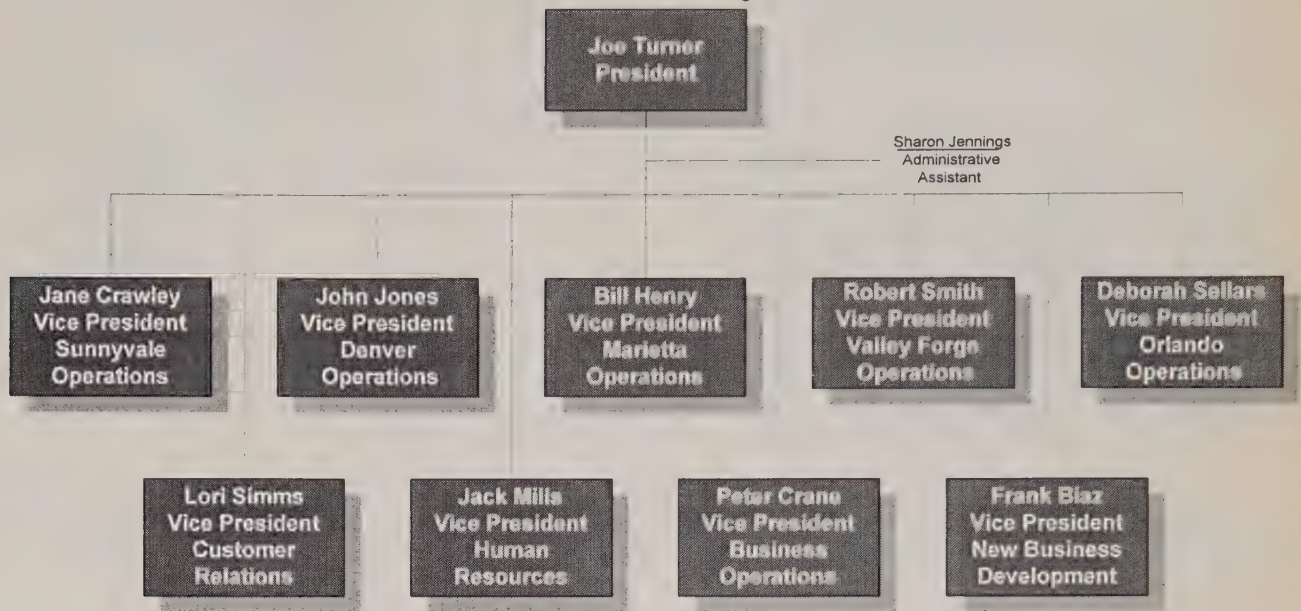
Sunnyvale, CA -	750 employees
Denver, CO -	675 employees
Valley Forge, PA -	1050 employees
Marietta, GA -	350 employees
Orlando, FL -	1175 employees

Each site is responsible for servicing and supporting the customers in their region as well as their internal employees.

A-1 IS has experienced a 25 % growth increase the past two years in terms of employees, and long range projections show that trend continuing for the next five years.

Organization Structure

A-1 Information Systems



Information Systems Facilities

The information architecture for A-1 IS standardizes on the following:

- Micro - Hewlett Packard Pentium Class.
 - 32 megabytes of RAM
 - 2 gig disk drive
 - 17" monitors
 - Operating system - MS Windows NT 4.0
 - Tools - MS Office suite. (97)
 - Internet Browser - Netscape 3.0
- Server - Hewlett Packard 9000 class
 - 256 megabytes of RAM
 - 25 gig available in hard disk storage
 - Operating system - MS Windows NT 4.0
 - Database Management System - Oracle 7.0
- Printers - Hewlett Packard LaserJet and PaintJet classes.
- Netscape's Commerce Server is used to provide data security through encryption.

Each site has comparable hardware and software and the employee to micro ratio is 1-to-1.

The Problem

Due to the tremendous growth the company has experienced in recent years, it has recognized that to ensure the continued success of servicing their internal; as well as external customers, it needed to develop a strategic plan and vision for the use and modernization of its computing resources. To further emphasize the need, was the challenges for creating common systems (centralized systems) across all five IS sites to support the business practices which are common if not identical across the sites.

In January of 1997 a strategic plan to modernize the company's resources was presented to executive management. This document included a multi-phased plan to reengineer the current systems to use state-of-the-art technology and provide a showcase of systems that eventually could be delivered across the whole corporation.

Phase one of the plan consisted of reengineering all Human Resource related systems, which included Employee Tracking, Staffing, Benefits, Compensation, and Payroll. Task one of this phase is the development of the Employee Self Service System (ESSS). A system that will house the repository containing the employee master data, which is the foundation for providing a common set of automated, integrated, platform independent system solutions for Human Resources. This system would provide the capability for each employee to maintain their own information in relation to address and telephone number changes, beneficiary changes, United Way deductions and Savings Bond deductions. Current practices now have each of these changes being processed by an extensive manual effort which includes the filling out of forms and the inputting of the data by Human Resource administrators. There is often a several day lag time before the employee's information was updated online once the forms were submitted by the employee. This caused several problems including, employees unable to quickly locate other employees, company mailings being sent to the wrong addresses, payroll checks unable to be delivered, paper-based company telephone books out of date almost as soon as they were printed, and less than ideal United Way and Savings Bonds contributions. By providing the capability for an employee to update their own data real-time, the problems mentioned above can be reduced, if not eliminated, plus the resource strain on the Human Resources department can be reduced by one person for each site.



MILESTONE 1 - SURVEYING THE PROJECT

Synopsis

The purpose of the survey phase is threefold. First, the survey phase answers the question, “Is this project worth looking at?” To answer this question, the survey phase must define the scope of the project and the perceived problems, opportunities, and directives that triggered the project.

In this milestone you will prepare a *Request for System Services*, which is the trigger for the Survey Phase. Also, you will use fact finding techniques to extract and analyze information from an interview to determine project scope, level of management commitment, and project feasibility for the Employee Self Service System (ESSS). With these facts and facts obtained from the Case Background, you will have the necessary information to complete the *Problem Statement Matrix* and construct the *Project Feasibility Assessment Report*.

Objectives

After completing this milestone, you should be able to:

- ⇒ Complete a *Request for System Services* form, which triggers the survey phase.
- ⇒ Analyze a user interview and extract pertinent facts, which can be used to assess project feasibility.
- ⇒ Complete a *Problem Statement Matrix* documenting the problems, opportunities, or directives of the project.

⇒ Prepare and understand the structure and content, of the *Project Feasibility Assessment Report*.

Prerequisites

Before starting this milestone the following topics should be covered:

1. The Survey Phase - Chapter 3, pages 84-87, Chapter 4, pages 128-137 of the SADM 4th ed. textbook.

Project

We have been contacted by Mr. Jack Mills, the president of Human Resources for A-1 Information Systems, to perform analysis and design of how they manage the employee information for A-1 IS. In this assignment you first need to assist Jack Mills in preparing a *Request for Systems Services*. Secondly, by analyzing the interview transcripts, determine the feasibility of the project, level of management commitment and project scope by using fact-finding techniques and the necessary communication skills to compose the *Problem Statement Matrix & Project Feasibility Assessment Report*.

Mr. Mills was gracious enough to allow us to record our interview session and Exhibit 1.1 is a copy of the transcripts. Refer to the **Case Background** found in the **Introduction** and the interview transcripts for the information necessary to complete the following activities.

Activities

1. To complete the *Request for System Services* form, use Figure 4.7, page 130 of the SADM 4th ed. Textbook as an example. Use information from the case background only when completing the form. Make assumptions where necessary. We have provided for your convenience a template of the form on the textbook's web page [<http://www.tech.purdue.edu/textbooks/sadm/>], as well as a hardcopy at the end of this milestone.
2. To complete the *Problem Statement Matrix*, use Figure 4.8, page 132 of the SADM 4th ed. textbook as an example. Use the interview with Jack Mills and the case background for the basis of your information. Make assumptions where necessary. Place yourself in the shoes of Jack Mills. Which problems do you believe are the most visible, have the highest visibility, and how should they be ranked? Try to determine the annual benefits. State assumptions and be prepared to justify your answers! Finally, what would be your proposed solution based on the facts you know now? We have provided for your

convenience a template of the matrix on the textbook's web page [<http://www.tech.purdue.edu/textbooks/sadm/>], as well as a hardcopy at the end of this milestone.

Deliverable format and software to be used are according to your instructor's specifications. Deliverables should be neatly packaged in a binder, separated with a tab divider labeled "Milestone 1" and accompanied with a Milestone Evaluation Sheet (appearing first) found in Appendix A.

References:**Case Background**

Workbook Introduction

Transcripts of Interview with Jack Mills

Exhibit 1.1

CS 1 - MILESTONE 1: SURVEYING THE PROJECT

Deliverables:

Request for System Services:

Due: __/__/__

Time: _____

Problem Statement Matrix:

Due: __/__/__

Time: _____

ADVANCED OPTION

For the advanced option, prepare a Project Feasibility Assessment Report. Use the information provided by the case background, the user interview, and the completed problem statement matrix from Activity 2. Use Figure 4.9, page 137 of the SADM 4th ed. textbook as a sample outline and a copy of the outline is also presented in exhibit 1.2. Be sure to include Gantt charts for the project schedules. Information on Gantt charts can be found on pages 602-609 of the SADM 4th ed. Textbook. Information on fact-finding techniques can be found on pages 623-639 of the SADM 4th ed. Textbook. And, information on feasibility analysis can be found on 643-659 of the SADM 4th ed. Textbook.

Project Feasibility Assessment Report:

Due: __/__/__

Time: _____

Milestone's Point Value:

The following is a copy of the transcripts of an interview between Mr. Jack Mills of A-1 Information Systems and Kira Webster, a computer information systems student. This was the initial interview with Mr. Mills and its goal was to obtain facts about the problems and opportunities which triggered the project request, plus other general information which could be used to prepare the *Problem Statement Matrix*.

Exhibit 1.1

Scene: <i>Kira Webster, computer information systems student, is meeting with Jack Mills, Vice President of Human Resources for A-1 Information Systems, at his office, room 4216, building 16. Ms. Webster scheduled the interview with Mr. Mills in response to his request for developing a new employee information system.</i> Jack: Good Morning! You must be Kira Webster. Kira: Yes, I am sir. Are you Mr. Jack Mills? Jack: Yes. Please call me Jack. I'm glad we could get together today. Things are quite chaotic around here. Kira: What do you mean? Jack: Our executive steering committee is very anxious to set in motion our plan for reengineering and modernizing our systems and computing resources. The first phase of that plan concentrates on	Human Resources and that is why you are here. Kira: That's sounds like an enormous task, but I love challenges. Jack: It is monumental, but we tried to simplify the task by breaking it up into smaller pieces. The first piece deals with the tracking and management of employee information. Kira: What is your current system like? Jack: My Employee Relations manager, Dotty Jones, explained the process in detail to me yesterday. I don't interact with it myself, but it appears there are a lot of inefficiencies and its operating costs are exorbitant. The system itself is a combination of manual and automated processes. The automated processes reside in both micro computers and legacy mainframe systems. Kira: Will this system replace the legacy systems?
---	---

Jack: Not initially. Interfaces will have to be built for the legacy systems, Payroll, Labor, and Benefits.

Kira: Could you please describe for me the business processes that will be included in the system?

Jack: I think the best way to explain the process is to start from the beginning. The first day an employee reports to work they go through an orientation program. During that program they are required to complete some personnel forms that include information such as: addresses, phone numbers, emergency contact information, and beneficiary information. Also, at that time they can elect to have a charitable deduction, plus elect to have money deducted from their paycheck to buy savings bonds. This data is then input into a mainframe system by one of my administrators. If there are any changes to be processed, the employee must complete a form and submit it to us to be input into the system.

Kira: You mentioned a micro application. Where does that come into play?

Jack: Oh. On a quarterly basis we produce and publish an employee telephone listing,

sort of a company telephone book, which consists of the employee's work telephone number and work location. This information resides on a microcomputer that is maintained by Alice Cockran, an administrator, who works for Dotty. Because we are such a dynamic and growing company, Alice spends 30 percent of her workweek maintaining the information. Each quarter we produce 5000 copies of the book and distribute them across the company. Currently this process costs the company \$27,000 a year. The sad thing is, the book is probably already out of date the day after its published.

Kira: What do you mean?

Jack: We are a large and growing company with sites all over the nation. Every day at least someone is hired, transferred, moves offices, or is terminated. These events all trigger a change to the listing.

Kira: It sounds like a losing battle.

Jack: We do our best. Besides you are going to help us win it. Right?

Kira: I'm going to do my best! Is there an automated interface

between the system that houses the employee information on the mainframe and the micro system?

Jack: I wish there were! Currently we have to input the data twice. Once into the mainframe and then into the micro application.

Kira: Let's switch gears and talk about the people that use the systems. You mentioned that Alice is responsible for the telephone listing system, who are the other people involved?

Jack: If you really think about it, everyone is affected or uses the system in some way. Whether it is directly or indirectly. Currently only Human Resources' employees can directly use the systems. But I would like to see that changed. There is no reason each employee can not be responsible for maintaining his or her own information. I mean why can't the employee get on the system and input his or her own address or telephone changes? The data would be more current and it reduces the labor support in my organization. Currently, to process each change in employee information it costs me \$15.00 in labor and computing costs. We processed

11,500 changes last year alone! That adds up to some substantial costs.

Kira: Before I forget to ask, may I have a copy of your organization chart and who of your organization will be working with me on this project?

Jack: Good questions. I will have my secretary send you a copy of my org. chart [*the organization chart appears following the interview*]. As for your direct contact, Dotty Jones will fulfill that role. But, I don't have a problem of you talking to anyone in the HR department if it will help you.

Kira: Thank you. For this project to be a success we must work as a team and you the user certainly must be involved. Now let's talk about the technology you are currently using. Does everyone use a PC?

Jack: Yes we do. We supposedly have the latest and greatest personal computers that come with the standard word processor, spreadsheet, and database. If you need detailed configuration information, I'm sure Dotty can provide that to you.

Kira: Do you have electronic mail and do you use the Internet?

Jack: We are heavy E-mail users. I think we use Microsoft Exchange. As for the Internet we use it too. There are volumes of HR related information on the web that has proven to be quite valuable, especially on legal issues.

Kira: What browser do you use.

Jack: Excuse me?

Kira: I'm sorry. What software do you use to enable you to access the web?

Jack: Oh, Netscape. That's called a browser? Interesting.

Kira: You mentioned the employee filled out forms to provide their information. May I get samples of those forms?

Jack: Sure. I will have Dotty get you some of the blank forms.

Kira: If you don't mind Mr. Mills, samples of completed forms can be quite valuable, plus they may indicate other opportunities where I can help.

Jack: I don't have a problem with that. But, you realize you will be dealing with sensitive data and you must make a conscious effort to respect that.

Kira: I am fully aware of the ethics involved Mr. Mills and I will

fulfill my professional responsibility to the best of my ability.

Jack: No doubt you will.

Kira: I have just a few more questions so hopefully we won't be much longer.

Jack: Shoot.

Kira: You stated earlier multiple sites existed, will all those sites be using the system?

Jack: We have five sites currently; here at Orlando, Denver, Sunnyvale, Marietta, and Valley Forge. All of those sites will use and currently use the system. Each site has a satellite HR staff that processes the transactions for their respective employees. All sites use the same type of equipment. When you buy in large volumes you get big discounts.

Kira: I see. Which site currently is responsible for technical support of the system?

Jack: Orlando.

Kira: Will that still be the case for the new system?

Jack: I have every reason to believe so unless the technical people tell me different. Right now all production employee data is housed in Orlando.

Kira: Can we talk a few minutes about the employee data on the mainframe and some of the problems associated with it?

Jack: Sure. First of all we have multiple copies of the same employee data across multiple systems. Whether it be the payroll system, labor system, or benefits system, each application has their own database of employee data. Shouldn't we have just one integrated database that all applications could access? This would eliminate the problems of the data being out of sync across the applications, excessive interfaces among systems, and current data not available real time.

Kira: Yes it would and that is definitely something I will look into. Mr. Mills what is your vision of the new system?

Jack: Good question. I see a system that is easy to use, intuitive, with a graphical user interface. It is accessible from every desktop in the company and the interface is consistent no matter what platform you are using. I have long had the idea that the interface should have a folder metaphor. In other words, if you were an employee entering the system, you would be

presented with a series of folders or tabs if you will. Each folder would consist of a certain type of information, such as: emergency information, beneficiary information, deductions, and in the future benefits, training, and so on.

Kira: That's a good vision. What else can you tell me?

Jack: Of course the system would have to have security to prevent unauthorized access of an employee's information. Only HR administrators and the employee's manager should be able to access the employee's information. Also, the system should contain an on-line company telephone book, containing phone numbers and work locations, so that every employee can easily locate and contact another employee no matter where they are located. The system should also be a place to view the employee's organization structure. Meaning, the system contains the information of who the employee's manager is, and other employees who report to the same manager.

Kira: Very good. Do you have a budget set for the development of this new system?

Jack: Yes, I do. The executive steering committee has given me a budget of \$225,000 to complete this phase of the project. I don't want to put any more money into it than is absolutely necessary. I expect this budget is more than enough to get the job done. In fact, I wouldn't be surprised if it can be done for less money.

Kira: Are there any timetables I should be aware of?

Jack: The executive steering committee and I would like to have the first phase of this project completed in six months. The success of the overall strategic plan is dependent on us meeting that date.

Kira: We will certainly see what we can do. We will know more once we have completed a detailed analysis and design of course. Is there anything else you would like to tell me about the system?

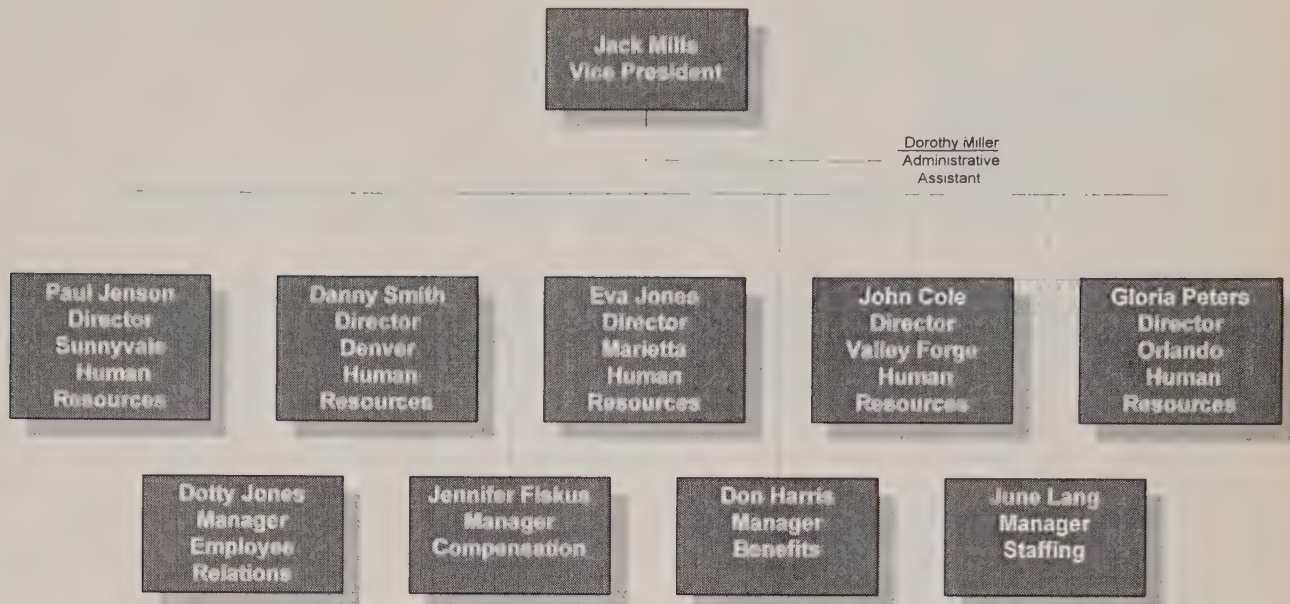
Jack: That's all I can think of right now.

Kira: OK. I'm going to review this information and give you my report of it early next week. Thank you for your time today, Mr. Mills.

Jack: You're welcome and please call me Jack. I'll see you next week.

Kira: Goodbye Mr. Mills. I mean Jack.

A-1 Information Systems Human Resources



CS 1 - MILESTONE 1: SURVEYING THE PROJECT



A-1 Information Systems

Phone: 838-9009 Fax: 838-1115

Internet: <http://www.a1.com>

Intranet: <http://www.a1.is.com/iss>

REQUEST FOR INFORMATION SYSTEM SERVICES

DATE OF REQUEST	SERVICE REQUESTED FOR DEPARTMENT(S)
	Human Resources

SUBMITTED BY (key user contact)	EXECUTIVE SPONSOR (funding authority)
Name Dotty Jones	Name Jack Mills
Title Employee Relations Manager	Title Vice President, Human Resources
Office 1180, Bldg. E7	Office 4216, Bldg. 16
Phone 306-2228	Phone 838-4456

TYPE OF SERVICE REQUESTED:

- | | |
|---|---|
| ☐ Information Strategy Planning | ☐ Existing Application Enhancement |
| ☐ Business Process Analysis and Redesign | ☐ Existing Application Maintenance (problem fix) |
| ☐ New Application Development | ☐ Not Sure |
| ☐ Other (please specify _____) | |

BRIEF STATEMENT OF PROBLEM, OPPORTUNITY, OR DIRECTIVE (attach additional documentation as necessary)

BRIEF STATEMENT OF EXPECTED SOLUTION

ACTION (ISS Office Use Only)

☐ Feasibility assessment approved

Assigned to _____

☐ Feasibility assessment waived

Approved Budget \$ _____

☐ Request delayed

Start Date _____ Deadline _____

Backlogged until date: _____

☐ Request rejected

Reason: _____

Authorized Signatures:

Chair, ISS Executive Steering Body

Project Executive Sponsor

PROBLEM STATEMENTS

PROJECT:	PROJECT MANAGER:
CREATED BY:	LAST UPDATED BY:
DATE CREATED:	DATE LAST UPDATED:

Brief Statements of Problem, Opportunity, or Directive	Urgency	Visibility	Annual Benefits	Priority or Rank	Proposed Solution
1.					
2.					
3.					
4.					
5.					
6.					
7.					
8.					

Exhibit 1.2

Project Feasibility Assessment Report

- I. Executive summary (1 page)
 - A. Summary of recommendation
 - B. Brief statement of anticipated benefits
 - C. Brief explanation of report contents
- II. Background information (1-2 pages)
 - A. Brief description of project request
 - B. Brief explanation of the survey phase activities
- III. Findings (2-3 pages)
 - A. Problems and analysis (optional: reference *problem statement matrix*)
 - B. Opportunities and analysis (optional: reference *problem statement matrix*)
 - C. Directives and implications
- IV. Detailed recommendation
 - A. Narrative recommendation (1 page)
 - 1. Immediate fixes
 - 2. Quick fixes
 - 3. Enhancements
 - 4. New systems development
 - B. Project Plan
 - 1. Initial project objectives
 - 2. Initial master project plan (phase level - Gantt chart)
 - 3. Detailed plan for the study or definition phase (Gantt chart)
- V. Appendices
 - A. Request for System Services
 - B. Problem Statements Matrix
 - C. (other documents as appropriate)



MILESTONE 2 - STUDYING THE CURRENT SYSTEM

Synopsis

There's an old saying that suggests, "Don't try to fix it unless you understand it." With those words of wisdom, the next milestone of our project is to study and analyze the existing system. There is always an existing business system, regardless of whether it currently uses a computer. The study phase provides the project team with a more thorough understanding of the problems, opportunities, and/or directives that triggered the project. Indeed, the analyst frequently uncovers new problems and opportunities. The study phase may answer the questions, "Are the problems worth solving?" and "Is a new system worth building?"

The purpose of the study phase is threefold. First and foremost, the project team must gain an appropriate understanding of the business problem domain. Second, we need to answer the question, "Are these problems (opportunities, and directives) worth solving"? Finally, we need to determine if the system is worth developing. The study phase provides the systems analyst and project team with a more thorough understanding of the problems, opportunities, and/or directives that triggered the project. In the process, they frequently uncover new problems and opportunities.

In this milestone you will perform *Cause-Effect Analysis* on the Employee Self Service System (ESSS) and document your findings using the *Problems, Opportunities, Objectives, and Constraints Matrix*. The PIECES framework, originally developed by James Wetherbe, and then adapted by the authors, can

serve as a useful tool to classify the various problems, opportunities, and directives identified in **Milestone 1**.

Objectives

After completing this milestone, you should be able to:

- ⇒ Perform *Cause-Effect Analysis* to be able to thoroughly understand a system's problems, opportunities, and/or directives that triggered the project.
- ⇒ Use and understand the PIECES framework for classifying problems, opportunities, and directives.
- ⇒ Complete the *Problems, Opportunities, Objectives, and Constraints Matrix*.

Prerequisites

Before starting this milestone the following topics should be covered:

1. The Study Phase - Chapter 3, pages 87-88, Chapter 4, pages 137-150 of the SADM 4th ed. textbook.
2. PIECES Framework - Chapter 3, pages 79-81 of the SADM 4th ed. Textbook.
3. Milestone 1 Solution

Project

Now that we have completed the survey of the system, and gained approval to proceed, we can now attempt to gain a better understanding of the current system. In this assignment we will use our results of Milestone 1, plus the Case background information and the user interview, in order to perform cause-effect analysis. The results of this activity will provide us a better understanding of the problems, opportunities, and constraints of the current system.

Activities

1. To complete the *Problems, Opportunities, Objectives, and Constraints Matrix*, use Figure 4.12, page 144 of the SADM 4th ed. textbook as an example. Make assumptions where necessary. Use the PIECES framework as a model to classify the problems, opportunities, and directives. Figure 3.4, pages 80-81 of the SADM 4th ed. textbook, may identify other areas that present problems or offer opportunities. We suggest you use it. It has proven to be a very valuable tool! We have provided for your convenience a copy of the matrix template on

the textbook's web page [<http://www.tech.purdue.edu/textbooks/sadm/>], as well as a hardcopy at the end of this milestone.

Deliverable format and software to be used are according to your instructor's specifications. Deliverables should be neatly packaged in a binder, separated with a tab divider labeled "Milestone 2" and accompanied with a Milestone Evaluation Sheet (appearing first) found in Appendix A.

References:

Case Background

Workbook Introduction

Transcripts of Interview with Jack Mills

Exhibit 1.1

Milestone 1 Solution

Provided by your instructor

Deliverables:

Problems, Opportunities, Objectives, and Constraints Matrix:

Due: ____/____/____

Time: _____

ADVANCED OPTION

This deliverable was not discussed in the narrative because students need to be exposed to modeling (data, process, interface, & geography), before this report can be completed. For those ambitious individuals who are familiar with those skills and wish to be challenged, use the detailed study report outline found in Figure 4.13, page 150 of the SADM 4th ed. textbook, as a guideline. We have also provided a copy of the outline in Exhibit 2.1

Study Report:

Due: ____/____/____

Time: _____

Milestone's Point Value:

PROBLEMS, OPPORTUNITIES, OBJECTIVES AND CONSTRAINTS MATRIX

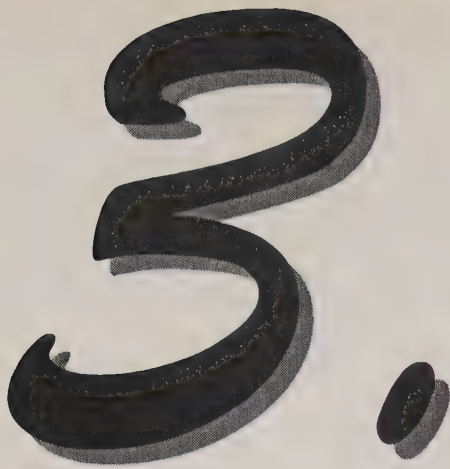
Project:	Project Manager:
Created by:	Last Updated by:
Date Created:	Date Last Updated:

CAUSE AND EFFECT ANALYSIS		SYSTEM IMPROVEMENT OBJECTIVES	
Problem or Opportunity	Causes and Effects	System Objective	System Constraint
1.	1.	1.	1.

Exhibit 2.1

Analysis of the Current _____ System

- I. Executive summary (approximately 2 pages)
 - A. Summary of recommendation
 - B. Summary of problems, opportunities, and directives
 - C. Brief statement of system improvement objectives
 - D. Brief explanation of report contents
- II. Background information (approximately 2 pages)
 - A. List of interviews and facilitated group meetings conducted
 - B. List of other sources of information that were exploited
 - C. Description of analytical techniques used
- III. Overview of the current system (approximately 5 pages)
 - A. Strategic implications (if the project is part of, or impacts an existing information systems strategic plan)
 - B. Models of the current system
 1. Interface model (showing project scope)
 2. Data model (showing project scope)
 3. Geographic models (showing project scope)
 4. Process model (showing functional decomposition only)
- IV. Analysis of the Current System (approximately 5-10 pages)
 - A. Performance problems, opportunities, and cause/effect analysis
 - B. Information problems, opportunities, and cause/effect analysis
 - C. Economic problems, opportunities, and cause/effect analysis
 - D. Control problems, opportunities, and cause/effect analysis
 - E. Efficiency problems, opportunities, and cause/effect analysis
 - F. Service problems, opportunities, and cause/effect analysis
- V. Detailed recommendations (approximately 5-10 pages)
 - A. System improvement objectives and priorities
 - B. Constraints
 - C. Project Plan
 1. Scope reassessment and refinement
 2. Revised master plan
 3. Detailed plan for the definition phase
- VI. Appendices
 - A. Any detailed system models
 - B. (other documents as appropriate)



MILESTONE 3 - MODELING THE SYSTEM'S DATA

Synopsis

The definition phase answers the question, 'What does the user need and want from a new system?' The definition phase is critical to the success of any new information system! In this milestone we need to identify what information systems requirements need to be defined from the system users' perspectives and draw graphical, logical, models to document the data requirements for a new and improved system.

Data modeling is a technique for organizing and documenting a system's Data. Data modeling is sometimes called database modeling because a data model is usually implemented as a database. Data is viewed as a resource to be shared by as many processes as possible. As a result, data must be organized in a way that is flexible and adaptable to unanticipated business requirements – and that is the purpose of data modeling.

In this milestone you will first discover those entities in the system that are or might be described by data. With each entity we identify, we will define it in respect to the business. Then, we will construct a *Context Data Model* that graphically depicts each of the entities and the relationships they have with each other. Next, we will refine the context data model to include primary and foreign keys. The resulting model is called a *Key-Based Data Model*. Finally, we refine

the key-based data model to include any hierarchies and attributes, and this model is referred to as the *Fully Attributed Data Model*.

Objectives

After completing this milestone, you should be able to:

- ⇒ Understand and perform the techniques for entity discovery.
- ⇒ Define each entity with respect to the business and complete an entity/definition matrix.
- ⇒ Perform the necessary data modeling techniques to organize and document the data requirements for the proposed system.
- ⇒ Construct the Context, Key-Based, and Fully Attributed data models.

Prerequisites

Before starting this milestone the following topics should be covered:

1. Data Modeling - Chapter 5, pages 170-202
of the SADM 4th ed. textbook.
2. Milestone 2 Solution

Project

Now that we have studied the current system and analyzed some of its problems and opportunities, plus gained approval to proceed, we can now start to identify the business data requirements and graphically model them. In this assignment we will use our results of the previous Milestone and transcripts of an interview with Dotty Jones, the Employee Relations Manager for A-1 Information Systems. The results of this activity will identify the business data requirements for the proposed system.

Ms. Jones was gracious enough to allow us to record our interview session and Exhibit 3.1 is a copy of the transcripts. Refer to the interview transcripts, sample forms, and results from Milestone 2, for the information necessary to complete the activities.

Activities

1. To complete the *Entity/Definition Matrix*, use Table 5.5, page 192 of the SADM 4th ed. textbook as an example. Analyze each of the forms referenced by the user interview plus any comments made by Dotty Jones. Make assumptions where necessary. We have provided for your convenience a

template of the matrix on the textbook's web page

[<http://www.tech.purdue.edu/textbooks/sadm/>], as well as a hardcopy in the back of this milestone.

2. To complete the *Context Data Model* use Figure 5.11, page 194 of the SADM 4th ed. textbook as an example. Construct the model to include each of the entities in **Activity 1** and depict the relationships that exist among the entities. Data model notation is according to your instructor's specifications, however the authors' prefer the notation depicted in Figure 5.3, page 180 of the SADM 4th ed. textbook.
3. To complete the *Key-Based Data Model* use Figure 5.12, page 194 of the SADM 4th ed. textbook as an example. Refine the Context Data Model from **Activity 2** to include associate entities that resolve many-to-many relationships, and include primary and foreign keys. Again data model notation is according to your instructor's specifications, however the authors' prefer the notation depicted in Figure 5.3, page 180 of the SADM 4th ed. textbook.
4. To complete the *Fully Attributed Data Model* use Figure 5.14, pages 200 & 201 of the SADM 4th ed. textbook as an example. Refine the Key-Based Data Model from **Activity 3** to include any generalization hierarchies, and add the data attributes for each entity.

Deliverable format and software to be used are according to your instructor's specifications. Deliverables should be neatly packaged in a binder, separated with a tab divider labeled "Milestone 3" and accompanied with a Milestone Evaluation Sheet (appearing first) found in Appendix A.

References:

Transcripts of Interview with Dotty Jones and Accompanying Sample Forms and Report

Exhibits 3.1-3.5

Milestone 2 Solution

Provided by your instructor

Deliverables:**Entity Definition Matrix:****Due:** __/__/__**Time:** _____**Context Data Model:****Due:** __/__/__**Time:** _____**Key-Based Data Model:****Due:** __/__/__**Time:** _____**Fully Attributed Data Model:****Due:** __/__/__**Time:** _____**ADVANCED OPTION**

For the advanced option, review the Case Background, the Request for System Services, the interview with Jack Mills, and other previously provided and submitted documents, to discover any additional entities and attributes. Construct a fully attributed data model to be able to handle employees quitting and then returning to the company, effectively having multiple hire and termination dates. Also, prepare the model to be able to distinguish employees that are actively employed, retired, terminated, on leave-of-absence, full-time and part-time.

Entity Definition Matrix:**Due:** __/__/__**Time:** _____**Fully Attributed Data Model:****Due:** __/__/__**Time:** _____**Milestone's Point Value:** _____

The following is a copy of the transcripts of an interview between Ms. Dotty Jones of A-1 Information Systems and Kira Webster, a computer information systems student. The goal of this interview was to obtain sample forms used for processing employee information and to be able to ask questions about them in order to discover data entities of the business system.

Exhibit 3.1

Scene: *Kira Webster, computer information systems student, is meeting with Dotty Jones, Manager of Employee Relations for A-1 Information Systems, at her office, room 1180, building E7. Ms. Webster scheduled the interview with Ms. Jones to obtain sample forms containing employee information.*

Dotty: Good Morning Kira! How are you today?

Kira: Fine thank you.

Dotty: Good. I don't have much time this morning because I have to conduct an Ethics training course, so lets get started.

Kira: That's fine. I appreciate you taking time out of your busy schedule to meet with me.

Dotty: Jack wanted me to be your main point of contact and be available as much as I could. Unfortunately, today I had this prior commitment I must attend to. The company takes employee ethics very seriously.

Kira: I see.

Dotty: You requested to Jack that you would like copies of sample forms and reports. I have made copies of three of the forms we use, plus I have made a copy of a page out of our employee telephone listing.

Kira: Great! That will be a big help.

Dotty: The first form is the Employee Information Form [Exhibit 3.2]. The employee completes this form on their first day of employment.
Dotty hands the form to Kira.
It is mandatory that the employee completes the employee profile block, but the emergency contact information is optional.

Kira: Is the date the date the form was filled out or the official date of hire?

Dotty: It's the official date of hire. Its possible the employee maybe ill on their first day of employment and not be able to come to work. Under those circumstances, when they do come to work and complete the form, that date reflects the day

they were supposed to start to work.

Kira: Okay. Do you only want to record an employee's middle initial?

Dotty: No. That's a weakness of the form. We should be capturing an employee's full name, including the middle name and any nicknames.

Kira: Good.

Dotty: Next, we have the Employee United Way Contribution Form [Exhibit 3.3].

Dotty hands the form to Kira.

New employees complete this form on their first day of employment. Current employees complete one of these forms every December. This gives them the opportunity to change their deductions or keep it the same.

Kira: It appears the top of the form is pre-typed?

Dotty: That's true. A form for every employee is previously typed with their name, ID number, department number, and manager name. The forms are then given to the managers for distribution. The managers ensure that the forms are completed and then return them back to us.

Kira: Is that true for new employees also?

Dotty: Yes, those forms are pre-typed also. We know ahead time what department a new employee will be working in, plus the manager they will be working for. That information is part of the Job Offer Letter that is sent to a prospective new employee. Once they accept the offer, by signing the letter and returning it, we assign them an Employee ID. So when they report for the first day of work, we have already captured information about them. That's why they only need to complete the remainder of the form during their orientation session.

Kira: I see. How are Employee ID's assigned?

Dotty: Sequentially. We never want to reuse an Employee ID. That's how we track an employee. Even if they leave the company, they might one day come back. And, quite possibly we might owe them medical, pension, or retirement benefits. Does that make sense?

Kira: Yes it does.

Dotty: Okay. The next form is the Employee United States Savings Bonds Contribution Form [Exhibit 3.4].

Dotty hands the form to Kira.
 New employees complete this form on their first day of employment. Current employees complete at least one of these forms every May. This gives them the opportunity to change their deductions or keep it the same.

Kira: You said they complete at least one. What do you mean by that?

Dotty: Some employees buy Bonds for each of their children plus themselves. For each Bond they buy that has a different owner, they must fill out a separate form.

Kira: Let me make sure I understand this. If I wanted to have a payroll deduction that would purchase bonds not only for me, but if I had a child for them also, I would need to complete two forms?

Dotty: That's right. It is required to have one form per bond owner.

Kira: Interesting.

Dotty: Finally, we have a copy of a page out of our Employee Telephone Listing.
[Exhibit 3.5].

Dotty hands the form to Kira.
 I believe it is pretty self-explanatory. You notice it contains employees from all our

five sites, plus the building and room they are located in.

Kira: What is a Mail Stop?

Dotty: Think of it as a company post office and that's the zip code. Every employee's mail is delivered to a centralized location identified as a mail stop. Usually it's located in a little room or an area of an office. There the employees can go and retrieve their own mail. We have too many employees to provide mail delivery to each individual office or cubicle. It wouldn't be very cost effective.

Kira: I see.

Dotty: Do you have any more questions or is there anything else I can help you with?

Kira: I don't believe so, not at this time. Dotty you have been a big help.

Dotty: Thank you. And now if you excuse me, I must run. Call me if you need anything else.
 Goodbye.

Kira: Goodbye Dotty.

Exhibit 3.2

A-1 Information Systems

Employee Information Form

Employee Profile

Date: 10/9/96

Last Name: Jacobs

First Name: John

M.I.: C

Social Security No.: 308 -57 - 1314

Home Phone: (407) 423-4598

Home Address: 1456 Forest Drive
Orlando, FL
32859-0032

Birth Date: 12/5/61

Marital Status: M: ☒ S: ☐

Emergency Contact Information

Primary Contact

First Name: Janet
Last Name: Jacobs
M.I.: L
Relationship: Spouse
Home Address: 1456 Forest Drive
Orlando, FL
32859-0032
Home Phone: (407) 423-4598
Work Phone: (407) 306-1508

Secondary Contact

First Name: Robert
Last Name: Jacobs
M.I.: W
Relationship: Father
Home Address: 2435 Doe Path Ln.
Tampa, FL
34566
Home Phone: (904) 859-7845
Work Phone: Retired

Exhibit 3.3

A-1 Information Systems		 Employee United Way Contribution Form
Employee ID: 77423	Employee Dept.: 6410	
Employee Name: Jacobs, John C	Employee Manager: William Stokes	
My contribution for the calendar year of 1997 is:		
☐ My Fair share One-time gift (5% of each payroll period (Amount to be deducted from first payroll) \$25. Other amount (Amount to be deducted from each payroll) 00 ☐ I do not wish to give at this time		
<u>John Jacobs</u> Signature <u>8/27/</u> Date		
To learn more about United Way, please visit their WWW site at: http://www.unitedway.org/		

Form 1599 - January 1997

Exhibit 3.4

A-1 Information Systems		Employee U.S. Savings Bond Contribution Form	
Employee ID: 77423		Employee Dept.: 6410	
Employee Name: Jacobs, John C		Employee Manager: William Stokes	
Bond Denomination ☐ \$100 ☒ \$200 ☐ \$500 ☐ \$1,000		Payroll Deduction Type ☐ One Time ☒ Regular	
		Deduction Amount: \$ <u>25.00</u> <i>(For regular deductions only.)</i>	
Bond Owner <i>(Last, First, Middle)</i> Jacobs, Katie, Lynn 315-70-8967		Bond Mailing Address: 1456 Forest Drive Orlando, FL 32859-0032	
☒ Co-Owner <i>(Last, First, Middle)</i>			
☐ Beneficiary Jacobs, John, Carl			
<u>John Jacobs</u> Signature		<u>8/27/</u> Date	
<i>To learn more about United States Savings Bonds, please visit their WWW site at: http://www.publicdebt.treas.gov/sav/sav.htm</i>			

Form 1099 - December 1996

Exhibit 3.5

<i>A-1 Information Systems</i>										Employee Telephone Listing
Revision Date: 7/17/97										Page 15
Last Name	First Name	M.I.	Emp. No.	Dept.	Phone No.	Site	Bldg.	Room	Mail Stop	
Jacob	Joh	C	7742	641	(407) 306-	ORL	E7	223	359	
Jame	Tereas	L	7645	642	(407) 306-	ORL	E7	246	358	
Janse	Beck	E	7322	650	(770) 595-	MAR	22	119	A12	
Jeffrie	Lora	L	7766	654	(770) 595-	MAR	22	117	A12	
Jenkin	Tom	A	7852	720	(215) 393-	VF	AA	560	69	
Johnso	Gus	L	7945	521	(408) 743-	SUN	X.2	8	MA	
Jone	Dott	L	7522	642	(407) 306-	ORL	E7	118	340	
Jone	Sam	A	7488	850	(303) 221-	DEN	C1	213	551	
Jone	Terri	L	7688	850	(303) 221-	DEN	C1	215	551	
Justic	Kevi	A	7733	854	(303) 221-	DEN	C1	311	561	
Kam	Thoma	C	7711	521	(408) 743-	SUN	X.1	1	MM	
Kapla	Davi	J	7655	650	(407) 306-	ORL	E8	115	240	
Kelle	Helle	E	7644	730	(215) 393-	VF	AB	310	72	
Kidwel	Brando	M	7711	520	(408) 743-	SUN	X.1	12	MB	
Klein	Brya	M	7822	643	(407) 306-	ORL	E7	356	410	

Entity/Definition Matrix

ENTITY	BUSINESS DEFINITION



MILESTONE 4 - MODELING THE SYSTEM'S PROCESSES

Synopsis

Process modeling is a technique for organizing and documenting the structure and flow of data through a system's processes and/or the logic, policies, and procedures to be implemented by a system's processes. In this milestone we focus on using and constructing data flow diagrams (DFDs) and decomposition diagrams to perform process modeling.

Data flow diagrams are tools that depict the flow of data through a system and the work or processing performed by that system. A decomposition diagram is a DFD planning tool that shows the top-down functional decomposition and structure of a system.

During this milestone you will first construct a *context diagram* to establish project scope and boundaries. Secondly, you will draw a *functional decomposition diagram* to partition the system into logical subsystems and/or functions. Thirdly, you will complete a partially completed *event list* that identifies and confirms the business events to which the system must respond. Once that task is completed, you will refine the decomposition diagram drawn earlier to include event-handling processes. Finally you will construct a *system data flow diagram* that shows the big picture of the system, and a *primitive data flow diagram* for a single event process.

Objectives

After completing this milestone, you should be able to:

- ⇒ Construct a context diagram to illustrate a system's interfaces with its environment.
- ⇒ Perform event partitioning and organize events in a functional decomposition diagram.
- ⇒ Identify external and temporal business events for a system.
- ⇒ Merge event diagrams into a system data flow diagram.
- ⇒ Draw primitive data flow diagrams.

Prerequisites

Before starting this milestone the following topics should be covered:

1. Process Modeling - Chapter 6, pages 205-254
of the SADM 4th ed. textbook.
2. Milestone 3 Solution

Project

As a systems analyst or knowledgeable end-user, you must learn how to draw decomposition and data flow diagrams to model business process requirements. The survey and study phases of the methodology have been completed and you understand the current system's strengths, weaknesses, limitations, problems, opportunities, and constraints. You have already built the data model (**Milestone 3**) to document business data requirements for the new system. You now need to build the corresponding process models.

Activities

1. To complete the *Context Diagram*, use Figure 6.20, page 242 of the SADM 4th ed. textbook as an example. Remember that the context diagram establishes the project scope and boundaries by depicting the system's interfaces with external parties. Refer to the narrative in Exhibit 4.1 to construct the diagram.
2. To complete the *Functional Decomposition Diagram*, use Figure 6.21, page 243 of the SADM 4th ed. textbook as an example. Assume that the system contains three subsystems with one of them being "Employee Tracking Subsystem". What would be the other two?

3. To complete the *Event – Response List*, use Figure 6.22, page 245 of the SADM 4th ed. textbook as an example. Exhibit 4.2 is a copy of a partially completed Event-Response List for the Employee Tracking Subsystem. Complete the list by providing data in the blank cells. Make assumptions where necessary. We have provided for your convenience a template of the Event-Response List on the textbook's web page [<http://www.tech.purdue.edu/textbooks/sadm/>].
4. To complete the *Event Decomposition Diagram*, use Figure 6.23, page 246 of the SADM 4th ed. textbook as an example. Refine the decomposition diagram constructed in **Activity 2** above, to include the event processes identified in **Activity 3** above. Focus only on the Employee Tracking Subsystem.
5. To complete the *System Diagram*, use Figure 6.27, pages 250 & 251 of the SADM 4th ed. textbook as an example. Exhibit 4.3 contains Event Diagrams for all the events within the Employee Tracking Subsystem. Merge these diagrams to produce the System Diagram.
6. To complete the *Primitive Diagram*, use Figure 6.28, page 252 of the SADM 4th ed. textbook as an example. Refer to the narrative in Exhibit 4.4 to construct the primitive diagram for the Employee Profile Changes Event.

Deliverable format and software to be used are according to your instructor's specifications. Deliverables should be neatly packaged in a binder, separated with a tab divider labeled "Milestone 4" and accompanied with a Milestone Evaluation Sheet (appearing first) found in Appendix A.

References:**Context Diagram Narrative**

Exhibit 4.1

Partially Completed Event-Response List

Exhibit 4.2

Event Diagrams

Exhibit 4.3

Primitive Diagram Narrative

Exhibit 4.4

Milestone 3 Solution

Provided by your instructor

Deliverables:

Context Diagram:

Due: __/__/__
Time: _____

Functional Decomposition Diagram:

Due: __/__/__
Time: _____

Event-Response List:

Due: __/__/__
Time: _____

Event Decomposition Diagram:

Due: __/__/__
Time: _____

System Diagram:

Due: __/__/__
Time: _____

Primitive Diagram:

Due: __/__/__
Time: _____

ADVANCED OPTION

For the advanced option, try to complete activities 3 through 6 for one of the other subsystems. In this option you will need to create the Event Diagrams before you can do the System Diagram. Make any assumptions as necessary.

Subsystem 2 or 3:

Due: __/__/__
Time: _____

Milestone's Point Value:

Exhibit 4.1

Use the following narrative to construct the Context Diagram for the Employee Self Service System (ESSS).

The purpose of the Employee Self Service System (ESSS) is to provide a single central repository of employee information. The Staffing Department provides new employee profile information and employee status changes. In response, the system generates staffing reports on a weekly, monthly, and yearly basis for the Staffing Department. The employees provide employee profile changes, as well as United Way and Savings Bonds contribution activity. In response, the system generates on request an Employee Participation Report for management, and Savings Bonds and United Way Contribution reports for the Employee Relations Department. The employee contribution activity records are sent to the Payroll System. Finally, an employee telephone listing can be generated for any employee who requests it.

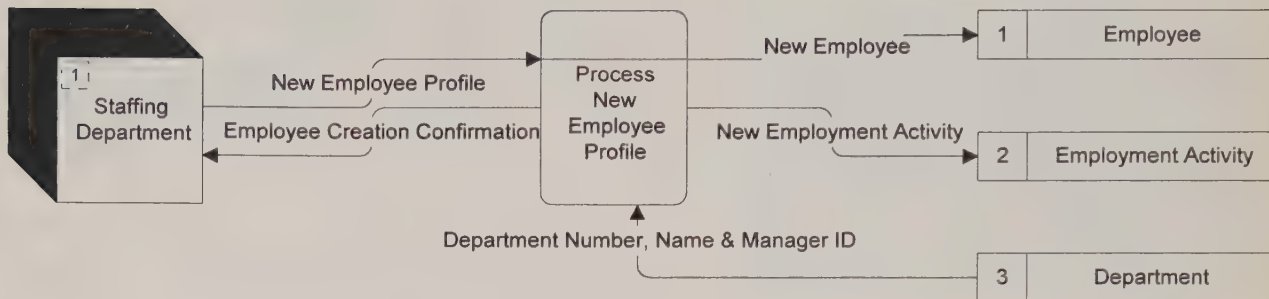
Exhibit 4.2

EVENT LIST

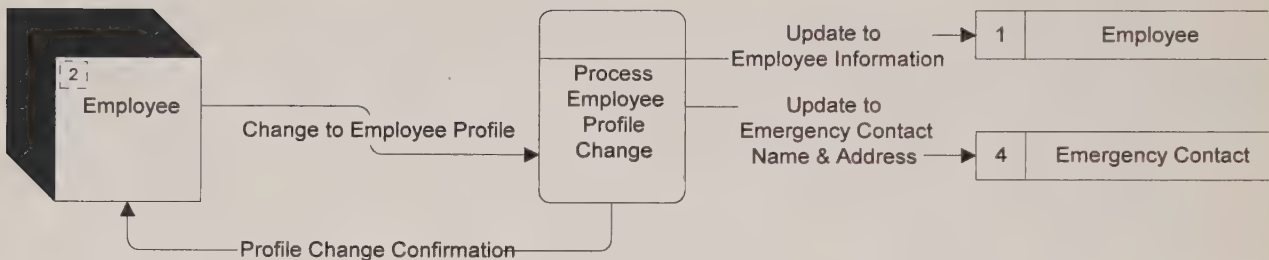
<i>Event Description</i>	Trigger (Inputs)	Responses (Outputs)
Staffing department receives acceptance letter from an individual accepting a job offer.	New Employee Profile	Create Employee Create Employment Activity
Employee provides changes to their personal information such as address or emergency contacts.		Update Employee Update Emergency Contact
Staffing department receives letter of resignation from employee.		
Request to produce listing of employee telephone numbers and locations.		
Time to analyze employee staffing.		Staffing Analysis Report

Exhibit 4.3

Event Diagram
Event: Staffing submits new employee profile



Event Diagram
Event: Employee changes profile information



Event Diagram
Event: Staffing submits employee status changes

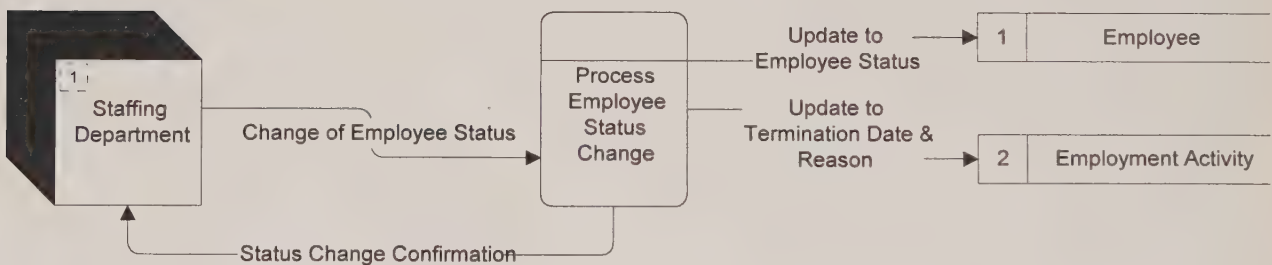


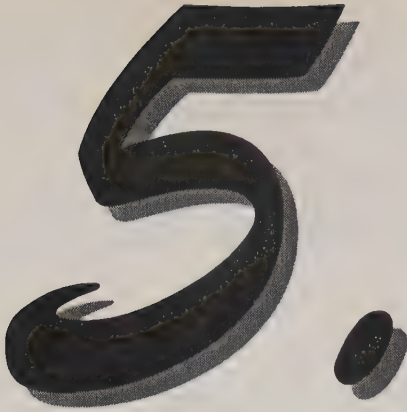
Exhibit 4.4

Use the following narrative to construct the Primitive Diagram for the Process Employee Profile Changes event.

Given an Employee ID, *validate* the employee. If the ID is not valid send a response to the employee. Verify that employee demographics are correct. If not, update them.

Verify employee emergency contact information. Verify that contact name and demographics are correct. If not correct them.

Given a validated employee and verified emergency contact information, *send* a confirmation to the employee that the change has been accepted.



MILESTONE 5 - MODELING THE NETWORK

Synopsis

As a systems analyst or knowledgeable end-user, you should learn how to draw network models to be able to deal with today's distributed business and computing environment. Network modeling is a diagrammatic technique used to document the shape of a business or information system in terms of its business locations.

In this milestone you will first construct a *Location Decomposition Diagram* to logically decompose and group locations. Next we will draw a *Location Connectivity Diagram* that depicts the shape of the system in terms of its user, process, data, and interface locations and the necessary interconnections between the locations.

Objectives

After completing this milestone, you should be able to:

- ⇒ Factor a system's or application's locations into component locations using a special location decomposition diagram.
- ⇒ Document the connections and essential data flow between locations using a special location decomposition diagram.

Prerequisites

Before starting this milestone the following topics should be covered:

1. Network Modeling - Chapter 7, pages 262-280
of the SADM 4th ed. textbook.
2. Case Study Introduction

Project

Now that we have completed the survey and study phases of the system development lifecycle, and we fully understand the system's strengths, weaknesses, limitations, problems, opportunities, and constraints, it is time to model the logical business network requirements.

Refer to the case study introduction to complete the following activities.

Activities

1. To complete the *Location Decomposition Diagram*, use Figure 7.8, page 277 of the SADM 4th ed. textbook as an example. Refer to the case study introduction to find the necessary information to construct diagram. Pay close attention to the organization chart and the location map. These can provide valuable information. Make assumptions where necessary.
2. To complete the *Location Connectivity Diagram*, use Figure 7.10, page 279 of the SADM 4th ed. textbook as an example. Refer to the case study introduction to find the necessary information to construct diagram. Again, pay close attention to the organization chart and the location map. These can provide valuable information. Make assumptions where necessary. Use a road atlas or other type of map to determine the mileage between sites.

Deliverable format and software to be used are according to your instructor's specifications. Deliverables should be neatly packaged in a binder, separated with a tab divider labeled "Milestone 5" and accompanied with a Milestone Evaluation Sheet (appearing first) found in Appendix A.

References:

Case Study Introduction

Deliverables:**Location Decomposition Diagram:****Due:** __/__/__**Time:** _____**Location Connectivity Diagram:****Due:** __/__/__**Time:** _____**ADVANCED OPTION**

For the advanced option, assume A-1 Information Systems has gone global and opened up a European headquarters in London, England, with regional offices in Paris, France, Rome, Italy, and Munich, Germany. Prepare a location decomposition diagram and a location connectivity diagram that includes these new locations.

Location Decomposition Diagram:**Due:** __/__/__**Time:** _____**Location Connectivity Diagram:****Due:** __/__/__**Time:** _____**Milestone's Point Value:**



MILESTONE 6 - DEFINING AND ANALYZING CANDIDATE SOLUTIONS

Synopsis

The purpose of the Configuration phase is to identify candidate solutions, analyze those candidate solutions, and recommend a target system that will be designed and implemented. Alternative solutions to be considered should be those that address the business requirements of the information system.

In this milestone you will complete a *Candidate Matrix*. This matrix will include three alternative candidate solutions that you have determined will meet the business requirements for the Employee Self Service System. Some of the solutions you will consider may have been posed from the design ideas and opinions from the system owners and users. Others may come from various sources including: systems analysts, system designers, technical consultants, and other IS professionals. Some technical choices may be limited by a predefined, approved technology architecture provided by system managers. When completing the matrix, it is not your intent to evaluate the candidates at this point, only to identify and define them.

Objectives

After completing this milestone, you should be able to:

⇒ Identify candidate solutions that meet the requirements of the information system and document those solutions in a candidate matrix.

Prerequisites

Before starting this milestone the following topics should be covered:

1. The Configuration Phase - Chapter 9, pages 319-335 of the SADM 4th ed. textbook.
2. Feasibility and Cost-Benefit Analysis Module C, pages 642-659 of the SADM 4th ed. Textbook.

Project

Given the business requirements established in the definition phase of systems analysis, you can now identify alternative candidate solutions.

Activities

1. To complete the *Candidate Matrix*, use Figure 9.5, page 322, and Figure C.7, page 658 of the SADM 4th ed. textbook as an example. Based on the standard workstation and server configuration defined in the Case Study Introduction, identify and document three solutions that could meet ESSS's needs. Assume that the system requires a typical client/server interface as well as a World Wide Web interface. Assume that all data will be housed in Orlando, but the other offices receive nightly downloads of the data for decision support reporting purposes. Also, assume that all employees have at least a 486 class PC or a comparable Macintosh. LAN printers provide printing services, usually one LAN printer per 10 employees. Make other assumptions where necessary. We have provided for your convenience a copy of the Candidate Matrix template on the textbook's web page [<http://www.tech.purdue.edu/textbooks/sadm/>], as well as a hardcopy at the end of this milestone.

Deliverable format and software to be used are according to your instructor's specifications. Deliverables should be neatly packaged in a binder, separated with a tab divider labeled "Milestone 6" and accompanied with a Milestone Evaluation Sheet (appearing first) found in Appendix A.

References:

Case Study Introduction

Deliverables:**Candidate Matrix:****Due:** __/__/__**Time:** _____**ADVANCED OPTION**

For the advanced option, perform feasibility analysis and complete a feasibility matrix for the three candidates identified above. This option will require you to do research on the cost of hardware and software. Assume that 125 new PCs and associated software will have to be purchased to use the new system. Also, assume that four new servers and associated software will be purchased one for each site except Orlando. Assume network connectivity and infrastructure already exists. Make the necessary assumptions to estimate development time, costs, and benefits. Use Figure 9.6, page 324, and Figure C.9, page 659 of the SADM 4th ed. textbook as an example. We have provided for your convenience a copy of the Feasibility Matrix template on the textbook's web page [<http://www.tech.purdue.edu/textbooks/sadm/>], as well as a hardcopy at the end of this milestone.

Feasibility Matrix:**Due:** __/__/__**Time:** _____**Milestone's Point Value:** _____

Candidate Matrix Template

Characteristics	Candidate 1	Candidate 2	Candidate 3
Portion of System Computerized Brief description of that portion of the system that would be computerized in this candidate.			
Benefits Brief description of the business benefits that would be realized for this candidate.			
Servers and Workstations A description of the servers and workstations needed to support this candidate.			
Software Tools Needed Software tools needed to design and build the candidate (e. g., database management system, emulators, operating systems, languages, etc.). Not generally applicable if applications software packages are to be purchased.			
Application Software A description of the software to be purchased, built, accessed, or some combination of these techniques.			
Method of Data Processing Generally some combination of: on-line, batch, deferred batch, remote batch, and real-time.			
Output Devices and Implications A description of output devices that would be used, special output requirements, (e.g. network, preprinted forms, etc.), and output considerations (e.g., timing constraints).			
Input Devices and Implications A description of Input methods to be used, input devices (e.g., keyboard, mouse, etc.), special input requirements, (e.g. new or revised forms from which data would be input), and input considerations (e.g., timing of actual inputs).			
Storage Devices and Implications Brief description of what data would be stored, what data would be accessed from existing stores, what storage media would be used, how much storage capacity would be needed, and how data would be organized.			

Feasibility Matrix Template

Feasibility Criteria	Wt.	Candidate 1	Candidate 2	Candidate 3
Operational Feasibility Functionality. A description of to what degree the candidate would benefit the organization and how well the system would work. Political. A description of how well received this solution would be from both user management, user, and organization perspective.	30%	Score:	Score:	Score:
Technical Feasibility Technology. An assessment of the maturity, availability (or ability to acquire), and desirability of the computer technology needed to support this candidate. Expertise. An assessment to the technical expertise needed to develop, operate, and maintain the candidate system.	30%	Score:	Score:	Score:
Economic Feasibility Cost to develop: Payback period (discounted): Net present value: Detailed calculations:	30%	Score:	Score:	Score:
Schedule Feasibility An assessment of how long the solution will take to design and implement.	10%	Score:	Score:	Score:
Ranking:	100%			



MILESTONE 7 – MODELING THE APPLICATION ARCHITECTURE

Synopsis

Just as we modeled business requirements during systems analysis, we should model technology architecture and requirements during systems design. The models serve as blueprints for system design, prototyping and construction.

In this milestone you will prepare a *Physical Data Flow Diagram*. Physical data flow diagrams model the technical and human design decisions to be implemented as part of an information system. They communicate technical and other design constraints to those who will actually implement the system—in other words, they serve as a technical blueprint for the implementation.

Objectives

After completing this milestone, you should be able to:

⇒ Draw physical data flow diagrams for an information system's architecture and processes.

Prerequisites

Before starting this milestone the following topics should be covered:

1. Application Architecture & Process Design

Chapter 10, pages 350-387
of the SADM 4th ed. Textbook.

Project

The goal of this project is to produce a high-level general design that can serve as an application architecture for the system and a general design for the processes that make up the system.

Activities

1. To construct the Physical Data Flow Diagram, use Figure 10.8, page 373, of the SADM 4th ed. textbook as an example. Prepare the DFD based on the narrative provided in Exhibit 7.1. Make assumptions where necessary.

Deliverable format and software to be used are according to your instructor's specifications. Deliverables should be neatly packaged in a binder, separated with a tab divider labeled "Milestone 7" and accompanied with a Milestone Evaluation Sheet (appearing first) found in Appendix A.

References:

Job Offer Business Process

Exhibit 7.1

Deliverables:

Physical Data Flow Diagram:

Due: __/__/__

Time: _____

ADVANCED OPTION

For the advanced option, construct a Network Topology Data Flow Diagram based on the five locations of A-1 Information Systems. Assume that Orlando is the central hub with network connections branching from there to the other four sites. Make the necessary assumptions for types of networks and line speeds. Use Figure 10.12, page 382 of the SADM 4th ed. textbook as an example.

Network Topology DFD:

Due: __/__/__

Time: _____

Milestone's Point Value: _____

Exhibit 7.1

Job Offer Business Process Narrative

The Staffing hiring managers forward the names and starting salary amounts of candidates they wish to extend job offers, via electronic memorandum to their clerical representatives. It is the responsibility of the clerical representatives to verify and validate the candidate's name and address, then prepare the formal job offers using a form letter in MS Word, and then send the completed job offer letters to the candidates.

Candidates can respond one of three ways to the letter. They can decline the offer, they can accept the offer, or they can make a counter offer. The response can be made by phone or through the mail. The Staffing Department evaluates the candidate response. If the candidate declines the offer or submits a counter offer, the Staffing Department may plan a new offer and send it (via E-mail) to the clerical staff for processing. For those candidates who rejected the offer or submitted a counter offer, but are being rejected by Staffing, those names are sent (via E-mail) to the clerical staff for processing. The clerical representatives verify and validate the candidate's name and address, then prepare a formal rejection letter using a form letter in MS Word. Once printed, the clerical staff sends the completed rejection letters to the candidates. Finally, if the candidate accepts the offer, or Staffing accepts the candidate's counter offer, those names are sent (via E-mail) to the clerical staff for processing. The clerical representatives verify and validate the candidate's name and address, then prepare a formal job acceptance letter that includes the candidate's starting date, using a form letter in MS Word. Once printed, the clerical staff sends the completed acceptance letters to the candidates.

Once a candidate accepts a position, the clerical staff inputs the new employee's name, address, and ID number in to the ESSS system where a new employee record is created. If the new employee was a former employee at one time, the old employee record is revised to reflect the new status.



MILESTONE 8 – DESIGNING THE DATABASE

Synopsis

All information systems create, read, update and delete data. This data is stored in files and databases. To fully exploit the advantages of database technology, a database must be carefully designed. Database design translates the data models that were developed for the system users during the definition phase into data structures supported by the chosen database technology. Subsequent to database design, system builders will construct those data structures using the language and tools of the chosen database technology.

In this milestone you will normalize the data model created in **Milestone 3**, along with addition data requirements, to be in third normal form. In the advanced option you will transform the logical data model into a physical model or database schema.

Objectives

After completing this milestone, you should be able to:

- ⇒ Normalize a logical data model to remove impurities that can make a database unstable, inflexible, and nonscalable.
- ⇒ Transform a logical data model into a physical, relational database schema.

Prerequisites

Before starting this milestone the following topics should be covered:

1. Database Design Chapter 11, pages 392-428
of the SADM 4th ed. Textbook.

Project

The goal of this project is to normalize our logical data model to remove impurities that can make a database unstable, inflexible, and nonscalable.

Activities

1. To construct an Entity Relationship Diagram to be in 3rd Normal form, follow the normalization procedure outlined on pages 409 to 418 of the SADM 4th ed. textbook. Use Figure 11.15, page 419, of the SADM 4th ed. textbook as an example. Use the ERD solution of Milestone 3 as well as the data dictionary provided at the end of this milestone to prepare the new data model. Make assumptions where necessary.

Deliverable format and software to be used are according to your instructor's specifications. Deliverables should be neatly packaged in a binder, separated with a tab divider labeled "Milestone 8" and accompanied with a Milestone Evaluation Sheet (appearing first) found in Appendix A.

References:

Milestone 3 Solution

Provided by your instructor

ESSS Data Dictionary

Provided at the end of this milestone

Deliverables:

Logical Data Model in 3rd Normal Form:

Due: __/__/__

Time: _____

ADVANCED OPTION

For the advanced option, transform your normalized data model created above into a physical database schema complete with referential integrity rules. Make any necessary assumptions. Use Figure 11.17, page 425 of the SADM 4th ed. textbook as an example.

Physical Database Schema:

Due: __/__/__

Time: _____

Milestone's Point Value:

CS 1 - MILESTONE 8: DESIGNING THE DATABASE

ESSS Data Dictionary

Employee Number	5 digit, alphanumeric, unique code assigned to every employee for identification purposes.
Employee Name	Field consisting of an employee's last, first, and middle names.
Employee Home Address	Field consisting of an employee's street address, city, state, and zip code.
Employee Home Phone	10 digit numeric field consisting of an employee's home telephone number.
Employee Social Security Number	9 digit numeric field unique to each employee.
Employee Birth Date	8 digit numeric field in MM/DD/YYYY format consisting of an employee's date of birth.
Employee Marital Status	1 digit field indicating an employee's marital status. Possible values are "M" for married, and "S" for single
Employee Job Code	10 digit alphanumeric field that relates to the employee's current job title. Staffing department maintains a standard table of Job Codes, related Job Titles, and permissible salary range for each job title.
Employee Job Title	Alphanumeric field that consists of the employee's current job title.
Employee Dept. Number	4 digit Numeric field indicating to what department the employee is assigned. Staffing department maintains a standard table of Department Numbers, Department Names, and Employee Number of the individual who is manager of the department.
Employee Dept. Name	Alphanumeric field indicating the department name of where the employee is assigned.
Employee Dept. Manager Number	Employee number of the individual who is the manager of the department.
Employee Current Annual Salary	9 digit field that includes two decimal points, consisting of a full-time employee's annual salary
Employee Hourly Rate	6 digit field that includes two decimal points, consisting of a part-time employee's hourly rate.
Number of Hours Scheduled	2-digit numeric field consisting of the number of hours a part-time employee is scheduled to work per week.
Employee Work Phone	10 digit numeric field consisting of an employee's work telephone number.
Employee Work Site	3 digit alphanumeric field consisting of an employee's work site. Valid values are "ORL", "MAR", "VF", "SUN", "DEN".
Employee Work Building	4 digit alphanumeric field consisting of the building code an employee works in.
Employee Work Room	4 digit alphanumeric field consisting of an employee's office or cubicle number.
Employee Work Mail Stop	4 digit alphanumeric field consisting of the location an employee's mail is to be delivered.
Employee Status	1 digit alphanumeric field consisting of the employee's current employment status. Valid codes are "A" for active, "R" for retired, "L" for on leave of absence, and "F" for former employee.
Employee Seniority Date	8 digit numeric field in MM/DD/YYYY format consisting of an employee's first date of hire.
Employee Employment Activity Code	1 digit alphanumeric field indicating the type of employee employment activity. Valid codes are "H" for hire, "T" for termination, "L" for on leave of absence, "R" for return from leave of absence, and "P" for promotion.
Employee Employment Activity Date	8 digit numeric field in MM/DD/YYYY format consisting of the date reflecting the employment activity.
Employee Employment Activity Reason	Alphanumeric free-from field consisting of the reason for the employment activity.

CS 1 - MILESTONE 8: DESIGNING THE DATABASE

Employee United Way Contribution Date	8 digit numeric field in MM/DD/YYYY format consisting of the date of the United Way Campaign.
Employee United Way Contribution Code	1 digit alphanumeric field indicating the type of deduction. Valid codes are "F" for fair share, "O" for one time gift, "T" for specific payroll amount, and "N" for non-participating.
Employee United Way Contribution Amount	7 digit field that includes two decimal points, consisting of the amount to be contributed.
Employee Savings Bonds Contribution Date	8 digit numeric field in MM/DD/YYYY format consisting of the date of the Savings Bonds Campaign.
Employee Savings Bonds Deduction Type Code	1 digit alphanumeric field indicating the type of deduction. Valid codes are "O" for one deduction and "R" for repeating payroll deduction.
Employee Savings Bonds Deduction Amount	7 digit field that includes two decimal points, consisting of the amount to be deducted from payroll period.
Savings Bond Denomination Code	1 digit alphanumeric field indicating the amount of bond. Valid codes are "1" for \$100, "2" for \$200, "3" for \$500, and "4" for \$1000.
Bond Owner Name	Field consisting of the bond owner's last, first, and middle names.
Bond Owner Home Address	Field consisting of the bond owner's street address, city, state, and zip code.
Bond Owner Social Security Number	9 digit numeric field unique to bond owner.
Employee Relationship to Owner	1 digit alphanumeric field indicating the relationship to bond owner. Valid codes are "S" for self, "C" for co-owner, and "B" for beneficiary.
Primary Contact Name	Field consisting of the primary contact's last, first, and middle names.
Primary Contact Home Address	Field consisting of the primary contact's street address, city, state, and zip code.
Primary Contact Home Phone	10 digit numeric field consisting of the primary contact's home telephone number.
Primary Contact Work Phone	10 digit numeric field consisting of the primary contact's work telephone number.
Primary Contact Relationship to Employee	Alphanumeric field consisting of the relationship to the employee.
Secondary Contact Name	Field consisting of the secondary contact's last, first, and middle names.
Secondary Contact Home Address	Field consisting of the secondary contact's street address, city, state, and zip code.
Secondary Contact Home Phone	10 digit numeric field consisting of the secondary contact's home telephone number.
Secondary Contact Work Phone	10 digit numeric field consisting of the secondary contact's work telephone number.
Secondary Contact Relationship to Employee	Alphanumeric field consisting of the relationship to the employee.



MILESTONE 9 – DESIGNING INPUTS AND OUTPUTS

Synopsis

Management and users make important decisions based on system outputs. These outputs are produced from data that is either input or retrieved from databases. And data in databases must have been input first. Input design serves an important goal. Capture and get the data into a format suitable for the computer. Outputs present information to system users. Outputs, the most visible component of a working information system, are the justification for the system.

In this milestone you will design both an input screen and an output report for the Employee Self Service System.

Objectives

After completing this milestone, you should be able to:

- ⇒ Design a GUI input screen that uses the proper screen-based controls.
- ⇒ Design a computer output.

Prerequisites

Before starting this milestone the following topics should be covered:

1. Input Design & Prototyping Chapter 12, pages 436-457
of the SADM 4th ed. Textbook.

2. Output Design & Prototyping Chapter 13, pages 460-477
of the SADM 4th ed. Textbook.

Project

The goal of this project is to design both a GUI input screen and an output report to track employee data.

Activities

1. To properly design a GUI input screen, follow the steps outlined on pages 450 to 457 of the SADM 4th ed. textbook. Use Figure 12.6, page 453, of the SADM 4th ed. textbook as an example. Prepare a screen that will be used by Staffing personnel to input employee profile data. Include such things as employee number, names, addresses, phone numbers, work location data, department assigned, job code, date of hire, emergency contacts, and other related data. Refer to previously supplied forms, narratives, and the ESSS data dictionary for additional ideas. Make assumptions where necessary.
2. To properly design an output report, follow the steps outlined on pages 471 to 477 of the SADM 4th ed. textbook. Use Figure 13.8, page 476, of the SADM 4th ed. textbook as an example. Prepare a report that is a listing of all A-1 Information Systems employees. Include such things as department name, manager name, employee name, job title, date of hire, and employee work location related data. Employees should be sorted by job title within department. Refer to previously supplied forms, narratives, and the ESSS data dictionary for additional ideas. Make assumptions where necessary

Deliverable format and software to be used are according to your instructor's specifications. Deliverables should be neatly packaged in a binder, separated with a tab divider labeled "Milestone 9" and accompanied with a Milestone Evaluation Sheet (appearing first) found in Appendix A.

References:

Previous Narratives and supplied forms

ESSS Data Dictionary

Deliverables:

GUI Input Screen Design:

Due: ____/____/____

Time: _____

Output Report Design:

Due: ____/____/____

Time: _____

ADVANCED OPTION

For the advanced option, design the GUI input screens for an employee to enter their Saving Bonds deduction information and their United Way contribution information. Refer to the Savings Bonds and United Way forms provided in Milestone 3.

Savings Bonds Input Screen Design:

Due: ____/____/____

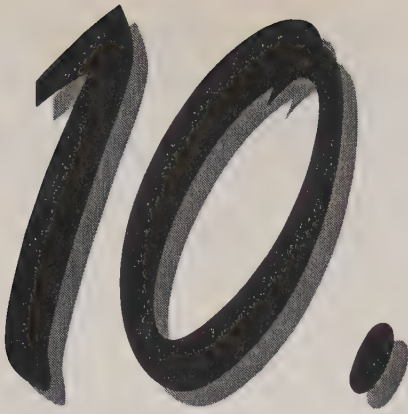
Time: _____

United Way Input Screen Design:

Due: ____/____/____

Time: _____

Milestone's Point Value:



MILESTONE 10 – DESIGNING THE SOFTWARE

Synopsis

You have designed the databases, inputs, outputs, and on-line user interfaces. You've selected appropriate computer equipment and packaged software (which has hopefully been delivered and installed during systems design). The final step involves software design. In software design we are concerned with how the programming specifications are presented to the computer programmer for implementation. To this end, we view software design as consisting of two components, modular design and packaging.

In this milestone you will use a technique called structure design that utilizes data flow diagrams and structure charts. The concept of structured design is simple. Design a program as a top-down hierarchy of modules. The top-down structure of these modules is developed according to various design rules and guidelines. The resulting hierarchy of modules can then be evaluated according to certain quality acceptance criteria to ensure the best modular design for the program. Upon completion, the modules are to be implemented using structured programming principles.

Structured design requires that data flow diagrams (DFDs) first be drawn for the program. You have already constructed some data flow diagrams in **Milestone 4**. However, because the audience was business users and management, not programmers, it is likely that those DFDs and their processes are not sufficiently detailed enough to be used to generate a structure chart. Thus, in this milestone you will refine the DFD created in **Milestone 4** to show more detail. In addition

you will perform transform analysis to identify which processes on the DFD perform input and editing, those that do processing or data transformation, and those that do output. In the advanced option you will create a structure chart based on the results of the transform analysis.

Objectives

After completing this milestone, you should be able to:

- ⇒ Revise a data flow diagram to reflect necessary program detail prior to program design.
- ⇒ Perform transform analysis to derive a program structure chart.
- ⇒ Construct a program structure chart.

Prerequisites

Before starting this milestone the following topics should be covered:

- | | |
|-------------------------|--|
| 1. Software Design | Chapter 15, pages 508-528
of the SADM 4 th ed. Textbook. |
| 2. Milestone 4 Solution | Provided by instructor |

Project

The goal of this project is to perform structure design that utilizes data flow diagrams and structure charts.

Activities

1. To create a data flow diagram with enough detail to produce a structure chart, reference the procedure outlined on pages 513 to 515 of the SADM 4th ed. textbook. Use Figure 15.4, page 515, of the SADM 4th ed. textbook as an example. Refine the Primitive DFD prepared in **Milestone 4** to include the necessary level of detail.
2. To perform transform analysis on the DFD of **Activity 1**, follow the steps outlined on pages 515 to 517 of the SADM 4th ed. textbook. Use Figure 15.5, page 516, of the SADM 4th ed. textbook as an example. Make a copy of the DFD prepared in **Activity 1**. Divide the DFD into the sections of Afferent, Central Transform, and Efferent. Label the sections appropriately as in Figure 15.5.

Deliverable format and software to be used are according to your instructor's specifications. Deliverables should be neatly packaged in a binder, separated with

a tab divider labeled "Milestone 10" and accompanied with a Milestone Evaluation Sheet (appearing first) found in Appendix A.

References:

Milestone 4 Solution

Provided by your instructor

Deliverables:

Refined Primitive DFD:

Due: __/__/__

Time: _____

Transform Analysis DFD:

Due: __/__/__

Time: _____

ADVANCED OPTION

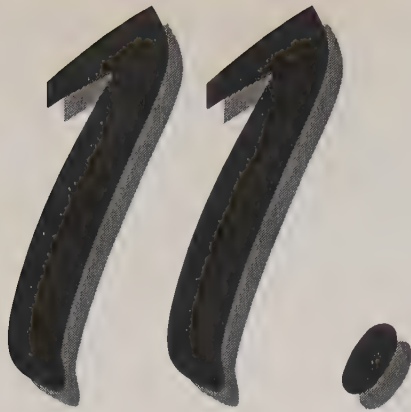
For the advanced option, use the results of the transform analysis to construct a structure chart. Follow the procedure outlined on pages 518 to 520 of SADM 4th ed. Textbook .Use figures 15.7, 15.8, and 15.9, pages 518 to 520 of SADM 4th ed. Textbook as examples.

Structure Chart:

Due: __/__/__

Time: _____

Milestone's Point Value:



MILESTONE 11 – MODELING THE OBJECTS

Synopsis

Object-oriented analysis techniques are best suited to projects that will implement systems using emerging object technologies to construct, manage, and assemble those objects into useful computer applications. The object-oriented approach is centered around a technique referred to as object modeling.

The object-oriented approach to system development is based on the concept of objects that exist within a system's environment. Objects are everywhere. Let's consider your environment, look around. What are some of the objects present within your environment? Perhaps you see a door, a window, or the room itself. What about this book – it's an object, and so is the very page you are reading. And of course, don't forget about you, you are an object. Perhaps you also have a student workbook with you and there are other individuals in the room. You may also see phone, a chair, and perhaps a table. All of these objects represent objects that may be clearly visible within your immediate environment.

In performing Object Oriented Analysis (OOA) like any other systems analysis method, the purpose is to gain a better understanding of the system and its functional requirements. In other words, OOA requires that we identify the objects, their data attributes, associated behavior, and relationships which support the required business system functionality. We perform object modeling to document the identified objects, the data and behavior they encapsulate, plus their relationships with other objects.

In this milestone you will first construct a context model used to communicate the scope and boundaries of the project. Secondly, you will perform use case modeling to document the business requirements and assist in identifying the necessary objects required by the system. Finally, you will construct an Object Association Model to graphically depict the object classes that make up the system as well as the objects' relationships with each other.

Objectives

After completing this milestone, you should be able to:

- ⇒ Construct a context model in order to communicate project scope and boundaries.
- ⇒ Prepare a Use Case Model Diagram to depict the scope and boundaries of a system in terms of actors and use cases.
- ⇒ Document a use case's course of events.
- ⇒ Construct an object association model to depict the objects of the system and how they are related to each other.

Prerequisites

Before starting this milestone the following topics should be covered:

1. Object Modeling Chapter 8, pages 284-304
 of the SADM 4th ed. Textbook.

Project

There are two general activities when performing object oriented analysis and they are as follows:

1. Finding and identifying the business objects
2. Organizing the objects and identifying their relationships

Activities

1. To construct the context model, use Figure 8.6, page 295, of the SADM 4th ed. textbook as an example. Refer to exhibit 11.1 which contains a narrative describing the scope of the Employee Self Service System. Use this narrative as the basis for your context model. Make assumptions where necessary.
2. To construct the use case model diagram, use Figure 8.8, page 297, of the SADM 4th ed. textbook as an example. Follow the steps 1 and 2 found on pages

295 and 296 of the SADM 4th ed. Textbook, to identify the actor s and use cases. Make assumptions where necessary.

3. To document a use case's course of events, use Figure 8.9, page 298, of the SADM 4th ed. textbook as an example. Follow the steps 3 and 5 found on pages 296 through 299 of the SADM 4th ed. Textbook. Refer to exhibit 11.2, which contains a narrative describing the event of an Employee Job Offer. Make assumptions where necessary. We have provided for your convenience a copy of the use case template on the textbook's web page [<http://www.tech.purdue.edu/textbooks/sadm/>] as well as a hardcopy at the end of this milestone.
4. To construct an Object Association Model, use Figure 8.16, page 304, of the SADM 4th ed. textbook as an example. Follow the steps 6 and 7 found on pages 299 through 303, and steps 1 through 4 found on pages 303 and 304 of the SADM 4th ed. Textbook. Use the use case course of events completed in Activity 3 above as the basis for the model. Make assumptions where necessary.

Deliverable format and software to be used are according to your instructor's specifications. Deliverables should be neatly packaged in a binder, separated with a tab divider labeled "Milestone 11" and accompanied with a Milestone Evaluation Sheet (appearing first) found in Appendix A.

References:**ESSS Scope Narrative**

Exhibit 11.1

ESSS Job Offer Narrative

Exhibit 11.2

Deliverables:

Context Model:

Due: __/__/__

Time: _____

Use Case Model:

Due: __/__/__

Time: _____

Job Offer Course Of Events:

Due: __/__/__

Time: _____

Object Association Model:

Due: __/__/__

Time: _____

ADVANCED OPTION

For the advanced option, construct a use case dependency diagram consisting of the use cases you included in preparing your use case model above. Make any necessary assumptions. Use Figure 8.10, page 299 of the SADM 4th ed. textbook as an example.

Use Case Dependency Diagram:

Due: __/__/__

Time: _____

Milestone's Point Value:

Exhibit 11.1

Use the following narrative to construct the Context Model for the Employee Self Service System (ESSS).

The purpose of the Employee Self Service System (ESSS) is to provide a single central repository of employee information. The Staffing Department requests that job offers be sent to potential employees who in turn can respond by accepting, declining or submitting a counter offer. Once an employee is hired, the Staffing Department submits the new employee profile information, The Staffing Department also provides any status changes for current employees. In response, the system generates staffing reports on a weekly, monthly, and yearly basis for the Staffing Department. The employees provide employee profile changes, as well as United Way and Savings Bonds contribution activity. In response, the system generates on request an Employee Participation Report for management, and Savings Bonds and United Way Contribution reports for the Employee Relations Department. The employee contribution activity records are sent to the Payroll System and ESSS accesses the Payroll database to retrieve the employee's current salary. Finally, an employee telephone listing can be generated for any employee who requests it.

Exhibit 11.2

Use the following narrative to construct the Context Model for the **Employee Self Service System (ESSS)**.

The Staffing hiring managers forward the names and starting salary amounts of candidates they wish to extend job offers, via electronic memorandum to their clerical representatives. It is the responsibility of the clerical representatives to verify and validate the candidate's name and address, then prepare the formal job offers using a form letter in MS Word, and then send the completed job offer letters to the candidates.

Candidates can respond one of three ways to the letter. They can decline the offer, they can accept the offer, or they can make a counter offer. The response can be made by phone or through the mail. The Staffing Department evaluates the candidate response. If the candidate declines the offer or submits a counter offer, the Staffing Department may plan a new offer and send it (via E-mail) to the clerical staff for processing. For those candidates who rejected the offer or submitted a counter offer, but are being rejected by Staffing, those names are sent (via E-mail) to the clerical staff for processing. The clerical representatives verify and validate the candidate's name and address, then prepare a formal rejection letter using a form letter in MS Word. Once printed, the clerical staff sends the completed rejection letters to the candidates. Finally, if the candidate accepts the offer, or Staffing accepts the candidate's counter offer, those names are sent (via E-mail) to the clerical staff for processing. The clerical representatives verify and validate the candidate's name and address, then prepare a formal job acceptance letter that includes the candidate's starting date, using a form letter in MS Word. Once printed, the clerical staff sends the completed acceptance letters to the candidates.

Once a candidate accepts a position, the clerical staff inputs the new employee's name, address, and ID number in to the

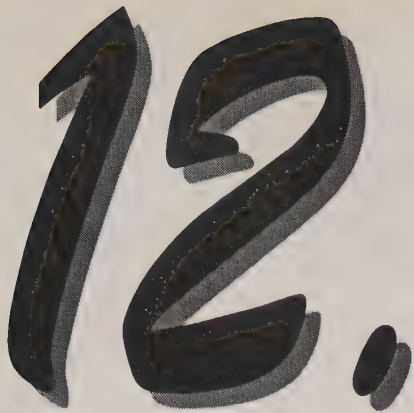
ESSS system where a new employee record is created. If the new employee was a former employee at one time, the old employee record is revised to reflect the new status.

USE CASE

Author: _____

Date: _____

USE CASE NAME:	
ACTOR:	
DESCRIPTION:	
NORMAL COURSE:	
ALTERNATE COURSE:	
PRE-CONDITION:	
POST-CONDITION:	
ASSUMPTIONS:	



MILESTONE 12 – DESIGNING THE OBJECTS

Synopsis

In performing object-oriented analysis we identified objects and use cases based on ideal conditions and independent of any hardware or software solution. During object-oriented design we want to refine those objects and use cases to reflect the actual environment of our proposed solution.

In this milestone you will first transform the use case course of events prepared in **Milestone 11** to be a design use case. Secondly, we update our object association model to include any additional attributes we have discovered. Finally, we will create an ideal object model diagram that models the object interactions for each use case.

Objectives

After completing this milestone, you should be able to:

- ⇒ Transform a requirements use case to a design use case.
- ⇒ Construct an ideal object model diagram to model the high-level object interactions for a use case.
- ⇒ Revise an object association diagram to include attributes and hierarchies.

Prerequisites

Before starting this milestone the following topics should be covered:

1. Object-Oriented Design Chapter 16, pages 532-550
 of the SADM 4th ed. Textbook.
2. Milestone 11 Solution Provided by your instructor.

Project

There are three general activities when performing object-oriented design and they are as follows:

1. Refining the use case model to reflect the implementation environment.
2. Modeling object interactions and behavior that support the use case scenarios.
3. Updating the object model to reflect the implementation environment.

Activities

1. To transform the *analysis use case* to a *design use case*, follow the steps on pages 537 and 538 of the SADM 4th ed. Textbook. Use Figure 16.2, page 538-539 of the SADM 4th ed. Textbook as an example. Use the completed analysis use case from **Milestone 11** to start from. Use **Exhibit 11.2** from **Milestone 11** to identify design items to include in the use case. Make assumptions where necessary.
2. To construct the *ideal model diagram*, follow the steps on page 540 of the SADM 4th ed. Textbook. Use Figure 16.5, page 541, of the SADM 4th ed. textbook as an example. Make assumptions where necessary.
3. To document additional identified attributes, use the *completed Object Association Diagram* prepared in **Milestone 11**. Make the necessary changes to this diagram to reflect the proposed environment

Deliverable format and software to be used are according to your instructor's specifications. Deliverables should be neatly packaged in a binder, separated with a tab divider labeled "Milestone 12" and accompanied with a Milestone Evaluation Sheet (appearing first) found in Appendix A.

References:

Milestone 11 Solution

Provided by your instructor.

ESSS Job Offer Narrative

Exhibit 11.2

Deliverables:

Design Use Case:

Due: __/__/__

Time: _____

Ideal Object Model:

Due: __/__/__

Time: _____

Refined Object Association Model:

Due: __/__/__

Time: _____

ADVANCED OPTION

For the advanced option, construct a Interaction Diagram for the Make Job Offer use case. Follow the steps to identify object behaviors and to prepare the interaction diagram. Make any necessary assumptions. Use Figure 16.10, page 546 of the SADM 4th ed. textbook as an example. Make any necessary assumptions.

Interaction Diagram:

Due: __/__/__

Time: _____

Milestone's Point Value:



ASSET TRACKING & MANAGEMENT SYSTEM

INTRODUCTION

In this section you will learn the necessary background information to be able to complete the upcoming milestones of this case study. This information includes a history of the business, a description of the business's current facilities, and the descriptions of the problems that triggered the project.

Case Background

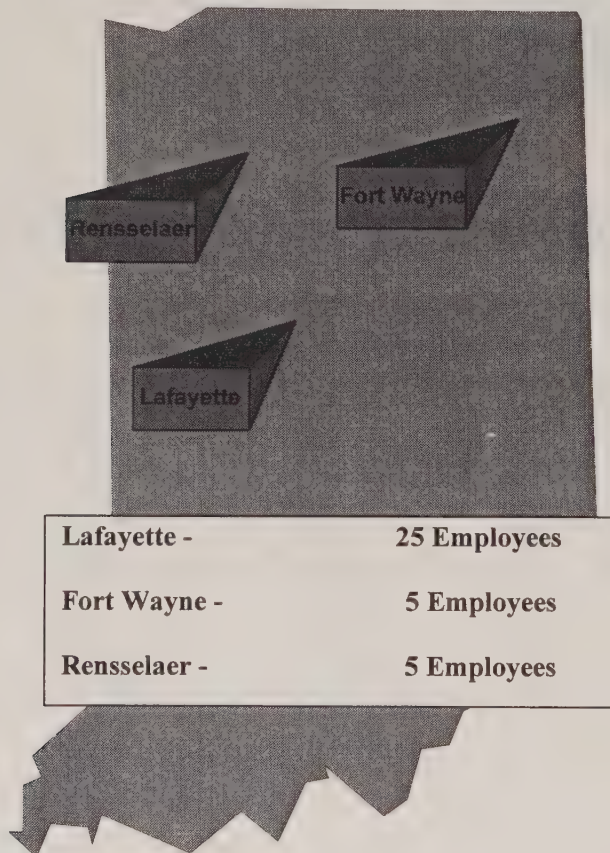
In 1947 Reid Lewis owned and operated a Marathon (gas), station and barely made a decent living to support his family. The station was a popular meeting place for the local teenagers, and Mr. Lewis seeing an opportunity to earn some additional income, allowed a local pinball operator to place a single machine in one corner of his building. The operator and Mr. Lewis agreed to evenly split the revenue generated by the pinball machine. At that time, a game of pinball would cost .05 cents and would consist of five plays. In comparison, today a game will cost .50 cents and consist of three plays. The machine was a big success and when the operator decided to retire from the business, Mr. Lewis offered to buy him out.

CASE STUDY 2 - ASSET TRACKING & MANAGEMENT SYSTEM

Mr. Lewis paid \$650 for the gentleman's route, which consisted of ten locations, each containing a single pinball machine.

Mr. Lewis's new business blossomed. All the profits earned from the machines were re-invested into the business to buy more machines. In 1948 he paid \$300 for his first jukebox, a used 1948 Seeburg, model number 1422 which held twenty, 78-RPM records. He continued to buy only used equipment until 1953 when he purchased his first new jukebox, a Rockola for \$700.00. Soon the growth of his new company forced him to sell the Marathon station and form R & S Entertainment Services.

R & S Entertainment Services has grown to be one of the largest amusement machine operators in North Central Indiana. Headquartered in Lafayette, Indiana, they have regional offices in Fort Wayne, IN and Rensselaer, IN. They employ thirty-five employees and service over five hundred establishments.

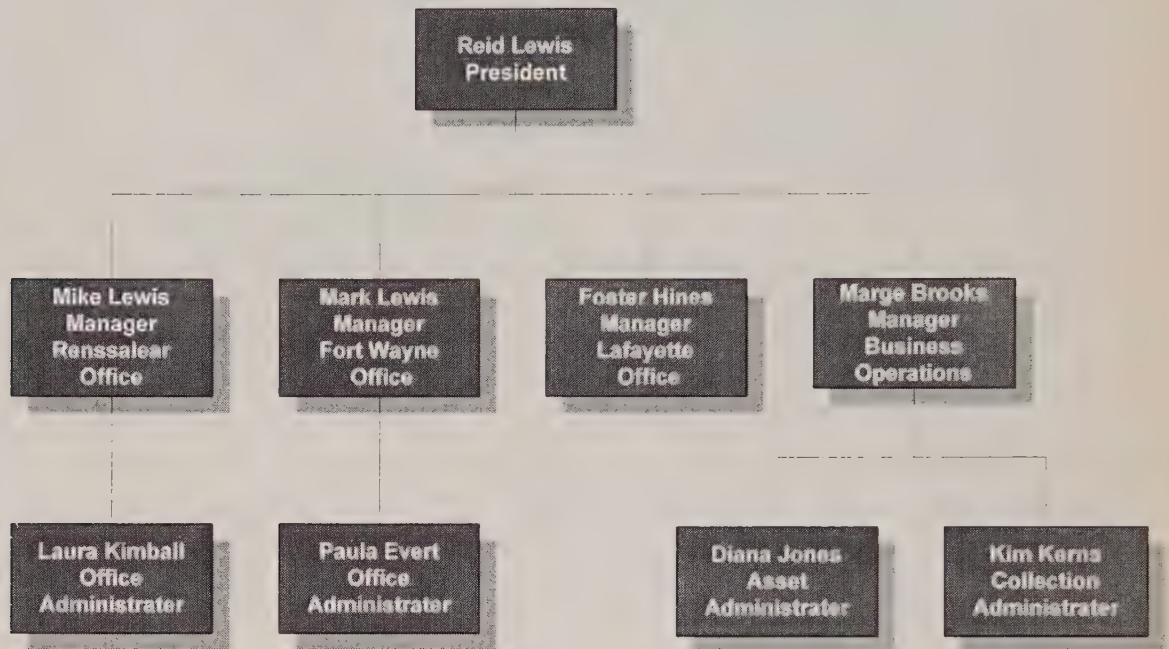


The types of establishments include arcades, restaurants, taverns, clubs, bowling alleys, campgrounds, and numerous others. Each of these establishments or

locations has installed one or more amusement machines from an inventory of 6500. The types of amusement machines include pinball machines, video games, pool tables, foosball tables, dart games, bowlers, air hockey tables, cranes, jukeboxes and change machines.

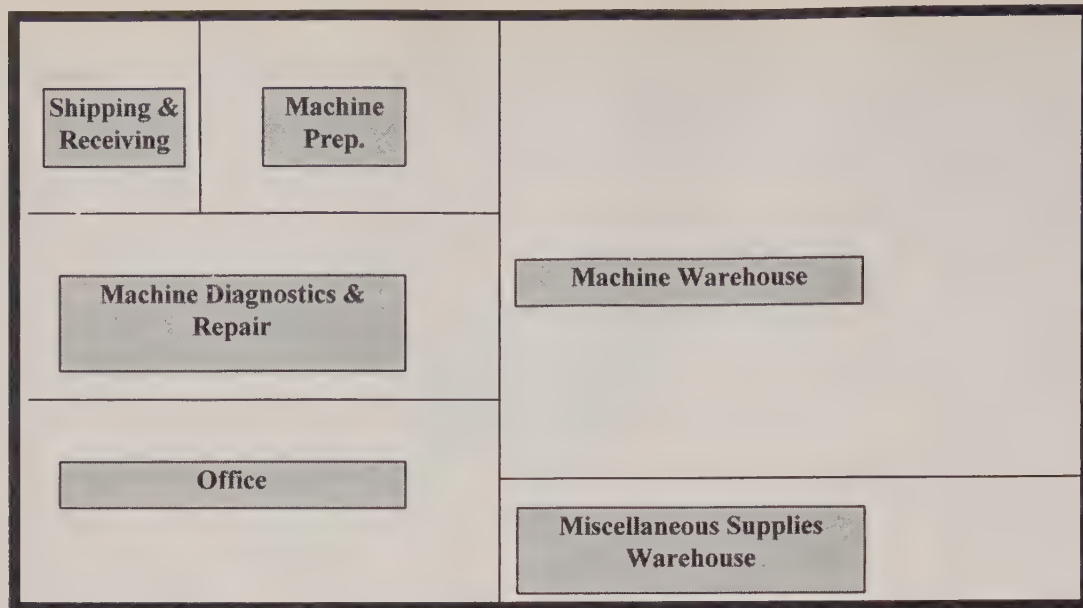
Organization Structure

R & S Entertainment Services



Physical Facilities

Facilities at R & S Entertainment Services headquarters include a 25,000 square foot building which houses the office, repair shop, prep shop, amusement machine warehouse, and miscellaneous supplies warehouse. Each region site contains 400 square feet of office space, plus 2000 square feet for machine storage, repair, and servicing.



Headquarters building layout.

Information Systems Facilities

The data processing facilities at R & S Entertainment Services are very limited. Currently, they have two stand-alone Pentium 100 PCs with 32 megabytes of memory for the business. One is dedicated for the use of sponsoring dart and pool tournaments and the other for word processing. Both of these machines are located at Lafayette site.

The configuration of the pool and dart tournament machine is: 1.5 gigabyte hard drive, both 5 1/4 and 3 1/2 inch floppy drives, 8x CDROM, 17" VGA color monitor, 28.8 baud Fax modem, HP Paintjet printer, HP Scanjet, MS Windows 95 and MS Office 95.

The configuration of the word processing machine is: 1.5 gigabyte hard drive, both 5 1/4 and 3 1/2 inch floppy drives, 8x CDROM, 17" VGA color monitor, 28.8 baud Fax modem, Iomega ZIP external disk drive, Epson LaserJet printer, MS Windows 95 and MS Office 95.

The Problem

R & S Entertainment Services currently has no automated amusement machine tracking system to manage their inventory of 6500 machines as well as tracking the revenue performance of each individual machine and business location.

The purchasing, installation, and disposal of machines is recorded manually and kept in binder or file cabinets. Because of this inadequate system, Mr. Lewis

cannot effectively make management decisions such as: when to order new machines, when and how to upgrade and service existing machines, and when to replace outdated machines with new ones. Also, Mr. Lewis needs to know information about his current inventory such as, quantity and type of machines in service, where the machines are currently located, the installation history of each machine as well as the business location, and accounting information for each machine (cost, current value, depreciation, etc.). Mr. Lewis believes he is not effectively utilizing his resources and thus losing considerable amounts of money, because he cannot effectively plan and make decisions. This is because he can't get the data he needs or he has very little confidence in the data he has. Numerous times he has purchased equipment he doesn't need, or in other cases has been asked to remove his machines from a business location because he couldn't or didn't effectively meet their needs.

Asset data needs to be available in real-time and presented in a user-friendly environment, in order for R & S Entertainment Services to respond to customer (business location) inquiries in a timely fashion. In the past, it has taken hours, even days, to respond to customer inquiries.



MILESTONE 1 - SURVEYING THE PROJECT

Synopsis

The purpose of the survey phase is threefold. First, the survey phase answers the question, “Is this project worth looking at?” To answer this question, the survey phase must define the scope of the project and the perceived problems, opportunities, and directives that triggered the project.

In this milestone you will prepare a *Request for System Services*, which is the trigger for the Survey Phase. Also, you will use fact finding techniques to extract and analyze information from an interview to determine project scope, level of management commitment, and project feasibility for the Asset Tracking and Management System (ATMS). With these facts and facts obtained from the Case Background, you will have the necessary information to complete the *Problem Statement Matrix* and construct the *Project Feasibility Assessment Report*.

Objectives

After completing this milestone, you should be able to:

- ⇒ Complete a *Request for System Services* form, which triggers the survey phase.
- ⇒ Analyze a user interview and extract pertinent facts, which can be used to assess project feasibility.
- ⇒ Complete a *Problem Statement Matrix* documenting the problems, opportunities, or directives of the project.
- ⇒ Prepare and understand the structure and content, of the *Project Feasibility Assessment Report*.

Prerequisites

Before starting this milestone the following topics should be covered:

1. The Survey Phase - Chapter 3, pages 84-87, Chapter 4, pages 128-137 of the SADM 4th ed. textbook.

Project

We have been contacted by Mr. Reid Lewis, the president of R & S Entertainment Services, to perform analysis and design of how they track and manage their amusement machine assets. In this assignment you first need to assist Reid Lewis in preparing a *Request for Systems Services*. Secondly, by analyzing the interview transcripts, determine the feasibility of the project, level of management commitment and project scope by using fact-finding techniques and the necessary communication skills to compose the *Problem Statement Matrix & Project Feasibility Assessment Report*.

Mr. Lewis was gracious enough to allow us to record our interview session and Exhibit 1.1 is a copy of the transcripts. Refer to the **Case Background** found in the **Introduction** and the interview transcripts for the information necessary to complete the following activities.

Activities

1. To complete the *Request for System Services* form, use Figure 4.7, page 130 of the SADM 4th ed. Textbook as an example. Use information from the case background only when completing the form. Make assumptions where necessary. We have provided for your convenience a template of the form on the textbook's web page [<http://www.tech.purdue.edu/textbooks/sadm/>], as well as a hardcopy at the end of this milestone.
2. To complete the *Problem Statement Matrix*, use Figure 4.8, page 132 of the SADM 4th ed. textbook as an example. Use the interview with Reid Lewis and the case background for the basis of your information. Make assumptions where necessary. Place yourself in the shoes of Reid Lewis. Which problems do you believe are the most visible, have the highest visibility, and how should they be ranked? Try to determine the annual benefits. State assumptions and be prepared to justify your answers! Finally, what would be your proposed solution based on the facts you know now? We have provided for your convenience a template of the matrix on the textbook's web page [<http://www.tech.purdue.edu/textbooks/sadm/>], as well as a hardcopy at the end of this milestone.

Deliverable format and software to be used are according to your instructor's specifications. Deliverables should be neatly packaged in a binder, separated with a tab divider labeled "Milestone 1" and accompanied with a Milestone Evaluation Sheet (appearing first) found in Appendix A.

References:

Case Background

Workbook Introduction

Transcripts of Interview with Reid Lewis

Exhibit 1.1

Deliverables:

Request for System Services:

Due: ____/____/____

Time: _____

Problem Statement Matrix:

Due: ____/____/____

Time: _____

ADVANCED OPTION

For the advanced option, prepare a Project Feasibility Assessment Report. Use the information provided by the case background, the user interview, and the completed problem statement matrix from Activity 2. Use Figure 4.9, page 137 of the SADM 4th ed. textbook as a sample outline and a copy of the outline is also presented in exhibit 1.2. Be sure to include Gantt charts for the project schedules. Information on Gantt charts can be found on pages 602-609 of the SADM 4th ed. Textbook. Information on fact-finding techniques can be found on pages 623-639 of the SADM 4th ed. Textbook. And, information on feasibility analysis can be found on 643-659 of the SADM 4th ed. Textbook.

Project Feasibility Assessment Report:

Due: ____/____/____

Time: _____

Milestone's Point Value:

The following is a copy of the transcripts of an interview between Mr. Reid Lewis of R & S Entertainment Services and Kira Webster, a computer information systems student. This was the initial interview with Mr. Lewis and its goal was to obtain facts about the problems and opportunities which triggered the project request, plus other general information which could be used to prepare the *Problem Statement Matrix*.

Exhibit 1.1

Scene: *Kira Webster, computer information systems student, is meeting with Reid Lewis, President of R & S Entertainment Services, at his office in Lafayette, Indiana. Ms. Webster scheduled the interview with Mr. Lewis in response to his request for developing a new asset tracking and management system.*

Reid: Good Morning! You must be Kira Webster.

Kira: Yes, I am sir. Are you Mr. Reid Lewis?

Reid: Yes. Please call me Reid. I'm glad we could get together today. Things are getting quite chaotic around here.

Kira: What do you mean?

Reid: My business operations manager, Marge Brooks, came into my office pleading for me to do something about the way we management our amusement machine inventory. She is basically at her wits end.

That is why I asked you to come see me.

Kira: I be glad to try and help. Can you tell me more about the problems that Marge is having?

Reid: I will try. She tried to explain them in detail to me this morning. I can't say I was much help. I don't get involved too much in what she does since she has done an excellent job in the past. Let me first say that anything that is related to the purchasing, tracking, and managing a machine is totally a manual process. Meaning none of the information is kept on the computer. It is all in mounds of paper that is stored in file cabinets. The closest thing we come to automation is faxing the purchase orders to the vendor.

Kira: I see.

Reid: The basis of my business is this. We are in the entertainment business. We provide amusement machines such as pinball machines, pool

tables, jukeboxes, video games, and the like, to various business locations for their customers to enjoy. The money earned by the machines is divided by the business location and us. That is our fee for us installing the machines.

Kira: Is the money divided evenly?

Reid: Depends on the contract. With every business location we deal with, we enter into a contract. Basically, it is an agreement between the business establishment and us, on what services we will provide, at what cost, and the time period of the contract. Depending on the nature of the business and the value of the machines factors into what the percentage split is.

Kira: Is the signing of a contract the trigger to buy new machines?

Reid: Sometimes it is. Sometimes we already have the machines in our warehouse that we can install. But most of the time we have to purchase new machines. That was the thing that Marge was so upset about. She really has no good way of keeping track of what we have, where it is installed at, and how well is it performing. I would wager that we spend thousands of dollars on buying new

equipment we really don't need. If we just had the information available to us to see what our inventory is, where it is installed, and how it is performing. We could make better decisions on the purchasing process. In a majority of the situations, instead of buying new machines, we could move machines from location to another location, especially if the machines were not performing well.

Kira: Who determines what machines will be purchased, and installed at the various locations?

Reid: Most of the time, it's the office managers at the three sites. If they need new machines, they give Marge a request on what they need. She prepares the purchase order and faxes it to the vendor who has the best price for that machine. When we receive the machines from the vendor, they are prepped, and the office manager decides on where they should be installed. Once the manager makes that decision, they fill out a move order and give to the technicians. The move order tells the technicians what machine to be installed (if any),

what machine is to be removed (if any), and from what business location.

Kira: I see. What happens to all the forms such as the purchase order, the machine request, and the move order after they have been processed?

Reid: You would have to ask Marge to be sure. But, I believe they are put in a file cabinet somewhere. Heck, that's half the problem. Nobody knows where the information is at, how it's stored, and it takes too long to find it. We need to have a history of every machine including when we bought it, from where, what it cost, how many repairs it has had, all the places it has been installed, and how well it has performed. We should be able to call that up and have it instantaneously.

Kira: I will certainly see what I can do. Do you receive anything else from the vendors?

Reid: Do you mean besides catalogs and the occasional junk mail?

Kira: Yes. (*laughing*) I mean like invoices, and shipping statements.

Reid: Sure we do. A shipper, that's what we call a shipping statement, accompanies every shipment that we receive. It

contains the serial number of the game, the name and the manufacturer. A few days later we receive the invoice through the mail. That's what Marge uses to pay the bill.

Kira: What happens to those?

Reid: Who knows? I assume they are probably filed away also.

Kira: Do you produce any reports or listings for anybody?

Reid: With what a typewriter? Just kidding. At the end of every year Marge creates a listing of all the assets we have purchased and disposed of, and gives that to our Accountant. She just writes the information down on paper.

Kira: Okay. Let's switch gears a little bit. I understand you have multiple offices? Do you want each office to have the capability to access the new system?

Reid: Yes, we also have offices in Fort Wayne and Rensselaer, and yes I want them to definitely be able to access the system. But, I want the data to reside here in Lafayette.

Kira: Do you want them to have access to the data once you make changes to it, or can they wait till the next day?

Reid: Yes, I want the managers to have access to the most current data.

Kira: Do all offices basically perform the same way?

Reid: No. The regional offices only handle machine moves and collections. All other business functions are handled here.

Kira: You mentioned collections.

Reid: That's a whole another can of worms. We will save that for the next project. Besides, I need this done in three months. You won't have time to tackle the collection problem also.

Kira: Okay. I will take your word for it! Lets talk about your computer technology. You currently have only two personal computers, is that correct?

Reid: Yes that's true. I believe one is used for the dart tournaments we sponsor and the other is used for word processing. They are both located here in Lafayette. We are probably not using them to their full potential. I think I have only two or three employees that actually know how to use a computer. I really would like to have all employees to have access to the system if they need it. And, yes I understand I

will need to purchase additional computers to be able to do that.

Kira: Good. So purchasing additional hardware and software if required will not be a problem?

Reid: If you can justify the need then I will make sure we get the hardware and software we need.

Kira: Would you expect us to train the employees that don't know how to use a computer?

Reid: Most definitely!

Kira: Before I forget to ask, may I have a copy of your organization chart and who of your organization will be working with me on this project?

Reid: Fair request. I will have Marge send you a copy of the company org. chart [*the organization chart appears in the case introduction*]. As for your direct contact, Marge will also fulfill that role. But, I don't have a problem of you talking to anyone in the company if it will help you.

Kira: Thank you. For this project to be a success we must work as a team and you the user certainly must be involved. Mr. Lewis would it also be possible for me

to have copies of the documents you use in your business?

Reid: Which ones?

Kira: Well if you don't mind, copies of the move order, purchase order, shipper, and invoice for starters.

Reid: Sure. I will have Marge provide those to you.

Kira: I see that my sixty minutes is about up, so hopefully we won't be much longer.

Reid: No problem. It is not often that I get to talk to a person such as yourself.

Kira: Why thank you Mr. Lewis, uh, I think. Lets stay on the subject and I have just a few more questions I need to ask. I'm sure you're a busy man so I don't want to continue this meeting needlessly. Mr. Lewis what do you think the new system should be like.?

Reid: Interesting question. Let me think. I see a system that is easy to use, intuitive, and has fields that can be clicked with the mouse so you don't to type every thing. It is accessible from every office and maybe even from my home, if that's possible. I should be able to get reports of the data I need and in the format that I want, when I want them.

Kira: That's a good vision. Are there parts of the system that should be secured, in that you wouldn't want everyone to be able to access those?

Reid: I believe there probably has to be, but I'm not sure right now what they are. I need to think on it somemore and discuss it with Marge.

Kira: Very good. Do you have a budget set for the development of this new system?

Reid: Not really. I understand we need to purchase new hardware and software and that can be costly. I would like to stay under \$100,000 if we could.

Kira: I will definitely do my best. We will know more once we have completed a detailed analysis and design of course. Is there anything else you would like to tell me about the system?

Reid: That's all I can think of right now.

Kira: OK. I'm going to review this information and give you my report of it early next week. Thank you for your time today, Mr. Lewis.

Reid: You're welcome. See you next week Kira.

Kira: Goodbye Mr. Lewis.



R & S Information Services

Phone: 743-7655 Fax: 743-8873

**REQUEST FOR
INFORMATION
SYSTEM SERVICES**

DATE OF REQUEST	SERVICE REQUESTED FOR DEPARTMENT(S)
	Business Operations

SUBMITTED BY (key user contact)		EXECUTIVE SPONSOR (funding authority)	
Name	Marge Brooks	Name	Reid Lewis
Title	Business Operations Manager	Title	President,
Office	Lafayette	Office	Lafayette
Phone	743-7655	Phone	743-7655

TYPE OF SERVICE REQUESTED:

- | | |
|---|---|
| ☐ Information Strategy Planning | ☐ Existing Application Enhancement |
| ☐ Business Process Analysis and Redesign | ☐ Existing Application Maintenance (problem fix) |
| ☒ New Application Development | ☐ Not Sure |
| ☐ Other (please specify) | |

BRIEF STATEMENT OF PROBLEM, OPPORTUNITY, OR DIRECTIVE (attach additional documentation as necessary)

BRIEF STATEMENT OF EXPECTED SOLUTION

ACTION (ISS Office Use Only)

☐ Feasibility assessment approved

Assigned to Professor K. C. Dittman

☐ Feasibility assessment waived

Approved Budget \$ 100,000

Start Date ASAP Deadline ASAP

☐ Request delayed

Backlogged until date: _____

☐ Request rejected

Reason: _____

Authorized Signatures:

Reid Lewis

Project Executive Sponsor

PROBLEM STATEMENTS

PROJECT:	PROJECT MANAGER:
CREATED BY:	LAST UPDATED BY:
DATE CREATED:	DATE LAST UPDATED:

Brief Statements of Problem, Opportunity, or Directive	Urgency	Visibility	Annual Benefits	Priority or Rank	Proposed Solution
1.					
2.					
3.					
4.					
5.					
6.					
7.					
8.					

Exhibit 1.2

Project Feasibility Assessment Report

- I. Executive summary (1 page)
 - A. Summary of recommendation
 - B. Brief statement of anticipated benefits
 - C. Brief explanation of report contents
- II. Background information (1-2 pages)
 - A. Brief description of project request
 - B. Brief explanation of the survey phase activities
- III. Findings (2-3 pages)
 - A. Problems and analysis (optional: reference *problem statement matrix*)
 - B. Opportunities and analysis (optional: reference *problem statement matrix*)
 - C. Directives and implications
- IV. Detailed recommendation
 - A. Narrative recommendation (1 page)
 - 1. Immediate fixes
 - 2. Quick fixes
 - 3. Enhancements
 - 4. New systems development
 - B. Project Plan
 - 1. Initial project objectives
 - 2. Initial master project plan (phase level - Gantt chart)
 - 3. Detailed plan for the study or definition phase (Gantt chart)
- V. Appendices
 - A. Request for System Services
 - B. Problem Statements Matrix
 - C. (other documents as appropriate)



MILESTONE 2 - STUDYING THE CURRENT SYSTEM

Synopsis

There's an old saying that suggests, "Don't try to fix it unless you understand it." With those words of wisdom, the next milestone of our project is to study and analyze the existing system. There is always an existing business system, regardless of whether it currently uses a computer. The study phase provides the project team with a more thorough understanding of the problems, opportunities, and/or directives that triggered the project. Indeed, the analyst frequently uncovers new problems and opportunities. The study phase may answer the questions, "Are the problems worth solving?" and "Is a new system worth building?"

The purpose of the study phase is threefold. First and foremost, the project team must gain an appropriate understanding of the business problem domain. Second, we need to answer the question, "Are these problems (opportunities, and directives) worth solving"? Finally, we need to determine if the system is worth developing. The study phase provides the systems analyst and project team with a more thorough understanding of the problems, opportunities, and/or directives that triggered the project. In the process, they frequently uncover new problems and opportunities.

In this milestone you will perform *Cause-Effect Analysis* using information obtained through an interview with Marge Brooks. You will document your findings using the *Problems, Opportunities, Objectives, and Constraints Matrix*. The PIECES framework, originally developed by James Wetherbe, and then

adapted by the authors, can serve as a useful tool to classify the various problems, opportunities, and directives identified in the user interview.

Objectives

After completing this milestone, you should be able to:

- ⇒ Perform *Cause-Effect Analysis* to be able to thoroughly understand a system's problems, opportunities, and/or directives that triggered the project.
- ⇒ Use and understand the PIECES framework for classifying problems, opportunities, and directives.
- ⇒ Complete the *Problems, Opportunities, Objectives, and Constraints Matrix*.

Prerequisites

Before starting this milestone the following topics should be covered:

1. The Study Phase - Chapter 3, pages 87-88,
Chapter 4, pages 137- 150
of the SADM 4th ed. textbook.
2. PIECES Framework - Chapter 3, pages 79-81 of the SADM 4th ed.
Textbook.

Project

Now that we have completed the survey of the system, and gained approval to proceed, we can now attempt to gain a better understanding of the current system. In this assignment we will use an interview with Marge Brooks, in order to perform cause-effect analysis. The results of this activity will provide us a better understanding of the problems, opportunities, and constraints of the current system.

Mrs. Brooks was gracious enough to allow us to record our interview session and Exhibit 2.1 is a copy of the transcripts. Refer to this interview only for the information necessary to complete the following activities.

Activities

1. To complete the *Problems, Opportunities, Objectives, and Constraints Matrix*, use Figure 4.12, page 144 of the SADM 4th ed. textbook as an example. Make assumptions where necessary. Use the PIECES framework as a model to classify the problems, opportunities, and directives. Figure 3.4, pages 80-81 of the SADM 4th ed. textbook, may identify other areas that present problems or

offer opportunities. We suggest you use it. It has proven to be a very valuable tool! We have provided for your convenience a copy of the matrix template on the textbook's web page [<http://www.tech.purdue.edu/textbooks/sadm/>], as well as a hardcopy at the end of this milestone.

Deliverable format and software to be used are according to your instructor's specifications. Deliverables should be neatly packaged in a binder, separated with a tab divider labeled "Milestone 2" and accompanied with a Milestone Evaluation Sheet (appearing first) found in Appendix A.

References:**Transcripts of Interview with Marge Brooks**

Exhibit 2.1

Deliverables:**Problems, Opportunities, Objectives, and Constraints Matrix:**

Due: __/__/__

Time: _____

ADVANCED OPTION

This deliverable was not discussed in the narrative because students need to be exposed to modeling (data, process, interface, & geography), before this report can be completed. For those ambitious individuals who are familiar with those skills and wish to be challenged, use the detailed study report outline found in Figure 4.13, page 150 of the SADM 4th ed. textbook, as a guideline. We have also provided a copy of the outline in Exhibit 2.1

Study Report:

Due: __/__/__

Time: _____

Milestone's Point Value: _____

The following is a copy of the transcripts of an interview between Mrs. Marge Brooks of R & S Entertainment Services and Kira Webster, a computer information systems student. This interview was scheduled with Mrs. Brooks to discuss R & S Entertainment Services' current business processes and the problems that have been occurring. The goal of this interview was to learn more about the current system and to obtain enough information to perform cause-effect analysis.

Exhibit 2.1

Scene: *Kira Webster, computer technology student at Purdue University, is meeting with Marge Brooks, Manager of Business Operations for R & S Entertainment Services, at her office. Miss Webster scheduled the interview with Mrs. Brooks in order to learn more about the current system and ask questions about the current business processes when dealing with assets.*

Marge: Good Morning Kira! How are you today?

Kira: Fine thank you. How are you Mrs. Brooks.

Marge: Very well thank you and please call me Marge. What can I do for you today.

Kira: Well, I want to learn as much as I can about your current system and its business processes.

Marge: We don't really have a current system. I mean we

don't utilize the computer for anything that we do.

Kira: Just because your processes are not automated doesn't mean you don't have a system. A system can be made up of manual processes using a typewriter, file cabinets and the like.

Marge: I see. I didn't know that. Where would you like me to start?

Kira: Let's start with the purchase of new equipment, the machines you install at a business location. What triggers the purchasing process and how is it performed?

Marge: The demand to purchase new machines is usually driven by the event of us getting a new location.

Kira: What do you mean?

Marge: When a business establishment such as a pizza parlor, wants us to install machines at their location, we sign a new contract with the business for us to install amusement machines for a percentage of the revenue that

the machine earns. If we don't have the necessary machines in our warehouse, or installed at another location we have to purchased them.

Kira: I see. Do you make that decision?

Marge: Not by myself. The office managers determine the need and submit a request to me.

Kira: Let's see if I understand this. If a new location needs machines installed, the office managers check the available inventory and if no such machines exist, they submit a request to you to purchase them.

Marge: That's correct. Also, we may purchase machines to replace outdated stock, machines that are broke, machines that have been stolen, or ones that are not performing the way they should. And of course we may purchase any new game that is released by the manufacturer because our customers always demand the "latest and greatest" machines.

Kira: Interesting. How do you actually do the purchasing?

Marge: Once we determine we need to place an order, I complete a

purchase order with the items we want to order and fax it to the vendor.

Kira: Is the purchase order a form that you complete by hand?

Marge: No, we have pre-printed forms that we buy from an office supply store and I use a typewriter to type in the information.

Kira: How long does it usually take to prepare a purchase order and how many purchase orders do you process in a month?

Marge: Well, once we determine what we want to order, I search the catalogs of the vendors or call them to get the best price. We have four vendors we typically deal with. Once I have that information, I fill out the purchase order. Depending on the number of items, it may take me two to four hours to complete a single purchase order. I do on an average, ten per week - that means about 40-50 per month.

Kira: Is taking two-four hours to do a purchase order acceptable to you?

Marge: By all means no! I despise typing. Honestly, we should be able to generate a purchase order in 15-30 minutes. That would give me time to do other things I need to do.

Kira: Have you ever found that you ordered something that you didn't need, or received and billed for something you didn't order?

Marge: Of course! One of the biggest problems we have around here is not accurately tracking our inventory of machines. We don't really have an accurate count of how many machines we have, where they are installed at, or how well they are doing. We order approximately \$1,500,000 worth of new equipment every year. I would estimate about 25% of that is unnecessary. If we knew the status and whereabouts of all our equipment, we could save ourselves a lot of money. In terms of receiving things we didn't order, yes, that does happen sometimes. When it does happen, a lot of times it too late too correct it.

Kira: What do you mean?

Marge: I mean the boys in the back have already unpacked it, prepped it, and put it on the truck ready to be installed somewhere. By that time it's too late to put it on a truck and ship it back. We tried to make the policy that all new shipments need to be

confirmed by Diana or myself, but it didn't work out. By the time Diana or I had the time to do it, found the correct paperwork, and reconciled all the numbers, two days have past. The boys in the back weren't very happy with us. Especially the managers, who had to deal with the unhappy customers.

Kira: What would be acceptable?

Marge: Well, as soon as the equipment is delivered, all the paperwork should be checked out and verified. It should take no longer than one hour.

Kira: Do you ever verify the equipment you receive?

Marge: Yes. Later when we have time, we take the shipment statement, that was shipped along with the equipment, and reconcile it with the purchase order or orders on which the equipment was ordered. We keep all pending purchase orders in a file. When a piece of equipment comes in, we attach the shipping statement to the proper purchase order, then write on the purchase order "received" next to the appropriate line item. When we receive the invoice, we do the same thing. And when we pay the bill, we make a copy of the check, and attach it also. That is our way of tracking the ordering

and receiving of a machine. Once all the equipment of a particular purchase order has been received and paid for, the purchase order and all of its attachments are filed away. You can see why it is such a nightmare and time consuming process to have to track anything down.

Kira: Yes, I can. Do you ever use the purchase order and all its attachments again?

Marge: At the end of the year we pull them all out and give them to our accountant for tax purposes.

Kira: What do they do with them?

Marge: I believe he enters all the information into his computer system so he can prepare our tax statements. I know he charges us an arm and a leg for doing it! It's probably for all the time he has to go through to make sense of our paperwork and then re-enter it into his system. He does the same thing with our ledger that contains a record for all our collections of revenues. If we could provide him the data in a better format, I'm sure we could save thousands of dollars in accountant fees.

Kira: I will see what I can do. Let's get back to the machines. Exactly how does the technicians know where to install a machine?

Marge: The site managers, Mike, Mark, or Foster will prepare a move order indicating where a particular machine is to be moved from (de-installed) or moved to (installed). There can be several of these install/de-installs on one order.

Kira: How does the manager know what machines to be moved?

Marge: Oh, they talk with the people that do the collections periodically and ask them which machines are doing poorly. Or, a repair technician may report that a machine is broke or needs servicing. Or, we may have a new business location that needs set, or one that has went out of business and we need to pick all our machines up. Finally, we may have a special request from a business location for a particular machine. To keep that customer happy we will try to honor the request.

Kira: I see. How often are these move orders created.

Marge: I would say each site goes through about five orders per day. Each move order has up to ten individual machine moves.

Once an individual move has been completed, the technician signs and date stamps that particular move item. This is our way of tracking where the equipment is coming from and going to, plus when it got done and who did it. Once an order has been completed it gets filed away. Unfortunately, it's not a very good process.

Kira: What do you mean?

Marge: Right now we don't keep a history of the machines that have been installed at a business location. Therefore, we rely on the memory of the manager, collector, and the technician as to whether a certain kind of machine has been installed at the location before. It's much too time consuming to search through all the move orders for a particular location. There could be thousands. Sometimes when we are ready to install the machine we discover it has already been installed there before and performed poorly, or the location already has a similar machine installed. At any rate we don't look good in the customer's eyes, and we wasted the labor cost of a

technician going up there and the gas and maintenance of the truck.

Kira: I see. Do you ever get rid of machines?

Marge: Yes we do. Periodically we sell used equipment to people who call up and want an authentic amusement machine for their recreation room or basement. We also have had machines stolen, destroyed by vandalism, and burnt up in fires. Once in awhile we will have a machine we can't sell and it is not good enough to be installed at a location. We usually just scavenge it for parts and then junk the rest. At any rate, we write down all the necessary information and give it to the accountant.

Kira: Good. Is there anything else you would like to tell me about how your system handles the amusement machines?

Marge: That's all I can think of right now.

Kira: OK. I'm going to review this information and give you my report of it next week. Thank you for your time today, Mrs. Brooks.

Marge: You're welcome. I'll see you next week.

PROBLEMS, OPPORTUNITIES, OBJECTIVES AND CONSTRAINTS MATRIX

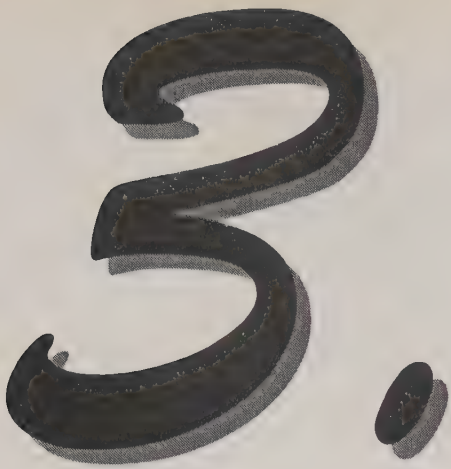
Project:	Project Manager:
Created by:	Last Updated by:
Date Created:	Date Last Updated:

CAUSE AND EFFECT ANALYSIS		SYSTEM IMPROVEMENT OBJECTIVES	
Problem or Opportunity	Causes and Effects	System Objective	System Constraint
1.	1.	1.	1.

Exhibit 2.1

Analysis of the Current _____ System

- I. Executive summary (approximately 2 pages)
 - A. Summary of recommendation
 - B. Summary of problems, opportunities, and directives
 - C. Brief statement of system improvement objectives
 - D. Brief explanation of report contents
- II. Background information (approximately 2 pages)
 - A. List of interviews and facilitated group meetings conducted
 - B. List of other sources of information that were exploited
 - C. Description of analytical techniques used
- III. Overview of the current system (approximately 5 pages)
 - A. Strategic implications (if the project is part of, or impacts an existing information systems strategic plan)
 - B. Models of the current system
 1. Interface model (showing project scope)
 2. Data model (showing project scope)
 3. Geographic models (showing project scope)
 4. Process model (showing functional decomposition only)
- IV. Analysis of the Current System (approximately 5-10 pages)
 - A. Performance problems, opportunities, and cause/effect analysis
 - B. Information problems, opportunities, and cause/effect analysis
 - C. Economic problems, opportunities, and cause/effect analysis
 - D. Control problems, opportunities, and cause/effect analysis
 - E. Efficiency problems, opportunities, and cause/effect analysis
 - F. Service problems, opportunities, and cause/effect analysis
- V. Detailed recommendations (approximately 5-10 pages)
 - A. System improvement objectives and priorities
 - B. Constraints
 - C. Project Plan
 1. Scope reassessment and refinement
 2. Revised master plan
 3. Detailed plan for the definition phase
- VI. Appendices
 - A. Any detailed system models
 - B. (other documents as appropriate)



MILESTONE 3 - MODELING THE SYSTEM'S DATA

Synopsis

The definition phase answers the question, ‘What does the user need and want from a new system?’ The definition phase is critical to the success of any new information system! In this milestone we need to identify what information systems requirements need to be defined from the system users’ perspectives and draw graphical, logical, models to document the data requirements for a new and improved system.

Data modeling is a technique for organizing and documenting a system’s Data. Data modeling is sometimes called database modeling because a data model is usually implemented as a database. Data is viewed as a resource to be shared by as many processes as possible. As a result, data must be organized in a way that is flexible and adaptable to unanticipated business requirements – and that is the purpose of data modeling.

In this milestone you will first discover those entities in the system that are or might be described by data. With each entity we identify, we will define it in respect to the business. Then, we will construct a *Context Data Model* that graphically depicts each of the entities and the relationships they have with each other. Next, we will refine the context data model to include primary and foreign keys. The resulting model is called a *Key-Based Data Model*. Finally, we refine the key-based data model to include any hierarchies and attributes, and this model is referred to as the *Fully Attributed Data Model*.

Objectives

After completing this milestone, you should be able to:

- ⇒ Understand and perform the techniques for entity discovery.
- ⇒ Define each entity with respect to the business and complete an entity/definition matrix.
- ⇒ Perform the necessary data modeling techniques to organize and document the data requirements for the proposed system.
- ⇒ Construct the Context, Key-Based, and Fully Attributed data models.

Prerequisites

Before starting this milestone the following topics should be covered:

1. Data Modeling - Chapter 5, pages 170-202
of the SADM 4th ed. textbook.

Project

Now that we have studied the current system and analyzed some of its problems and opportunities, plus gained approval to proceed, we can now start to identify the business data requirements and graphically model them. In this assignment we will use the business forms provided by Marge Brooks. The results of this activity will identify some of the business data requirements for the proposed system.

Activities

1. To complete the *Entity/Definition Matrix*, use Table 5.5, page 192 of the SADM 4th ed. textbook as an example. Analyze each of the forms provided by Marge Brooks. Make assumptions where necessary. We have provided for your convenience a template of the matrix on the textbook's web page [<http://www.tech.purdue.edu/textbooks/sadm/>], as well as a hardcopy in the back of this milestone.
2. To complete the *Context Data Model* use Figure 5.11, page 194 of the SADM 4th ed. textbook as an example. Construct the model to include each of the entities in **Activity 1** and depict the relationships that exist among the entities. Data model notation is according to your instructor's specifications, however the authors' prefer the notation depicted in Figure 5.3, page 180 of the SADM 4th ed. textbook.

3. To complete the *Key-Based Data Model* use Figure 5.12, page 194 of the SADM 4th ed. textbook as an example. Refine the Context Data Model from **Activity 2** to include associate entities that resolve many-to-many relationships, and include primary and foreign keys. Again data model notation is according to your instructor's specifications, however the authors' prefer the notation depicted in Figure 5.3, page 180 of the SADM 4th ed. textbook.
4. To complete the *Fully Attributed Data Model* use Figure 5.14, pages 200 & 201 of the SADM 4th ed. textbook as an example. Refine the Key-Based Data Model from **Activity 3** to include any generalization hierarchies, and add the data attributes for each entity.

Deliverable format and software to be used are according to your instructor's specifications. Deliverables should be neatly packaged in a binder, separated with a tab divider labeled "Milestone 3" and accompanied with a Milestone Evaluation Sheet (appearing first) found in Appendix A.

References:

Sample business forms provided by Marge Brooks.

Exhibits 3.1-3.5

Deliverables:

Entity Definition Matrix:

Due: __/__/__

Time: _____

Context Data Model:

Due: __/__/__

Time: _____

Key-Based Data Model:

Due: __/__/__

Time: _____

Fully Attributed Data Model:

Due: __/__/__

Time: _____

ADVANCED OPTION

For the advanced option, review the Case Background, the Request for System Services, and the previous interviews with Reid Lewis and Marge Brooks to discover any additional entities and attributes. Construct a fully attributed data model containing the additional entities and attributes that you have found.

Entity Definition Matrix:**Due:** __/__/__**Time:** _____**Fully Attributed Data Model:****Due:** __/__/__**Time:** _____**Milestone's Point Value:**

Exhibit 3.1

The following business form is a purchase order form that R&S Entertainment Services uses to order amusement machines from their vendors.

00045

PURCHASE ORDER**Date:** 05/28/97

**R & S
ENTERTAINMENT SERVICES**
412 Main Street
Lafayette, IN 47905
(765) 743-7655

VENDOR:

One Stop Amusement
Distributing Company
P. O. Box 20046
Pittsburgh, PA 15230
(412) 398-7789
Fax: (412) 398-0003

BUYER: Marge Brooks

PRODUCT NO.	NAME -MANUF.	QTY	UNIT PRICE	TOTAL PRICE
66578	Jurassic Park pinball - Williams	2	\$3250.00	\$6500.00
J55642	CD150F jukebox - Rowe	1	\$3700.00	\$3700.00
P1000564	8E Maple pool table - Valley	4	\$1750.00	\$7000.00
TOTAL ORDER AMOUNT				\$17200.00

Exhibit 3.3

R & S Entertainment Services records the name and address of each business location in a rolodex. Following is an example of the data they would record.



*John's Pizza King
2356 18th Street
Lafayette, TN
(765) 474-2212*

Date of Contract 05/08/95 – 05/08/2000

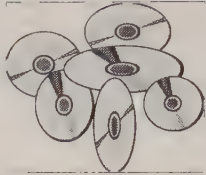
Owner – John Higgins (765) 233-8976

Manager – none

*Picked up – 2/23/97 – Restaurant
was sold.*

Exhibit 3.4

The following business form is used by the managers to request the purchase of machines.

MACHINE PURCHASE REQUEST			
			DATE: <u>6/1/97</u>
LOCATION:	<i>John's Pizza King</i> <i>2356 187th Street</i> <i>Lafayette, IN</i> <i>(765) 474-2212</i>	R&S ENTERTAINMENT SERVICES 412 Main Street Lafayette, IN 47905 (765) 743-7655	
Name		Qty	
<i>William's Jurassic Park pinball</i>		<i>2</i>	
<i>Rowe CD1507 jukebox</i>		<i>1</i>	
<i>Valley 8E Maple pool tables</i>		<i>4</i>	
COMMENTS <i>Need ASAP!</i> <i><u>Foster Hines</u></i>			

The following business form is used by managers to issue orders to install and remove machines.

Exhibit 3.5

00986

MACHINE INSTALL/REMOVE ORDER

Date: 06/13/97

FOR: David Foster

MACHINE NAME	INSTALL LOCATION	REMOVE LOCATION	DATE/ TIME
<i>World Cup pinball</i> <i>SN - 567893</i>		<i>John's Pizza King</i>	<i>6/13/97</i> <i>1:00</i>
<i>Jurassic Park pinball</i> <i>SN - 303558765</i>	<i>John's Pizza King</i>		<i>6/13/97</i> <i>1:10</i>

Entity/Definition Matrix

ENTITY	BUSINESS DEFINITION



MILESTONE 4 - MODELING THE SYSTEM'S PROCESSES

Synopsis

Process modeling is a technique for organizing and documenting the structure and flow of data through a system's processes and/or the logic, policies, and procedures to be implemented by a system's processes. In this milestone we focus on using and constructing data flow diagrams (DFDs) and decomposition diagrams to perform process modeling.

Data flow diagrams are tools that depict the flow of data through a system and the work or processing performed by that system. A decomposition diagram is a DFD planning tool that shows the top-down functional decomposition and structure of a system.

During this milestone you will first construct a *context diagram* to establish project scope and boundaries. Secondly, you will draw a *functional decomposition diagram* to partition the system into logical subsystems and/or functions. Thirdly, you will complete a partially completed *event list* that identifies and confirms the business events to which the system must respond. Once that task is completed, you will refine the decomposition diagram drawn earlier to include event-handling processes. Finally you will construct a *system data flow diagram* that shows the big picture of the system, and a *primitive data flow diagram* for a single event process.

Objectives

After completing this milestone, you should be able to:

- ⇒ Construct a context diagram to illustrate a system's interfaces with its environment.
- ⇒ Perform event partitioning and organize events in a functional decomposition diagram.
- ⇒ Identify external and temporal business events for a system.
- ⇒ Merge event diagrams into a system data flow diagram.
- ⇒ Draw primitive data flow diagrams.

Prerequisites

Before starting this milestone the following topics should be covered:

1. Process Modeling - Chapter 6, pages 205-254
of the SADM 4th ed. textbook.
2. Milestone 3 Solution

Project

As a systems analyst or knowledgeable end-user, you must learn how to draw decomposition and data flow diagrams to model business process requirements. The survey and study phases of the methodology have been completed and you understand the current system's strengths, weaknesses, limitations, problems, opportunities, and constraints. You have already built the data model (**Milestone 3**) to document business data requirements for the new system. You now need to build the corresponding process models.

Activities

1. To complete the *Context Diagram*, use Figure 6.20, page 242 of the SADM 4th ed. textbook as an example. Remember that the context diagram establishes the project scope and boundaries by depicting the system's interfaces with external parties. Refer to the narrative in Exhibit 4.1 to construct the diagram.
2. To complete the *Functional Decomposition Diagram*, use Figure 6.21, page 243 of the SADM 4th ed. textbook as an example. Assume that the system contains two subsystems with one of them being "Machine Purchasing". What could be the other one?

3. To complete the *Event – Response List*, use Figure 6.22, page 245 of the SADM 4th ed. textbook as an example. Exhibit 4.2 is a copy of a partially completed Event-Response List for the Machine Purchasing Subsystem. Complete the list by providing data in the blank cells. Make assumptions where necessary. We have provided for your convenience a template of the Event-Response List on the textbook's web page [<http://www.tech.purdue.edu/textbooks/sadm/>].
4. To complete the *Event Decomposition Diagram*, use Figure 6.23, page 246 of the SADM 4th ed. textbook as an example. Refine the decomposition diagram constructed in **Activity 2** above, to include the event processes identified in **Activity 3** above. Focus only on the Machine Purchasing Subsystem.
5. To complete the *System Diagram*, use Figure 6.27, pages 250 & 251 of the SADM 4th ed. textbook as an example. Exhibit 4.3 contains two Event Diagrams for the events within the Machine Purchasing Subsystem. Create the other event diagrams then merge all event diagrams together to produce the System Diagram.
6. To complete the *Primitive Diagram*, use Figure 6.28, page 252 of the SADM 4th ed. textbook as an example. Refer to the narrative in Exhibit 4.4 to construct the primitive diagram for the Employee Profile Changes Event.

Deliverable format and software to be used are according to your instructor's specifications. Deliverables should be neatly packaged in a binder, separated with a tab divider labeled "Milestone 4" and accompanied with a Milestone Evaluation Sheet (appearing first) found in Appendix A.

References:

Context Diagram Narrative

Exhibit 4.1

Partially Completed Event-Response List

Exhibit 4.2

Event Diagrams

Exhibit 4.3

Primitive Diagram Narrative

Exhibit 4.4

Milestone 3 Solution

Provided by your instructor

Deliverables:**Context Diagram:****Due:** __/__/__**Time:** _____**Functional Decomposition Diagram:****Due:** __/__/__**Time:** _____**Event-Response List:****Due:** __/__/__**Time:** _____**Event Decomposition Diagram:****Due:** __/__/__**Time:** _____**System Diagram:****Due:** __/__/__**Time:** _____**Primitive Diagram:****Due:** __/__/__**Time:** _____**ADVANCED OPTION**

For the advanced option, try to complete activities 3 through 6 for one of the other subsystems. In this option you will need to create the Event Diagrams before you can do the System Diagram. Make any assumptions as necessary.

Subsystem 2 or 3: _____**Due:** __/__/__**Time:** _____**Milestone's Point Value:** _____

Exhibit 4.1

Use the following narrative to construct the Context Diagram for the Asset Tracking and Management System (ATMS).

The purpose of the Asset Tracking and Management System (ATMS) is to provide a single central repository of machine acquisition, movement, performance, and disposal history. Management submits purchase requests and in response the system generates purchase orders, which are faxed to the vendor. In response the vendor ships amusement machines along with a shipping statement. The vendor also sends an invoice, which triggers a payment to be sent back to the vendor. Business locations provide a signed contract and in response machines are installed at the location. Management also provides machine move order requests, which generate move orders for the technicians to carry out. Management can request a machine history report, an inventory report, and a location machine inventory and history report.

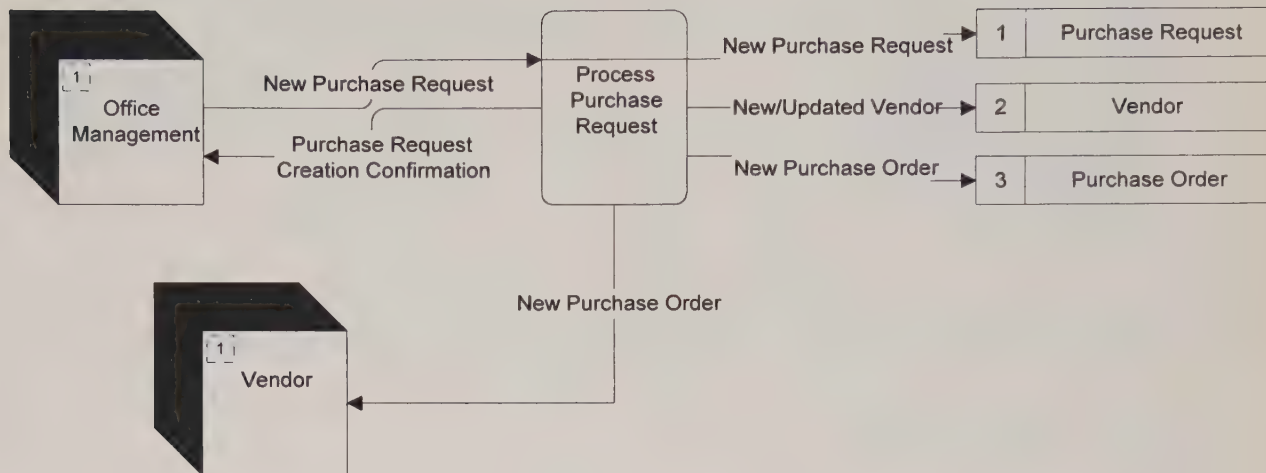
Exhibit 4.2

EVENT LIST

<i>Event Description</i>	Trigger (Inputs)	Responses (Outputs)
Office managers submit request to purchase machines.	Purchase Request	Create/update Vendor Create Purchase Order Generate Purchase Order form
Vendor ships ordered machine.	Shipper	Create Machine Create Shipper
Vendor sends Invoice		
Time to provide new machine purchases and machine disposals to the accountant.		

Exhibit 4.3

Event Diagram
Event: Management submits purchase request



Event Diagram
Event: Vendor Ships Product

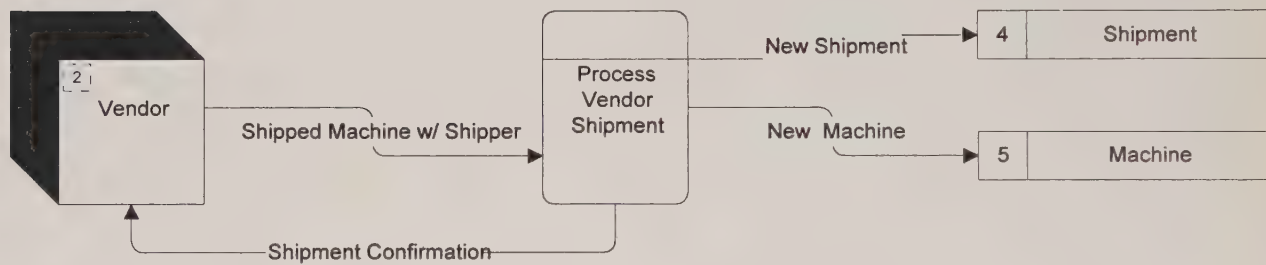


Exhibit 4.4

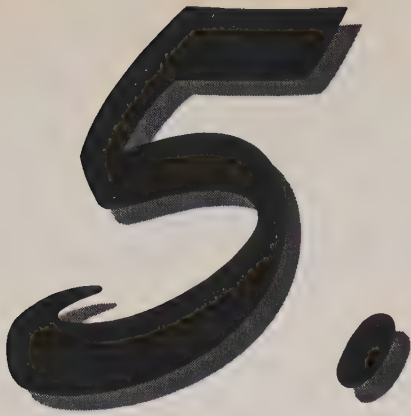
Use the following narrative to construct the Primitive Diagram for the Process Employee Profile Changes event.

Given an Vendor Invoice *validate* the vendor address and telephone information. If what is on file is different, update what is on file.

Verify machines on invoice against stored purchase orders and current machines. If machine was not ordered or machine has not yet been received, send notification to vendor.

Once the invoice has been verified, store invoice then *prepare* payment for the amount due.

Once payment has been prepared, store payment then *send* payment to vendor.



MILESTONE 5 - MODELING THE NETWORK

Synopsis

As a systems analyst or knowledgeable end-user, you should learn how to draw network models to be able to deal with today's distributed business and computing environment. Network modeling is a diagrammatic technique used to document the shape of a business or information system in terms of its business locations.

In this milestone you will first construct a *Location Decomposition Diagram* to logically decompose and group locations. Next we will draw a *Location Connectivity Diagram* that depicts the shape of the system in terms of its user, process, data, and interface locations and the necessary interconnections between the locations.

Objectives

After completing this milestone, you should be able to:

- ⇒ Factor a system's or application's locations into component locations using a special location decomposition diagram.
- ⇒ Document the connections and essential data flow between locations using a special location decomposition diagram.

Prerequisites

Before starting this milestone the following topics should be covered:

1. Network Modeling - Chapter 7, pages 262-280
of the SADM 4th ed. textbook.

2. Case Study Introduction

Project

Now that we have completed the survey and study phases of the system development lifecycle, and we fully understand the system's strengths, weaknesses, limitations, problems, opportunities, and constraints, it is time to model the logical business network requirements.

Refer to the case study introduction to complete the following activities.

Activities

1. To complete the *Location Decomposition Diagram*, use Figure 7.8, page 277 of the SADM 4th ed. textbook as an example. Refer to the case study introduction to find the necessary information to construct diagram. Pay close attention to the organization chart and the location map. These can provide valuable information. Make assumptions where necessary.
2. To complete the *Location Connectivity Diagram*, use Figure 7.10, page 279 of the SADM 4th ed. textbook as an example. Refer to the case study introduction to find the necessary information to construct diagram. Again, pay close attention to the organization chart and the location map. These can provide valuable information. Make assumptions where necessary. Use a road atlas or other type of map to determine the mileage between sites.

Deliverable format and software to be used are according to your instructor's specifications. Deliverables should be neatly packaged in a binder, separated with a tab divider labeled "Milestone 5" and accompanied with a Milestone Evaluation Sheet (appearing first) found in Appendix A.

References:

Case Study Introduction

Deliverables:**Location Decomposition Diagram:****Due:** __/__/__**Time:** _____**Location Connectivity Diagram:****Due:** __/__/__**Time:** _____**ADVANCED OPTION**

For the advanced option, assume R & S Entertainment Services now has the technology of communicating via cell phone and laptop computers from the technician's trucks. Each site has this technology. Prepare a location decomposition diagram and a location connectivity diagram that includes these new features.

Location Decomposition Diagram:**Due:** __/__/__**Time:** _____**Location Connectivity Diagram:****Due:** __/__/__**Time:** _____**Milestone's Point Value:**



MILESTONE 6 - DEFINING AND ANALYZING CANDIDATE SOLUTIONS

Synopsis

The purpose of the Configuration phase is to identify candidate solutions, analyze those candidate solutions, and recommend a target system that will be designed and implemented. Alternative solutions to be considered should be those that address the business requirements of the information system.

In this milestone you will complete a *Candidate Matrix*. This matrix will include three alternative candidate solutions that you have determined will meet the business requirements for the Asset Tracking and Management System. Some of the solutions you will consider may have been posed from the design ideas and opinions from the system owners and users. Others may come from various sources including: systems analysts, system designers, technical consultants, and other IS professionals. Some technical choices may be limited by a predefined, approved technology architecture provided by system managers. When completing the matrix, it is not your intent to evaluate the candidates at this point, only to identify and define them.

Objectives

After completing this milestone, you should be able to:

- ⇒ Identify candidate solutions that meet the requirements of the information system and document those solutions in a candidate matrix.

Prerequisites

Before starting this milestone the following topics should be covered:

1. The Configuration Phase - Chapter 9, pages 319-335 of the SADM 4th ed. textbook.
2. Feasibility and Cost-Benefit Analysis Module C, pages 642-659 of the SADM 4th ed. Textbook.

Project

Given the business requirements established in the definition phase of systems analysis, you can now identify alternative candidate solutions.

Activities

1. To complete the *Candidate Matrix*, use Figure 9.5, page 322, and Figure C.7, page 658 of the SADM 4th ed. textbook as an example. Based on the current workstation configuration defined in the Case Study Introduction, identify and document three solutions that could meet ATMS's needs. Assume that the system requires a typical client/server interface as well as a World Wide Web interface. Assume that all data will be housed in Lafayette. Also, assume LAN printers will provide printing services, usually one LAN printer per 5 employees. Make other assumptions where necessary. We have provided for your convenience a copy of the Candidate Matrix template on the textbook's web page [<http://www.tech.purdue.edu/textbooks/sadm/>], as well as a hardcopy at the end of this milestone.

Deliverable format and software to be used are according to your instructor's specifications. Deliverables should be neatly packaged in a binder, separated with a tab divider labeled "Milestone 6" and accompanied with a Milestone Evaluation Sheet (appearing first) found in Appendix A.

References:

Case Study Introduction

Deliverables:**Candidate Matrix:****Due:** __/__/__**Time:** _____**ADVANCED OPTION**

For the advanced option, perform feasibility analysis and complete a feasibility matrix for the three candidates identified above. This option will require you to do research on the cost of hardware and software. Assume that 14 new PCs and associated software will have to be purchased to use the new system. Also, assume that one new server and associated software will be purchased. Assume inter-office connectivity will be provided by the internet, but Lafayette will require their own Ethernet based network. Make the necessary assumptions to estimate development time, costs, and benefits. Use Figure 9.6, page 324, and Figure C.9, page 659 of the SADM 4th ed. textbook as an example. We have provided for your convenience a copy of the Feasibility Matrix template on the textbook's web page [<http://www.tech.purdue.edu/textbooks/sadm/>], as well as a hardcopy at the end of this milestone.

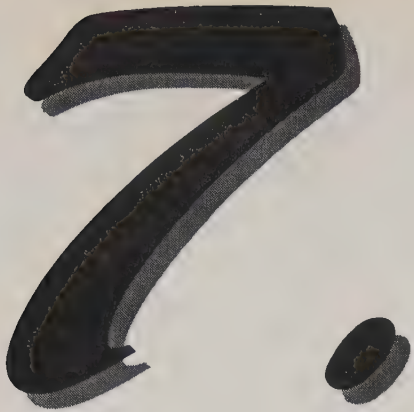
Feasibility Matrix:**Due:** __/__/__**Time:** _____**Milestone's Point Value:** _____

Candidate Matrix Template

Characteristics	Candidate 1	Candidate 2	Candidate 3
Portion of System Computerized Brief description of that portion of the system that would be computerized in this candidate.			
Benefits Brief description of the business benefits that would be realized for this candidate.			
Servers and Workstations A description of the servers and workstations needed to support this candidate.			
Software Tools Needed Software tools needed to design and build the candidate (e. g., database management system, emulators, operating systems, languages, etc.). Not generally applicable if applications software packages are to be purchased.			
Application Software A description of the software to be purchased, built, accessed, or some combination of these techniques.			
Method of Data Processing Generally some combination of: on-line, batch, deferred batch, remote batch, and real-time.			
Output Devices and Implications A description of output devices that would be used, special output requirements, (e.g. network, preprinted forms, etc.), and output considerations (e.g., timing constraints).			
Input Devices and Implications A description of Input methods to be used, input devices (e.g., keyboard, mouse, etc.), special input requirements, (e.g. new or revised forms from which data would be input), and input considerations (e.g., timing of actual inputs).			
Storage Devices and Implications Brief description of what data would be stored, what data would be accessed from existing stores, what storage media would be used, how much storage capacity would be needed, and how data would be organized.			

Feasibility Matrix Template

Feasibility Criteria	Wt.	Candidate 1	Candidate 2	Candidate 3
Operational Feasibility Functionality. A description of to what degree the candidate would benefit the organization and how well the system would work. Political. A description of how well received this solution would be from both user management, user, and organization perspective.	30%	Score:	Score:	Score:
Technical Feasibility Technology. An assessment of the maturity, availability (or ability to acquire), and desirability of the computer technology needed to support this candidate. Expertise. An assessment to the technical expertise needed to develop, operate, and maintain the candidate system.	30%	Score:	Score:	Score:
Economic Feasibility Cost to develop: Payback period (discounted): Net present value: Detailed calculations:	30%	Score:	Score:	Score:
Schedule Feasibility An assessment of how long the solution will take to design and implement.	10%	Score:	Score:	Score:
Ranking:	100%			



MILESTONE 7 – MODELING THE APPLICATION ARCHITECTURE

Synopsis

Just as we modeled business requirements during systems analysis, we should model technology architecture and requirements during systems design. The models serve as blueprints for system design, prototyping and construction.

In this milestone you will prepare a *Physical Data Flow Diagram*. Physical data flow diagrams model the technical and human design decisions to be implemented as part of an information system. They communicate technical and other design constraints to those who will actually implement the system—in other words, they serve as a technical blueprint for the implementation.

Objectives

After completing this milestone, you should be able to:

⇒ Draw physical data flow diagrams for an information system's architecture and processes.

Prerequisites

Before starting this milestone the following topics should be covered:

1. Application Architecture & Process Design

Chapter 10, pages 350-387
of the SADM 4th ed. Textbook.

** Project**

The goal of this project is to produce a high-level general design that can serve as an application architecture for the system and a general design for the processes that make up the system.

** Activities**

1. To construct the Physical Data Flow Diagram, use Figure 10.8, page 373, of the SADM 4th ed. textbook as an example. Prepare the DFD based on the process of receiving a purchasing request and preparing the purchase order. Refer to the interview with Marge Brooks in **Milestone 2** for the necessary data to prepare the DFD. Make assumptions where necessary.

Deliverable format and software to be used are according to your instructor's specifications. Deliverables should be neatly packaged in a binder, separated with a tab divider labeled "Milestone 7" and accompanied with a Milestone Evaluation Sheet (appearing first) found in Appendix A.

References:**Processing a Purchase Request**

Marge Brooks Interview – Milestone 2.

Deliverables:

Physical Data Flow Diagram:

Due: __/__/__

Time: _____

ADVANCED OPTION

For the advanced option, construct a Network Topology Data Flow Diagram based on the three sites of R & S Entertainment Services. Assume that Lafayette is the central hub with network connections branching from there to the other two sites. Make the necessary assumptions for types of networks and line speeds. Use Figure 10.12, page 382 of the SADM 4th ed. textbook as an example.

Network Topology DFD:**Due:** __/__/__**Time:** _____**Milestone's Point Value:**



MILESTONE 8 – DESIGNING THE DATABASE

Synopsis

All information systems create, read, update and delete data. This data is stored in files and databases. To fully exploit the advantages of database technology, a database must be carefully designed. Database design translates the data models that were developed for the system users during the definition phase into data structures supported by the chosen database technology. Subsequent to database design, system builders will construct those data structures using the language and tools of the chosen database technology.

In this milestone you will normalize the data model created in **Milestone 3**, along with addition data requirements, to be in third normal form. In the advanced option you will transform the logical data model into a physical model or database schema.

Objectives

After completing this milestone, you should be able to:

- ⇒ Normalize a logical data model to remove impurities that can make a database unstable, inflexible, and nonscalable.
- ⇒ Transform a logical data model into a physical, relational database schema.

 Prerequisites

Before starting this milestone the following topics should be covered:

1. Database Design Chapter 11, pages 392-428
of the SADM 4th ed. Textbook.

 Project

The goal of this project is to normalize our logical data model to remove impurities that can make a database unstable, inflexible, and nonscalable.

 Activities

1. To construct an Entity Relationship Diagram to be in 3rd Normal form, follow the normalization procedure outlined on pages 409 to 418 of the SADM 4th ed. textbook. Use Figure 11.15, page 419, of the SADM 4th ed. textbook as an example. Use the ERD solution of Milestone 3 to prepare the new data model. Make assumptions where necessary.

Deliverable format and software to be used are according to your instructor's specifications. Deliverables should be neatly packaged in a binder, separated with a tab divider labeled "Milestone 8" and accompanied with a Milestone Evaluation Sheet (appearing first) found in Appendix A.

References:**Milestone 3 Solution**

Provided by your instructor

Deliverables:

Logical Data Model in 3rd Normal Form:

Due: __/__/__

Time: _____

ADVANCED OPTION

For the advanced option, transform your normalized data model created above into a physical database schema complete with referential integrity rules. Make any necessary assumptions. Use Figure 11.17, page 425 of the SADM 4th ed. textbook as an example.

Physical Database Schema:**Due:** __/__/__**Time:** _____**Milestone's Point Value:** _____



MILESTONE 9 – DESIGNING INPUTS AND OUTPUTS

Synopsis

Management and users make important decisions based on system outputs. These outputs are produced from data that is either input or retrieved from databases. And data in databases must have been input first. Input design serves an important goal. Capture and get the data into a format suitable for the computer. Outputs present information to system users. Outputs, the most visible component of a working information system, are the justification for the system.

In this milestone you will design both an input screen and an output report for the Employee Self Service System.

Objectives

After completing this milestone, you should be able to:

- ⇒ Design a GUI input screen that uses the proper screen-based controls.
- ⇒ Design a computer output.

Prerequisites

Before starting this milestone the following topics should be covered:

- | | |
|--------------------------------|--|
| 1. Input Design & Prototyping | Chapter 12, pages 436-457
of the SADM 4 th ed. Textbook. |
| 2. Output Design & Prototyping | Chapter 13, pages 460-477
of the SADM 4 th ed. Textbook. |

** Project**

The goal of this project is to design both a GUI input screen and an output report to amusement machine data.

** Activities**

1. To properly design a GUI input screen, follow the steps outlined on pages 450 to 457 of the SADM 4th ed. textbook. Use Figure 12.6, page 453, of the SADM 4th ed. textbook as an example. Prepare a screen that will be used by the business office to prepare a purchase order. Refer to the form in **Milestone 3** for the necessary fields. Make assumptions where necessary.
2. To properly design an output report, follow the steps outlined on pages 471 to 477 of the SADM 4th ed. textbook. Use Figure 13.8, page 476, of the SADM 4th ed. textbook as an example. Prepare a report that shows the history of all the machines that have been installed at a particular business location. Include the machine ID, machine name, machine type, date installed and date removed. Refer to previously supplied forms on **Milestone 3** for additional ideas. Make assumptions where necessary

Deliverable format and software to be used are according to your instructor's specifications. Deliverables should be neatly packaged in a binder, separated with a tab divider labeled "Milestone 9" and accompanied with a Milestone Evaluation Sheet (appearing first) found in Appendix A.

References:**Previous Narratives and supplied forms****Deliverables:****GUI Input Screen Design:****Due:** __/__/__**Time:** _____**Output Report Design:****Due:** __/__/__**Time:** _____

ADVANCED OPTION

For the advanced option, design the GUI input screens for recording an invoice and creating a new business location. Make assumptions where necessary. Refer to the Invoice and Rolodex forms provided in Milestone 3.

Invoice Input Screen Design:**Due:** __/__/__**Time:** _____**Business Location Input Screen Design:****Due:** __/__/__**Time:** _____**Milestone's Point Value:**



MILESTONE 10 – DESIGNING THE SOFTWARE

Synopsis

You have designed the databases, inputs, outputs, and on-line user interfaces. You've selected appropriate computer equipment and packaged software (which has hopefully been delivered and installed during systems design). The final step involves software design. In software design we are concerned with how the programming specifications are presented to the computer programmer for implementation. To this end, we view software design as consisting of two components, modular design and packaging.

In this milestone you will use a technique called structure design that utilizes data flow diagrams and structure charts. The concept of structured design is simple. Design a program as a top-down hierarchy of modules. The top-down structure of these modules is developed according to various design rules and guidelines. The resulting hierarchy of modules can then be evaluated according to certain quality acceptance criteria to ensure the best modular design for the program. Upon completion, the modules are to be implemented using structured programming principles.

Structured design requires that data flow diagrams (DFDs) first be drawn for the program. You have already constructed some data flow diagrams in **Milestone 4**. However, because the audience was business users and management, not programmers, it is likely that those DFDs and their processes are not sufficiently detailed enough to be used to generate a structure chart. Thus, in this milestone you will refine the DFD created in **Milestone 4** to show more detail. In addition you will perform transform analysis to identify which processes on the DFD

perform input and editing, those that do processing or data transformation, and those that do output. In the advanced option you will create a structure chart based on the results of the transform analysis.

Objectives

After completing this milestone, you should be able to:

- ⇒ Revise a data flow diagram to reflect necessary program detail prior to program design.
- ⇒ Perform transform analysis to derive a program structure chart.
- ⇒ Construct a program structure chart.

Prerequisites

Before starting this milestone the following topics should be covered:

- | | |
|-------------------------|--|
| 1. Software Design | Chapter 15, pages 508-528
of the SADM 4 th ed. Textbook. |
| 2. Milestone 4 Solution | Provided by instructor |

Project

The goal of this project is to perform structure design that utilizes data flow diagrams and structure charts.

Activities

1. To create a data flow diagram with enough detail to produce a structure chart, reference the procedure outlined on pages 513 to 515 of the SADM 4th ed. textbook. Use Figure 15.4, page 515, of the SADM 4th ed. textbook as an example. Refine the Primitive DFD prepared in **Milestone 4** to include the necessary level of detail.
2. To perform transform analysis on the DFD of **Activity 1**, follow the steps outlined on pages 515 to 517 of the SADM 4th ed. textbook. Use Figure 15.5, page 516, of the SADM 4th ed. textbook as an example. Make a copy of the DFD prepared in **Activity 1**. Divide the DFD into the sections of Afferent, Central Transform, and Efferent. Label the sections appropriately as in Figure 15.5.

Deliverable format and software to be used are according to your instructor's specifications. Deliverables should be neatly packaged in a binder, separated with

a tab divider labeled “Milestone 10” and accompanied with a Milestone Evaluation Sheet (appearing first) found in Appendix A.

References:

Milestone 4 Solution

Provided by your instructor

Deliverables:

Refined Primitive DFD:

Due: __/__/__

Time: _____

Transform Analysis DFD:

Due: __/__/__

Time: _____

ADVANCED OPTION

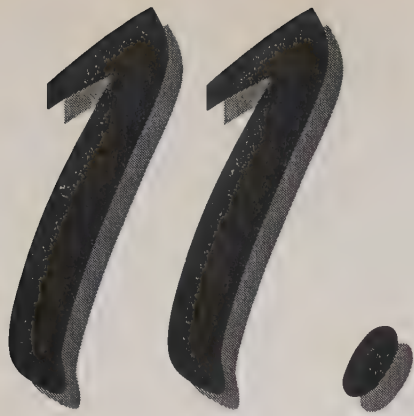
For the advanced option, use the results of the transform analysis to construct a structure chart. Follow the procedure outlined on pages 518 to 520 of SADM 4th ed. Textbook .Use figures 15.7, 15.8, and 15.9, pages 518 to 520 of SADM 4th ed. Textbook as examples.

Structure Chart:

Due: __/__/__

Time: _____

Milestone's Point Value: _____



MILESTONE 11 – MODELING THE OBJECTS

Synopsis

Object-oriented analysis techniques are best suited to projects that will implement systems using emerging object technologies to construct, manage, and assemble those objects into useful computer applications. The object-oriented approach is centered around a technique referred to as object modeling.

The object-oriented approach to system development is based on the concept of objects that exist within a system's environment. Objects are everywhere. Let's consider your environment, look around. What are some of the objects present within your environment? Perhaps you see a door, a window, or the room itself. What about this book – it's an object, and so is the very page you are reading. And of course, don't forget about you, you are an object. Perhaps you also have a student workbook with you and there are other individuals in the room. You may also see phone, a chair, and perhaps a table. All of these objects represent objects that may be clearly visible within your immediate environment.

In performing Object Oriented Analysis (OOA) like any other systems analysis method, the purpose is to gain a better understanding of the system and its functional requirements. In other words, OOA requires that we identify the objects, their data attributes, associated behavior, and relationships which support the required business system functionality. We perform object modeling to document the identified objects, the data and behavior they encapsulate, plus their relationships with other objects.

In this milestone you will first construct a context model used to communicate the scope and boundaries of the project. Secondly, you will perform use case modeling to document the business requirements and assist in identifying the necessary objects required by the system. Finally, you will construct an Object Association Model to graphically depict the object classes that make up the system as well as the objects' relationships with each other.

Objectives

After completing this milestone, you should be able to:

- ⇒ Construct a context model in order to communicate project scope and boundaries.
- ⇒ Prepare a Use Case Model Diagram to depict the scope and boundaries of a system in terms of actors and use cases.
- ⇒ Document a use case's course of events.
- ⇒ Construct an object association model to depict the objects of the system and how they are related to each other.

Prerequisites

Before starting this milestone the following topics should be covered:

1. Object Modeling Chapter 8, pages 284-304
 of the SADM 4th ed. Textbook.

Project

There are two general activities when performing object oriented analysis and they are as follows:

1. Finding and identifying the business objects
2. Organizing the objects and identifying their relationships

Activities

1. To construct the context model, use Figure 8.6, page 295, of the SADM 4th ed. textbook as an example. Refer to exhibit 11.1, which contains a narrative describing the scope of the Asset Tracking and Management System. Use this narrative as the basis for your context model. Make assumptions where necessary.
2. To construct the use case model diagram, use Figure 8.8, page 297, of the SADM 4th ed. textbook as an example. Follow the steps 1 and 2 found on pages

295 and 296 of the SADM 4th ed. Textbook, to identify the actor s and use cases. Make assumptions where necessary.

3. To document a use case's course of events, use Figure 8.9, page 298, of the SADM 4th ed. textbook as an example. Follow the steps 3 and 5 found on pages 296 through 299 of the SADM 4th ed. Textbook. Refer to the interview with Marge Brooks in **Milestone 2** for the necessary information to document the course of events for processing a purchasing request. Make assumptions where necessary. We have provided for your convenience a copy of the use case template on the textbook's web page [<http://www.tech.purdue.edu/textbooks/sadm/>] as well as a hardcopy at the end of this milestone.
4. To construct an Object Association Model, use Figure 8.16, page 304, of the SADM 4th ed. textbook as an example. Follow the steps 6 and 7 found on pages 299 through 303, and steps 1 through 4 found on pages 303 and 304 of the SADM 4th ed. Textbook. Use the use case course of events completed in Activity 3 above as the basis for the model. Make assumptions where necessary.

Deliverable format and software to be used are according to your instructor's specifications. Deliverables should be neatly packaged in a binder, separated with a tab divider labeled "Milestone 11" and accompanied with a Milestone Evaluation Sheet (appearing first) found in Appendix A.

References:

ESSS Scope Narrative

Exhibit 11.1

ESSS Job Offer Narrative

Exhibit 11.2

Deliverables:

Context Model:

Due: __/__/__

Time: _____

Use Case Model:

Due: __/__/__

Time: _____

Job Offer Course Of Events:

Due: __/__/__

Time: _____

Object Association Model:

Due: __/__/__

Time: _____

ADVANCED OPTION

For the advanced option, construct a use case dependency diagram consisting of the use cases you included in preparing your use case model above. Make any necessary assumptions. Use Figure 8.10, page 299 of the SADM 4th ed. textbook as an example.

Use Case Dependency Diagram:

Due: __/__/__

Time: _____

Milestone's Point Value:

Exhibit 11.1

Use the following narrative to construct the Context Model for the Asset Tracking and Management System (ATMS).

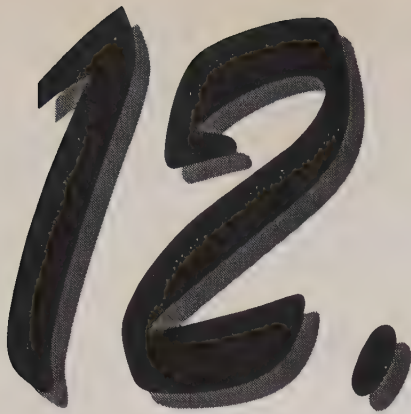
The purpose of the Asset Tracking and Management System (ATMS) is to provide a single central repository of machine acquisition, movement, performance, and disposal history. Management submits purchase requests and in response the system generates purchase orders, which are faxed to the vendor. In response the vendor ships amusement machines along with a shipping statement. The vendor also sends an invoice, which triggers a payment to be sent back to the vendor. Business locations provide a signed contract and in response machines are installed at the location. Management also provides machine move order requests, which generate move orders for the technicians to carry out. Management can request a machine history report, an inventory report, and a location machine inventory and history report.

USE CASE

Author: _____

Date: _____

USE CASE NAME:	
ACTOR:	
DESCRIPTION:	
NORMAL COURSE:	
ALTERNATE COURSE:	
PRE-CONDITION:	
POST-CONDITION:	
ASSUMPTIONS:	



MILESTONE 12 – DESIGNING THE OBJECTS

Synopsis

In performing object-oriented analysis we identified objects and use cases based on ideal conditions and independent of any hardware or software solution. During object-oriented design we want to refine those objects and use cases to reflect the actual environment of our proposed solution.

In this milestone you will first transform the use case course of events prepared in **Milestone 11** to be a design use case. Secondly, we update our object association model to include any additional attributes we have discovered. Finally, we will create an ideal object model diagram that models the object interactions for each use case.

Objectives

After completing this milestone, you should be able to:

- ⇒ Transform a requirements use case to a design use case.
- ⇒ Construct an ideal object model diagram to model the high-level object interactions for a use case.
- ⇒ Revise an object association diagram to include attributes and hierarchies.

Prerequisites

Before starting this milestone the following topics should be covered:

1. Object-Oriented Design Chapter 16, pages 532-550
of the SADM 4th ed. Textbook.

2. Milestone 11 Solution Provided by your instructor.

Project

There are three general activities when performing object-oriented design and they are as follows:

1. Refining the use case model to reflect the implementation environment.
2. Modeling object interactions and behavior that support the use case scenarios.
3. Updating the object model to reflect the implementation environment.

Activities

1. To transform the *analysis use case* to a *design use case*, follow the steps on pages 537 and 538 of the SADM 4th ed. Textbook. Use Figure 16.2, page 538-539 of the SADM 4th ed. Textbook as an example. Use the completed analysis use case from **Milestone 11** to start from. Use the Marge Brooks interview from **Milestone 2** to identify design items to include in the use case. Make assumptions where necessary.
2. To construct the *ideal model diagram*, follow the steps on page 540 of the SADM 4th ed. Textbook. Use Figure 16.5, page 541, of the SADM 4th ed. textbook as an example. Make assumptions where necessary.
3. To document additional identified attributes, use the *completed Object Association Diagram* prepared in **Milestone 11**. Make the necessary changes to this diagram to reflect the proposed environment

Deliverable format and software to be used are according to your instructor's specifications. Deliverables should be neatly packaged in a binder, separated with a tab divider labeled "Milestone 12" and accompanied with a Milestone Evaluation Sheet (appearing first) found in Appendix A.

References:

Milestone 11 Solution

Provided by your instructor.

Marge Brooks Interview

Milestone 2

Deliverables:**Design Use Case:****Due:** __/__/__**Time:** _____**Ideal Object Model:****Due:** __/__/__**Time:** _____**Refined Object Association Model:****Due:** __/__/__**Time:** _____**ADVANCED OPTION**

For the advanced option, construct a Interaction Diagram for the Process Purchase Request use case. Follow the steps to identify object behaviors and to prepare the interaction diagram. Make any necessary assumptions. Use Figure 16.10, page 546 of the SADM 4th ed. textbook as an example. Make any necessary assumptions.

Interaction Diagram:**Due:** __/__/__**Time:** _____**Milestone's Point Value:**

BUILD YOUR OWN CASE

[BYOC]

INTRODUCTION

In this alternative, you select a business and application from your personal background. While it might be nice to select this business and application from a real data processing experience, it is not a requirement. Most of you have work experience outside of data processing and, in some sense, that work experience may provide a better foundation on which to base a business and application.

After your business and application have been approved, you will build a case around them. In the early stages of this project, you will probably tend to make your scope too large, not realizing the implications until later in the project, usually too late. To prevent you from becoming overwhelmed, we will control the size of your case study so that it both demonstrates all of the characteristics of a modern information system and remains small enough that you get an opportunity to practice all of the necessary systems analysis and design skills.

The advantage of this case study approach is that the business and application are at least somewhat familiar to you. You don't waste time trying to understand an instructor's or author's business and application. At the same time, you are not burdened with the politics and behavioral components of a "live" project.



MILESTONE 1 - SURVEYING THE PROJECT

Synopsis

The purpose of the survey phase is threefold. First, the survey phase answers the question, “Is this project worth looking at?” To answer this question, the survey phase must define the scope of the project and the perceived problems, opportunities, and directives that triggered the project.

In this milestone you will prepare a *Request for System Services*, which is the trigger for the Survey Phase. Also, you will use fact finding techniques to extract and analyze information about the system you have selected to determine project scope, level of management commitment, and project feasibility. With the facts you have acquired, you will have the necessary information to complete the *Problem Statement Matrix* and construct the *Project Feasibility Assessment Report*.

Objectives

After completing this milestone, you should be able to:

- ⇒ Complete a *Request for System Services* form, which triggers the survey phase.
- ⇒ Complete a *Problem Statement Matrix* documenting the problems, opportunities, or directives of the project.
- ⇒ Prepare and understand the structure and content, of the *Project Feasibility Assessment Report*.

Prerequisites

Before starting this milestone the following topics should be covered:

1. The Survey Phase - Chapter 3, pages 84-87,
Chapter 4, pages 128-137
of the SADM 4th ed. textbook.

Project

Meet with your instructor to get their agreement on the system you have selected. Obtain the necessary system documentation or interview key individuals who are responsible for the system, in order to obtain the necessary information to complete not only the *Request for Systems Services*, but also the *Problem Statement Matrix*.

Activities

1. To complete the *Request for System Services* form, use Figure 4.7, page 130 of the SADM 4th ed. Textbook as an example. Use information from the system documentation you have obtained or interviews that you have performed to complete the form. Make assumptions where necessary. We have provided for your convenience a template of the form on the textbook's web page [<http://www.tech.purdue.edu/textbooks/sadm/>], as well as a hardcopy at the end of this milestone.
2. To complete the *Problem Statement Matrix*, use Figure 4.8, page 132 of the SADM 4th ed. textbook as an example. Based on the information you have obtained so far, which problems do you believe are the most visible, have the highest visibility, and how should they be ranked? Try to determine the annual benefits. State assumptions and be prepared to justify your answers! Finally, what would be your proposed solution based on the facts you know now? We have provided for your convenience a template of the matrix on the textbook's web page [<http://www.tech.purdue.edu/textbooks/sadm/>], as well as a hardcopy at the end of this milestone.

Deliverable format and software to be used are according to your instructor's specifications. Deliverables should be neatly packaged in a binder, separated with a tab divider labeled "Milestone 1" and accompanied with a Milestone Evaluation Sheet (appearing first) found in Appendix A.

References:

Case Background

Workbook Introduction

Deliverables:

Request for System Services:

Due: __/__/__

Time: _____

Problem Statement Matrix:

Due: __/__/__

Time: _____

ADVANCED OPTION

For the advanced option, prepare a Project Feasibility Assessment Report.. Use Figure 4.9, page 137 of the SADM 4th ed. textbook as a sample outline and a copy of the outline is also presented in exhibit 1.2. Be sure to include Gantt charts for the project schedules. Information on Gantt charts can be found on pages 602-609 of the SADM 4th ed. Textbook. Information on fact-finding techniques can be found on pages 623-639 of the SADM 4th ed. Textbook. And, information on feasibility analysis can be found on 643-659 of the SADM 4th ed. Textbook.

Project Feasibility Assessment Report:

Due: __/__/__

Time: _____

Milestone's Point Value:

CS 3 - MILESTONE 1: SURVEYING THE PROJECT

REQUEST FOR INFORMATION SYSTEM SERVICES

Phone: Fax:
Internet: <http://www.>
Intranet: <http://www.>

DATE OF REQUEST	SERVICE REQUESTED FOR DEPARTMENT(S)
	Human Resources

SUBMITTED BY (key user contact)	EXECUTIVE SPONSOR (funding authority)
Name Title Office Phone	Name Title Office Phone

TYPE OF SERVICE REQUESTED:

- | | |
|---|---|
| ☐ Information Strategy Planning | ☐ Existing Application Enhancement |
| ☐ Business Process Analysis and Redesign | ☐ Existing Application Maintenance (problem fix) |
| ☐ New Application Development | ☐ Not Sure |
| ☐ Other (please specify _____) | |

BRIEF STATEMENT OF PROBLEM, OPPORTUNITY, OR DIRECTIVE (attach additional documentation as necessary)

BRIEF STATEMENT OF EXPECTED SOLUTION

ACTION (ISS Office Use Only)

☐ Feasibility assessment approved

Assigned to _____

☐ Feasibility assessment waived

Approved Budget \$ _____

☐ Request delayed

Start Date _____ Deadline _____

Backlogged until date: _____

☐ Request rejected

Reason: _____

Authorized Signatures:

Chair, ISS Executive Steering Body

Project Executive Sponsor

PROBLEM STATEMENTS

PROJECT:	PROJECT MANAGER:
CREATED BY:	LAST UPDATED BY:
DATE CREATED:	DATE LAST UPDATED:

Brief Statements of Problem, Opportunity, or Directive	Urgency	Visibility	Annual Benefits	Priority or Rank	Proposed Solution
1.					
2.					
3.					
4.					
5.					
6.					
7.					
8.					

Exhibit 1.2

Project Feasibility Assessment Report

- I. Executive summary (1 page)
 - A. Summary of recommendation
 - B. Brief statement of anticipated benefits
 - C. Brief explanation of report contents
- II. Background information (1-2 pages)
 - A. Brief description of project request
 - B. Brief explanation of the survey phase activities
- III. Findings (2-3 pages)
 - A. Problems and analysis (optional: reference *problem statement matrix*)
 - B. Opportunities and analysis (optional: reference *problem statement matrix*)
 - C. Directives and implications
- IV. Detailed recommendation
 - A. Narrative recommendation (1 page)
 - 1. Immediate fixes
 - 2. Quick fixes
 - 3. Enhancements
 - 4. New systems development
 - B. Project Plan
 - 1. Initial project objectives
 - 2. Initial master project plan (phase level - Gantt chart)
 - 3. Detailed plan for the study or definition phase (Gantt chart)
- V. Appendices
 - A. Request for System Services
 - B. Problem Statements Matrix
 - C. (other documents as appropriate)



MILESTONE 2 - STUDYING THE CURRENT SYSTEM

Synopsis

There's an old saying that suggests, "Don't try to fix it unless you understand it." With those words of wisdom, the next milestone of our project is to study and analyze the existing system. There is always an existing business system, regardless of whether it currently uses a computer. The study phase provides the project team with a more thorough understanding of the problems, opportunities, and/or directives that triggered the project. Indeed, the analyst frequently uncovers new problems and opportunities. The study phase may answer the questions, "Are the problems worth solving?" and "Is a new system worth building?"

The purpose of the study phase is threefold. First and foremost, the project team must gain an appropriate understanding of the business problem domain. Second, we need to answer the question, "Are these problems (opportunities, and directives) worth solving"? Finally, we need to determine if the system is worth developing. The study phase provides the systems analyst and project team with a more thorough understanding of the problems, opportunities, and/or directives that triggered the project. In the process, they frequently uncover new problems and opportunities.

In this milestone you will perform *Cause-Effect Analysis* on the system you have selected and document your findings using the *Problems, Opportunities, Objectives, and Constraints Matrix*. The PIECES framework, originally developed by James Wetherbe, and then adapted by the authors, can serve as a useful tool to

classify the various problems, opportunities, and directives identified in **Milestone 1**.

Objectives

After completing this milestone, you should be able to:

- ⇒ Perform *Cause-Effect Analysis* to be able to thoroughly understand a system's problems, opportunities, and/or directives that triggered the project.
- ⇒ Use and understand the PIECES framework for classifying problems, opportunities, and directives.
- ⇒ Complete the *Problems, Opportunities, Objectives, and Constraints Matrix*.

Prerequisites

Before starting this milestone the following topics should be covered:

1. The Study Phase - Chapter 3, pages 87-88,
Chapter 4, pages 137-150
of the SADM 4th ed. textbook.
2. PIECES Framework - Chapter 3, pages 79-81 of the SADM 4th ed.
Textbook.
3. Milestone 1 Solution

Project

Now that we have completed the survey of the system, and gained approval to proceed, we can now attempt to gain a better understanding of the current system. In this assignment we will need to perform more fact-finding (research, observation, user interviews) to gain a detailed understanding of the system. The results of this activity will provide us a better understanding of the problems, opportunities, and constraints of the current system.

Activities

1. To complete the *Problems, Opportunities, Objectives, and Constraints Matrix*, use Figure 4.12, page 144 of the SADM 4th ed. textbook as an example. Make assumptions where necessary. Use the PIECES framework as a model to classify the problems, opportunities, and directives. Figure 3.4, pages 80-81 of the SADM 4th ed. textbook, may identify other areas that present problems or offer opportunities. We suggest you use it. It has proven to be a very valuable tool! We have provided for your convenience a copy of the matrix template on

the textbook's web page [<http://www.tech.purdue.edu/textbooks/sadm/>], as well as a hardcopy at the end of this milestone.

Deliverable format and software to be used are according to your instructor's specifications. Deliverables should be neatly packaged in a binder, separated with a tab divider labeled "Milestone 2" and accompanied with a Milestone Evaluation Sheet (appearing first) found in Appendix A.

References:

Milestone 1 Solution

Deliverables:

Problems, Opportunities, Objectives, and Constraints Matrix:

Due: ____/____/____

Time: _____

ADVANCED OPTION

This deliverable was not discussed in the narrative because students need to be exposed to modeling (data, process, interface, & geography), before this report can be completed. For those ambitious individuals who are familiar with those skills and wish to be challenged, use the detailed study report outline found in Figure 4.13, page 150 of the SADM 4th ed. textbook, as a guideline. We have also provided a copy of the outline in Exhibit 2.1

Study Report:

Due: ____/____/____

Time: _____

Milestone's Point Value:

PROBLEMS, OPPORTUNITIES, OBJECTIVES AND CONSTRAINTS MATRIX

Project:	Project Manager:
Created by:	Last Updated by:
Date Created:	Date Last Updated:

CAUSE AND EFFECT ANALYSIS		SYSTEM IMPROVEMENT OBJECTIVES	
Problem or Opportunity	Causes and Effects	System Objective	System Constraint
1.	1.	1.	1.

Exhibit 2.1

Analysis of the Current _____ System

- I. Executive summary (approximately 2 pages)
 - A. Summary of recommendation
 - B. Summary of problems, opportunities, and directives
 - C. Brief statement of system improvement objectives
 - D. Brief explanation of report contents
- II. Background information (approximately 2 pages)
 - A. List of interviews and facilitated group meetings conducted
 - B. List of other sources of information that were exploited
 - C. Description of analytical techniques used
- III. Overview of the current system (approximately 5 pages)
 - A. Strategic implications (if the project is part of, or impacts an existing information systems strategic plan)
 - B. Models of the current system
 1. Interface model (showing project scope)
 2. Data model (showing project scope)
 3. Geographic models (showing project scope)
 4. Process model (showing functional decomposition only)
- IV. Analysis of the Current System (approximately 5-10 pages)
 - A. Performance problems, opportunities, and cause/effect analysis
 - B. Information problems, opportunities, and cause/effect analysis
 - C. Economic problems, opportunities, and cause/effect analysis
 - D. Control problems, opportunities, and cause/effect analysis
 - E. Efficiency problems, opportunities, and cause/effect analysis
 - F. Service problems, opportunities, and cause/effect analysis
- V. Detailed recommendations (approximately 5-10 pages)
 - A. System improvement objectives and priorities
 - B. Constraints
 - C. Project Plan
 1. Scope reassessment and refinement
 2. Revised master plan
 3. Detailed plan for the definition phase
- VI. Appendices
 - A. Any detailed system models
 - B. (other documents as appropriate)

3.

MILESTONE 3 - MODELING THE SYSTEM'S DATA

Synopsis

The definition phase answers the question, ‘What does the user need and want from a new system?’ The definition phase is critical to the success of any new information system! In this milestone we need to identify what information systems requirements need to be defined from the system users’ perspectives and draw graphical, logical, models to document the data requirements for a new and improved system.

Data modeling is a technique for organizing and documenting a system’s Data. Data modeling is sometimes called database modeling because a data model is usually implemented as a database. Data is viewed as a resource to be shared by as many processes as possible. As a result, data must be organized in a way that is flexible and adaptable to unanticipated business requirements – and that is the purpose of data modeling.

In this milestone you will first discover those entities in the system that are or might be described by data. With each entity we identify, we will define it in respect to the business. Then, we will construct a *Context Data Model* that graphically depicts each of the entities and the relationships they have with each other. Next, we will refine the context data model to include primary and foreign keys. The resulting model is called a *Key-Based Data Model*. Finally, we refine the key-based data model to include any hierarchies and attributes, and this model is referred to as the *Fully Attributed Data Model*.

Objectives

After completing this milestone, you should be able to:

- ⇒ Understand and perform the techniques for entity discovery.
- ⇒ Define each entity with respect to the business and complete an entity/definition matrix.
- ⇒ Perform the necessary data modeling techniques to organize and document the data requirements for the proposed system.
- ⇒ Construct the Context, Key-Based, and Fully Attributed data models.

Prerequisites

Before starting this milestone the following topics should be covered:

1. Data Modeling - Chapter 5, pages 170-202
of the SADM 4th ed. textbook.
2. Milestone 2 Solution

Project

Now that we have studied the current system and analyzed some of its problems and opportunities, plus gained approval to proceed, we can now start to identify the business data requirements and graphically model them. In this assignment try to obtain sample forms that are used with the system or even a data dictionary if one exists. Any summary, detailed, or exception reports would be good too in order to identify data requirements.

Activities

1. To complete the *Entity/Definition Matrix*, use Table 5.5, page 192 of the SADM 4th ed. textbook as an example. Analyze each of the forms, reports, or the dictionary you have obtained. Make assumptions where necessary. We have provided for your convenience a template of the matrix on the textbook's web page [<http://www.tech.purdue.edu/textbooks/sadm/>], as well as a hardcopy in the back of this milestone.
2. To complete the *Context Data Model* use Figure 5.11, page 194 of the SADM 4th ed. textbook as an example. Construct the model to include each of the entities in **Activity 1** and depict the relationships that exist among the entities. Data model notation is according to your instructor's specifications, however

the authors' prefer the notation depicted in Figure 5.3, page 180 of the SADM 4th ed. textbook.

3. To complete the *Key-Based Data Model* use Figure 5.12, page 194 of the SADM 4th ed. textbook as an example. Refine the Context Data Model from **Activity 2** to include associate entities that resolve many-to-many relationships, and include primary and foreign keys. Again data model notation is according to your instructor's specifications, however the authors' prefer the notation depicted in Figure 5.3, page 180 of the SADM 4th ed. textbook.
4. To complete the *Fully Attributed Data Model* use Figure 5.14, pages 200 & 201 of the SADM 4th ed. textbook as an example. Refine the Key-Based Data Model from **Activity 3** to include any generalization hierarchies, and add the data attributes for each entity.

Deliverable format and software to be used are according to your instructor's specifications. Deliverables should be neatly packaged in a binder, separated with a tab divider labeled "Milestone 3" and accompanied with a Milestone Evaluation Sheet (appearing first) found in Appendix A.

References:

Milestone 2 Solution

Deliverables:

Entity Definition Matrix:

Due: ____/____/____
Time: _____

Context Data Model:

Due: ____/____/____
Time: _____

Key-Based Data Model:

Due: ____/____/____
Time: _____

Fully Attributed Data Model:

Due: ____/____/____
Time: _____

ADVANCED OPTION

For the advanced option, include other entities and attributes that you believe should be part of the system. Construct a fully attributed data model with the new items you have identified.

Entity Definition Matrix:

Due: __/__/__

Time: _____

Fully Attributed Data Model:

Due: __/__/__

Time: _____

Milestone's Point Value:

Entity/Definition Matrix

ENTITY	BUSINESS DEFINITION



MILESTONE 4 - MODELING THE SYSTEM'S PROCESSES

Synopsis

Process modeling is a technique for organizing and documenting the structure and flow of data through a system's processes and/or the logic, policies, and procedures to be implemented by a system's processes. In this milestone we focus on using and constructing data flow diagrams (DFDs) and decomposition diagrams to perform process modeling.

Data flow diagrams are tools that depict the flow of data through a system and the work or processing performed by that system. A decomposition diagram is a DFD planning tool that shows the top-down functional decomposition and structure of a system.

During this milestone you will first construct a *context diagram* to establish project scope and boundaries. Secondly, you will draw a *functional decomposition diagram* to partition the system into logical subsystems and/or functions. Thirdly, you will complete a partially completed *event list* that identifies and confirms the business events to which the system must respond. Once that task is completed, you will refine the decomposition diagram drawn earlier to include event-handling processes. Finally you will construct a *system data flow diagram* that shows the big picture of the system, and a *primitive data flow diagram* for a single event process.

Objectives

After completing this milestone, you should be able to:

- ⇒ Construct a context diagram to illustrate a system's interfaces with its environment.
- ⇒ Perform event partitioning and organize events in a functional decomposition diagram.
- ⇒ Identify external and temporal business events for a system.
- ⇒ Merge event diagrams into a system data flow diagram.
- ⇒ Draw primitive data flow diagrams.

Prerequisites

Before starting this milestone the following topics should be covered:

1. Process Modeling - Chapter 6, pages 205-254
of the SADM 4th ed. textbook.
2. Milestone 3 Solution

Project

As a systems analyst or knowledgeable end-user, you must learn how to draw decomposition and data flow diagrams to model business process requirements. The survey and study phases of the methodology have been completed and you understand the current system's strengths, weaknesses, limitations, problems, opportunities, and constraints. You have already built the data model (**Milestone 3**) to document business data requirements for the new system. You now need to build the corresponding process models.

Activities

1. To complete the *Context Diagram*, use Figure 6.20, page 242 of the SADM 4th ed. textbook as an example. Remember that the context diagram establishes the project scope and boundaries by depicting the system's interfaces with external parties.
2. To complete the *Functional Decomposition Diagram*, use Figure 6.21, page 243 of the SADM 4th ed. textbook as an example. Try to break the system you have selected into subsystems of related events (transactions)
3. To complete the *Event – Response List*, use Figure 6.22, page 245 of the SADM 4th ed. textbook as an example. Try to document ten business events of

your target system. What were the triggers that initiated the event are what were the generated responses? We have provided for your convenience a template of the Event-Response List on the textbook's web page [<http://www.tech.purdue.edu/textbooks/sadm/>].

4. To complete the *Event Decomposition Diagram*, use Figure 6.23, page 246 of the SADM 4th ed. textbook as an example. Refine the decomposition diagram constructed in **Activity 2** above, to include the event processes identified in **Activity 3** above. If your target system is quite large, focus on only one subsystem.
5. To complete the *System Diagram*, use Figure 6.27, pages 250 & 251 of the SADM 4th ed. textbook as an example. For the ten events identified above, first prepare an event DFD for each one. Once completed, merge the DFD's to produce the System Diagram.
6. To complete the *Primitive Diagram*, use Figure 6.28, page 252 of the SADM 4th ed. textbook as an example. Take one of the events identified above and model it in detail.

Deliverable format and software to be used are according to your instructor's specifications. Deliverables should be neatly packaged in a binder, separated with a tab divider labeled "Milestone 4" and accompanied with a Milestone Evaluation Sheet (appearing first) found in Appendix A.

References:

Milestone 3 Solution

Deliverables:

Context Diagram:

Due: __/__/__
Time: _____

Functional Decomposition Diagram:

Due: __/__/__
Time: _____

Event-Response List:

Due: __/__/__
Time: _____

Event Decomposition Diagram:

Due: __/__/__
Time: _____

System Diagram:

Due: __/__/__
Time: _____

Primitive Diagram:

Due: __/__/__
Time: _____

ADVANCED OPTION

For the advanced option, try to complete activities 3 through 6 for one of the other subsystems. In this option you will need to create the Event Diagrams before you can do the System Diagram. Make any assumptions as necessary.

Subsystem diagrams:

Due: __/__/__
Time: _____

Milestone's Point Value:

5.

MILESTONE 5 - MODELING THE NETWORK

Synopsis

As a systems analyst or knowledgeable end-user, you should learn how to draw network models to be able to deal with today's distributed business and computing environment. Network modeling is a diagrammatic technique used to document the shape of a business or information system in terms of its business locations.

In this milestone you will first construct a *Location Decomposition Diagram* to logically decompose and group locations. Next we will draw a *Location Connectivity Diagram* that depicts the shape of the system in terms of its user, process, data, and interface locations and the necessary interconnections between the locations.

Objectives

After completing this milestone, you should be able to:

- ⇒ Factor a system's or application's locations into component locations using a special location decomposition diagram.
- ⇒ Document the connections and essential data flow between locations using a special location decomposition diagram.

Prerequisites

Before starting this milestone the following topics should be covered:

1. Network Modeling - Chapter 7, pages 262-280
of the SADM 4th ed. textbook.

Project

Now that we have completed the survey and study phases of the system development lifecycle, and we fully understand the system's strengths, weaknesses, limitations, problems, opportunities, and constraints, it is time to model the logical business network requirements.

If your target system is not distributed across multiple locations, you can emulate that environment to complete this milestone by thinking of each user of the system as a separate location.

Activities

1. To complete the *Location Decomposition Diagram*, use Figure 7.8, page 277 of the SADM 4th ed. textbook as an example.
2. To complete the *Location Connectivity Diagram*, use Figure 7.10, page 279 of the SADM 4th ed. textbook as an example.

Deliverable format and software to be used are according to your instructor's specifications. Deliverables should be neatly packaged in a binder, separated with a tab divider labeled "Milestone 5" and accompanied with a Milestone Evaluation Sheet (appearing first) found in Appendix A.

References:

None

Deliverables:

Location Decomposition Diagram:

Due: __/__/__

Time: _____

Location Connectivity Diagram:

Due: __/__/__

Time: _____

ADVANCED OPTION

For the advanced option, assume your system will be installed at additional sites. Prepare a location decomposition diagram and a location connectivity diagram that includes these new locations.

Location Decomposition Diagram:**Due:** __/__/__**Time:** _____**Location Connectivity Diagram:****Due:** __/__/__**Time:** _____**Milestone's Point Value:**



MILESTONE 6 - DEFINING AND ANALYZING CANDIDATE SOLUTIONS

Synopsis

The purpose of the Configuration phase is to identify candidate solutions, analyze those candidate solutions, and recommend a target system that will be designed and implemented. Alternative solutions to be considered should be those that address the business requirements of the information system.

In this milestone you will complete a *Candidate Matrix*. This matrix will include three alternative candidate solutions that you have determined will meet the business requirements for your target system. Some of the solutions you will consider may have been posed from the design ideas and opinions from the system owners and users. Others may come from various sources including: systems analysts, system designers, technical consultants, and other IS professionals. Some technical choices may be limited by a predefined, approved technology architecture provided by system managers. When completing the matrix, it is not your intent to evaluate the candidates at this point, only to identify and define them.

Objectives

After completing this milestone, you should be able to:

- ⇒ Identify candidate solutions that meet the requirements of the information system and document those solutions in a candidate matrix.

Prerequisites

Before starting this milestone the following topics should be covered:

1. The Configuration Phase - Chapter 9, pages 319-335
of the SADM 4th ed. textbook.
2. Feasibility and Cost-Benefit Analysis Module C, pages 642-659
of the SADM 4th ed. Textbook.

Project

Given the business requirements established in the definition phase of systems analysis, you can now identify alternative candidate solutions.

Activities

1. To complete the *Candidate Matrix*, use Figure 9.5, page 322, and Figure C.7, page 658 of the SADM 4th ed. textbook as an example. We have provided for your convenience a copy of the Candidate Matrix template on the textbook's web page [<http://www.tech.purdue.edu/textbooks/sadm/>], as well as a hardcopy at the end of this milestone.

Deliverable format and software to be used are according to your instructor's specifications. Deliverables should be neatly packaged in a binder, separated with a tab divider labeled "Milestone 6" and accompanied with a Milestone Evaluation Sheet (appearing first) found in Appendix A.

References:

None

Deliverables:

Candidate Matrix:

Due: __/__/__

Time: _____

ADVANCED OPTION

For the advanced option, perform feasibility analysis and complete a feasibility matrix for the three candidates identified above. This option will require you to do research on the cost of hardware and software. Make the necessary assumptions to estimate development time, costs, and benefits. Use Figure 9.6, page 324, and Figure C.9, page 659 of the SADM 4th ed. textbook as an example. We have provided for your convenience a copy of the Feasibility Matrix template on the textbook's web page [<http://www.tech.purdue.edu/textbooks/sadm/>], as well as a hardcopy at the end of this milestone.

Feasibility Matrix:**Due:** ____/____/____**Time:** _____**Milestone's Point Value:** _____

Candidate Matrix Template

Characteristics	Candidate 1	Candidate 2	Candidate 3
Portion of System Computerized Brief description of that portion of the system that would be computerized in this candidate.			
Benefits Brief description of the business benefits that would be realized for this candidate.			
Servers and Workstations A description of the servers and workstations needed to support this candidate.			
Software Tools Needed Software tools needed to design and build the candidate (e. g., database management system, emulators, operating systems, languages, etc.). Not generally applicable if applications software packages are to be purchased.			
Application Software A description of the software to be purchased, built, accessed, or some combination of these techniques.			
Method of Data Processing Generally some combination of: on-line, batch, deferred batch, remote batch, and real-time.			
Output Devices and Implications A description of output devices that would be used, special output requirements, (e.g. network, preprinted forms, etc.), and output considerations (e.g., timing constraints).			
Input Devices and Implications A description of Input methods to be used, input devices (e.g., keyboard, mouse, etc.), special input requirements, (e.g. new or revised forms from which data would be input), and input considerations (e.g., timing of actual inputs).			
Storage Devices and Implications Brief description of what data would be stored, what data would be accessed from existing stores, what storage media would be used, how much storage capacity would be needed, and how data would be organized.			

Feasibility Matrix Template

Feasibility Criteria	Wt.	Candidate 1	Candidate 2	Candidate 3
Operational Feasibility Functionality. A description of to what degree the candidate would benefit the organization and how well the system would work. Political. A description of how well received this solution would be from both user management, user, and organization perspective.	30%	Score:	Score:	Score:
Technical Feasibility Technology. An assessment of the maturity, availability (or ability to acquire), and desirability of the computer technology needed to support this candidate. Expertise. An assessment to the technical expertise needed to develop, operate, and maintain the candidate system.	30%	Score:	Score:	Score:
Economic Feasibility Cost to develop: Payback period (discounted): Net present value: Detailed calculations:	30%	Score:	Score:	Score:
Schedule Feasibility An assessment of how long the solution will take to design and implement.	10%	Score:	Score:	Score:
Ranking:	100%			



MILESTONE 7 – MODELING THE APPLICATION ARCHITECTURE

Synopsis

Just as we modeled business requirements during systems analysis, we should model technology architecture and requirements during systems design. The models serve as blueprints for system design, prototyping and construction.

In this milestone you will prepare a *Physical Data Flow Diagram*. Physical data flow diagrams model the technical and human design decisions to be implemented as part of an information system. They communicate technical and other design constraints to those who will actually implement the system—in other words, they serve as a technical blueprint for the implementation.

Objectives

After completing this milestone, you should be able to:

⇒ Draw physical data flow diagrams for an information system's architecture and processes.

Prerequisites

Before starting this milestone the following topics should be covered:

1. Application Architecture & Process Design

Chapter 10, pages 350-387
of the SADM 4th ed. Textbook.

Project

The goal of this project is to produce a high-level general design that can serve as an application architecture for the system and a general design for the processes that make up the system.

Activities

1. To construct the Physical Data Flow Diagram, use Figure 10.8, page 373, of the SADM 4th ed. textbook as an example. Research one of the processes within your target system. Follow and document the process from start to finish. Use the physical DFD to model the flow of data and how it is handled and by whom.

Deliverable format and software to be used are according to your instructor's specifications. Deliverables should be neatly packaged in a binder, separated with a tab divider labeled "Milestone 7" and accompanied with a Milestone Evaluation Sheet (appearing first) found in Appendix A.

References:

Job Offer Business Process

None

Deliverables:

Physical Data Flow Diagram:

Due: __/__/__

Time: _____

ADVANCED OPTION

For the advanced option, construct a Network Topology Data Flow Diagram based on the locations identified in Milestone 5. Make the necessary assumptions for types of networks and line speeds if that data is not available. Use Figure 10.12, page 382 of the SADM 4th ed. textbook as an example.

Network Topology DFD:

Due: __/__/__

Time: _____

Milestone's Point Value: _____



MILESTONE 8 – DESIGNING THE DATABASE

Synopsis

All information systems create, read, update and delete data. This data is stored in files and databases. To fully exploit the advantages of database technology, a database must be carefully designed. Database design translates the data models that were developed for the system users during the definition phase into data structures supported by the chosen database technology. Subsequent to database design, system builders will construct those data structures using the language and tools of the chosen database technology.

In this milestone you will normalize the data model created in **Milestone 3**, along with any additional data requirements, to be in third normal form. In the advanced option you will transform the logical data model into a physical model or database schema.

Objectives

After completing this milestone, you should be able to:

- ⇒ Normalize a logical data model to remove impurities that can make a database unstable, inflexible, and nonscalable.
- ⇒ Transform a logical data model into a physical, relational database schema.

Prerequisites

Before starting this milestone the following topics should be covered:

- | | |
|--------------------|--|
| 1. Database Design | Chapter 11, pages 392-428
of the SADM 4 th ed. Textbook. |
|--------------------|--|

Project

The goal of this project is to normalize our logical data model to remove impurities that can make a database unstable, inflexible, and nonscalable.

Activities

1. To construct an Entity Relationship Diagram to be in 3rd Normal form, follow the normalization procedure outlined on pages 409 to 418 of the SADM 4th ed. textbook. Use Figure 11.15, page 419, of the SADM 4th ed. textbook as an example. Use the ERD solution of **Milestone 3** as well as any additional data requirements you have discovered since completing Milestone 3.

Deliverable format and software to be used are according to your instructor's specifications. Deliverables should be neatly packaged in a binder, separated with a tab divider labeled "Milestone 8" and accompanied with a Milestone Evaluation Sheet (appearing first) found in Appendix A.

References:

Milestone 3 Solution

Deliverables:

Logical Data Model in 3rd Normal Form:

Due: __/__/__

Time: _____

ADVANCED OPTION

For the advanced option, transform your normalized data model created above into a physical database schema complete with referential integrity rules. Make any necessary assumptions. Use Figure 11.17, page 425 of the SADM 4th ed. textbook as an example.

Physical Database Schema:

Due: __/__/__

Time: _____

Milestone's Point Value: _____



MILESTONE 9 – DESIGNING INPUTS AND OUTPUTS

Synopsis

Management and users make important decisions based on system outputs. These outputs are produced from data that is either input or retrieved from databases. And data in databases must have been input first. Input design serves an important goal. Capture and get the data into a format suitable for the computer. Outputs present information to system users. Outputs, the most visible component of a working information system, are the justification for the system.

In this milestone you will design both an input screen and an output report for your target system.

Objectives

After completing this milestone, you should be able to:

- ⇒ Design a GUI input screen that uses the proper screen-based controls.
- ⇒ Design a computer output.

Prerequisites

Before starting this milestone the following topics should be covered:

- | | |
|--------------------------------|--|
| 1. Input Design & Prototyping | Chapter 12, pages 436-457
of the SADM 4 th ed. Textbook. |
| 2. Output Design & Prototyping | Chapter 13, pages 460-477
of the SADM 4 th ed. Textbook. |

** Project**

The goal of this project is to design both a GUI input screen and an output report for your target system.

** Activities**

1. To properly design a GUI input screen, follow the steps outlined on pages 450 to 457 of the SADM 4th ed. textbook. Use Figure 12.6, page 453, of the SADM 4th ed. textbook as an example.
2. To properly design an output report, follow the steps outlined on pages 471 to 477 of the SADM 4th ed. textbook. Use Figure 13.8, page 476, of the SADM 4th ed. textbook as an example.

Deliverable format and software to be used are according to your instructor's specifications. Deliverables should be neatly packaged in a binder, separated with a tab divider labeled "Milestone 9" and accompanied with a Milestone Evaluation Sheet (appearing first) found in Appendix A.

References:

None

Deliverables:

GUI Input Screen Design:

Due: __/__/__

Time: _____

Output Report Design:

Due: __/__/__

Time: _____

ADVANCED OPTION

For the advanced option, design the GUI input screens or reports for any additional functions.

GUI Input Screen Design:

Due: __/__/__

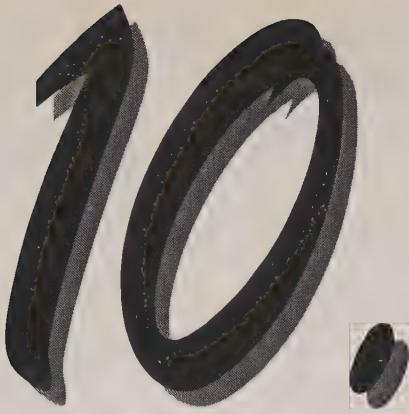
Time: _____

Report Design:

Due: __/__/__

Time: _____

Milestone's Point Value:



MILESTONE 10 – DESIGNING THE SOFTWARE

Synopsis

You have designed the databases, inputs, outputs, and on-line user interfaces. You've selected appropriate computer equipment and packaged software (which has hopefully been delivered and installed during systems design). The final step involves software design. In software design we are concerned with how the programming specifications are presented to the computer programmer for implementation. To this end, we view software design as consisting of two components, modular design and packaging.

In this milestone you will use a technique called structure design that utilizes data flow diagrams and structure charts. The concept of structured design is simple. Design a program as a top-down hierarchy of modules. The top-down structure of these modules is developed according to various design rules and guidelines. The resulting hierarchy of modules can then be evaluated according to certain quality acceptance criteria to ensure the best modular design for the program. Upon completion, the modules are to be implemented using structured programming principles.

Structured design requires that data flow diagrams (DFDs) first be drawn for the program. You have already constructed some data flow diagrams in **Milestone 4**. However, because the audience was business users and management, not programmers, it is likely that those DFDs and their processes are not sufficiently detailed enough to be used to generate a structure chart. Thus, in this milestone you will refine the DFD created in **Milestone 4** to show more detail. In addition you will perform transform analysis to identify which processes on the DFD

perform input and editing, those that do processing or data transformation, and those that do output. In the advanced option you will create a structure chart based on the results of the transform analysis.

Objectives

After completing this milestone, you should be able to:

- ⇒ Revise a data flow diagram to reflect necessary program detail prior to program design.
- ⇒ Perform transform analysis to derive a program structure chart.
- ⇒ Construct a program structure chart.

Prerequisites

Before starting this milestone the following topics should be covered:

1. Software Design Chapter 15, pages 508-528
of the SADM 4th ed. Textbook.
2. Milestone 4 Solution

Project

The goal of this project is to perform structure design that utilizes data flow diagrams and structure charts.

Activities

1. To create a data flow diagram with enough detail to produce a structure chart, reference the procedure outlined on pages 513 to 515 of the SADM 4th ed. textbook. Use Figure 15.4, page 515, of the SADM 4th ed. textbook as an example. Refine the Primitive DFD prepared in **Milestone 4** to include the necessary level of detail.
2. To perform transform analysis on the DFD of **Activity 1**, follow the steps outlined on pages 515 to 517 of the SADM 4th ed. textbook. Use Figure 15.5, page 516, of the SADM 4th ed. textbook as an example. Make a copy of the DFD prepared in **Activity 1**. Divide the DFD into the sections of Afferent, Central Transform, and Efferent. Label the sections appropriately as in Figure 15.5.

Deliverable format and software to be used are according to your instructor's specifications. Deliverables should be neatly packaged in a binder, separated with

a tab divider labeled “Milestone 10” and accompanied with a Milestone Evaluation Sheet (appearing first) found in Appendix A.

References:

Milestone 4 Solution

Deliverables:

Refined Primitive DFD:

Due: __/__/__

Time: _____

Transform Analysis DFD:

Due: __/__/__

Time: _____

ADVANCED OPTION

For the advanced option, use the results of the transform analysis to construct a structure chart. Follow the procedure outlined on pages 518 to 520 of SADM 4th ed. Textbook .Use figures 15.7, 15.8, and 15.9, pages 518 to 520 of SADM 4th ed. Textbook as examples.

Structure Chart:

Due: __/__/__

Time: _____

Milestone’s Point Value:



MILESTONE 11 – MODELING THE OBJECTS

Synopsis

Object-oriented analysis techniques are best suited to projects that will implement systems using emerging object technologies to construct, manage, and assemble those objects into useful computer applications. The object-oriented approach is centered around a technique referred to as object modeling.

The object-oriented approach to system development is based on the concept of objects that exist within a system's environment. Objects are everywhere. Let's consider your environment, look around. What are some of the objects present within your environment? Perhaps you see a door, a window, or the room itself. What about this book – it's an object, and so is the very page you are reading. And of course, don't forget about you, you are an object. Perhaps you also have a student workbook with you and there are other individuals in the room. You may also see phone, a chair, and perhaps a table. All of these objects represent objects that may be clearly visible within your immediate environment.

In performing Object Oriented Analysis (OOA) like any other systems analysis method, the purpose is to gain a better understanding of the system and its functional requirements. In other words, OOA requires that we identify the objects, their data attributes, associated behavior, and relationships which support the required business system functionality. We perform object modeling to document the identified objects, the data and behavior they encapsulate, plus their relationships with other objects.

In this milestone you will first construct a context model used to communicate the scope and boundaries of the project. Secondly, you will perform use case modeling to document the business requirements and assist in identifying the necessary objects required by the system. Finally, you will construct an Object Association Model to graphically depict the object classes that make up the system as well as the objects' relationships with each other.

Objectives

After completing this milestone, you should be able to:

- ⇒ Construct a context model in order to communicate project scope and boundaries.
- ⇒ Prepare a Use Case Model Diagram to depict the scope and boundaries of a system in terms of actors and use cases.
- ⇒ Document a use case's course of events.
- ⇒ Construct an object association model to depict the objects of the system and how they are related to each other.

Prerequisites

Before starting this milestone the following topics should be covered:

1. Object Modeling Chapter 8, pages 284-304
of the SADM 4th ed. Textbook.

Project

There are two general activities when performing object oriented analysis and they are as follows:

1. Finding and identifying the business objects
2. Organizing the objects and identifying their relationships

Activities

1. To construct the context model, use Figure 8.6, page 295, of the SADM 4th ed. textbook as an example. This model should be based on your target system.
2. To construct the use case model diagram, use Figure 8.8, page 297, of the SADM 4th ed. textbook as an example. Follow the steps 1 and 2 found on pages 295 and 296 of the SADM 4th ed. Textbook, to identify the actor s and use cases. Make assumptions where necessary.

3. To document a use case's course of events, use Figure 8.9, page 298, of the SADM 4th ed. textbook as an example. Follow the steps 3 and 5 found on pages 296 through 299 of the SADM 4th ed. Textbook. Select one use case of your target system and document it's course of events. We have provided for your convenience a copy of the use case template on the textbook's web page [<http://www.tech.purdue.edu/textbooks/sadm/>] as well as a hardcopy at the end of this milestone.
4. To construct an Object Association Model, use Figure 8.16, page 304, of the SADM 4th ed. textbook as an example. Follow the steps 6 and 7 found on pages 299 through 303, and steps 1 through 4 found on pages 303 and 304 of the SADM 4th ed. Textbook. Use the use case course of events completed in Activity 3 above as the basis for the model. Make assumptions where necessary.

Deliverable format and software to be used are according to your instructor's specifications. Deliverables should be neatly packaged in a binder, separated with a tab divider labeled "Milestone 11" and accompanied with a Milestone Evaluation Sheet (appearing first) found in Appendix A.

References:

None

Deliverables:

Context Model:

Due: __/__/__

Time: _____

Use Case Model:

Due: __/__/__

Time: _____

Job Offer Course Of Events:

Due: __/__/__

Time: _____

Object Association Model:

Due: __/__/__

Time: _____

ADVANCED OPTION

For the advanced option, construct a use case dependency diagram consisting of the use cases you included in preparing your use case model above. Make any necessary assumptions. Use Figure 8.10, page 299 of the SADM 4th ed. textbook as an example.

Use Case Dependency Diagram:

Due: __/__/__

Time: _____

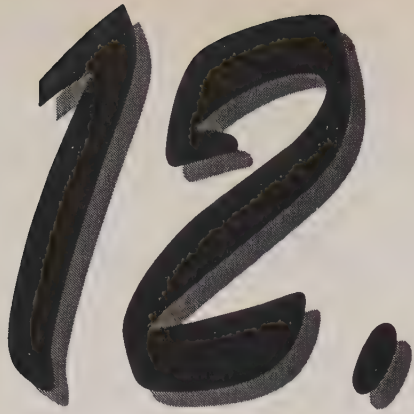
Milestone's Point Value:

USE CASE

Author: _____

Date: _____

USE CASE NAME:	
ACTOR:	
DESCRIPTION:	
NORMAL COURSE:	
ALTERNATE COURSE:	
PRE-CONDITION:	
POST-CONDITION:	
ASSUMPTIONS:	



MILESTONE 12 – DESIGNING THE OBJECTS

Synopsis

In performing object-oriented analysis we identified objects and use cases based on ideal conditions and independent of any hardware or software solution. During object-oriented design we want to refine those objects and use cases to reflect the actual environment of our proposed solution.

In this milestone you will first transform the use case course of events prepared in **Milestone 11** to be a design use case. Secondly, we update our object association model to include any additional attributes we have discovered. Finally, we will create an ideal object model diagram that models the object interactions for each use case.

Objectives

After completing this milestone, you should be able to:

- ⇒ Transform a requirements use case to a design use case.
- ⇒ Construct an ideal object model diagram to model the high-level object interactions for a use case.
- ⇒ Revise an object association diagram to include attributes and hierarchies.

Prerequisites

Before starting this milestone the following topics should be covered:

1. Object-Oriented Design Chapter 16, pages 532-550
of the SADM 4th ed. Textbook.

2. Milestone 11 Solution

Project

There are three general activities when performing object-oriented design and they are as follows:

1. Refining the use case model to reflect the implementation environment.
2. Modeling object interactions and behavior that support the use case scenarios.
3. Updating the object model to reflect the implementation environment.

Activities

1. To transform the *analysis use case* to a *design use case*, follow the steps on pages 537 and 538 of the SADM 4th ed. Textbook. Use Figure 16.2, page 538-539 of the SADM 4th ed. Textbook as an example. Use the completed analysis use case from **Milestone 11** to start from.
2. To construct the *ideal model diagram*, follow the steps on page 540 of the SADM 4th ed. Textbook. Use Figure 16.5, page 541, of the SADM 4th ed. textbook as an example. Make assumptions where necessary.
3. To document additional identified attributes, use the *completed Object Association Diagram* prepared in **Milestone 11**. Make the necessary changes to this diagram to reflect the proposed environment

Deliverable format and software to be used are according to your instructor's specifications. Deliverables should be neatly packaged in a binder, separated with a tab divider labeled "Milestone 12" and accompanied with a Milestone Evaluation Sheet (appearing first) found in Appendix A.

References:

Milestone 11 Solution

Deliverables:

Design Use Case:

Due: __/__/__

Time: _____

Ideal Object Model:

Due: __/__/__

Time: _____

Refined Object Association Model:

Due: __/__/__

Time: _____

ADVANCED OPTION

For the advanced option, construct a Interaction Diagram for your completed design use case. Follow the steps to identify object behaviors and to prepare the interaction diagram. Make any necessary assumptions. Use Figure 16.10, page 546 of the SADM 4th ed. textbook as an example. Make any necessary assumptions.

Interaction Diagram:

Due: __/__/__

Time: _____

Milestone's Point Value:



APPENDIX A - MILESTONE EVALUATION SHEET

The following form should be included with each milestone submission. It will be the mechanism for your instructor to communicate the score and any associated comments to you. It can also be used for you to communicate any assumptions you may have to your instructor.

MILESTONE EVALUATION SHEET

MILESTONE NO.

--

DATE:	COURSE NO.:
	INSTRUCTOR:

TEAM NO. OR NAME	TEAM MEMBERS

Student comments:

Instructor comments:

Score:

--

NOTES

NOTES

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